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SBML Model Report

Model name: “Wodke2013 - Genome-scale constraint-based model of M.pneumoniae energy metabolism (iJW145)”



June 18, 2013

1 General Overview

This is a document in SBML Level 2 Version 1 format. Table 1 provides an overview of the quantities of all components of this model.

Table 1: Number of components in this model, which are described in the following sections.

Element	Quantity	Element	Quantity
compartment types	0	compartments	2
species types	0	species	324
events	0	constraints	0
reactions	306	function definitions	0
global parameters	0	unit definitions	1
rules	0	initial assignments	0

Model Notes

Wodke2013 - Genome-scale constraint-based model of M.pneumoniae energy metabolism (iJW145)

A new genome-scale metabolic reconstruction of M.pneumoniae is used in combination with external metabolite measurement and protein abundance measurements to quantitatively explore the energy metabolism of this genome-reduce human pathogen.

This model is described in the article: [Dissecting the energy metabolism in Mycoplasma pneumoniae through genome-scale metabolic modeling](#). Wodke JA, Puchaka J, Lluch-Senar M, Marcos J, Yus E, Godinho M, Gutierrez-Gallego R, dos Santos VA, Serrano L, Klipp E, Maier T. Mol Syst Biol. 2013;9:653.

Abstract:

Mycoplasma pneumoniae, a threatening pathogen with a minimal genome, is a model organism for bacterial systems biology for which substantial experimental information is available. With the goal of understanding the complex interactions underlying its metabolism, we analyzed and characterized the metabolic network of *M. pneumoniae* in great detail, integrating data from different omics analyses under a range of conditions into a constraint-based model backbone. Iterating model predictions, hypothesis generation, experimental testing, and model refinement, we accurately curated the network and quantitatively explored the energy metabolism. In contrast to other bacteria, *M. pneumoniae* uses most of its energy for maintenance tasks instead of growth. We show that in highly linear networks the prediction of flux distributions for different growth times allows analysis of time-dependent changes, albeit using a static model. By performing an in silico knock-out study as well as analyzing flux distributions in single and double mutant phenotypes, we demonstrated that the model accurately represents the metabolism of *M. pneumoniae*. The experimentally validated model provides a solid basis for understanding its metabolic regulatory mechanisms.

This model is hosted on [BioModels Database](#) and identified by: [MODEL1301290000](#).

To cite BioModels Database, please use: [BioModels Database: An enhanced, curated and annotated resource for published quantitative kinetic models](#).

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2 Unit Definitions

This is an overview of six unit definitions of which five are predefined by SBML and not mentioned in the model.

2.1 Unit `mmol_per_gDW_per_h`

Definition $\text{mmol} \cdot \text{g}^{-1} \cdot (2.7777 \cdot 10^{-4} \text{ s})^{-1}$

2.2 Unit `substance`

Notes Mole is the predefined SBML unit for substance.

Definition mol

2.3 Unit `volume`

Notes Litre is the predefined SBML unit for volume.

Definition l

2.4 Unit area

Notes Square metre is the predefined SBML unit for area since SBML Level 2 Version 1.

Definition m²

2.5 Unit length

Notes Metre is the predefined SBML unit for length since SBML Level 2 Version 1.

Definition m

2.6 Unit time

Notes Second is the predefined SBML unit for time.

Definition s

3 Compartments

This model contains two compartments.

Table 2: Properties of all compartments.

Id	Name	SBO	Spatial	Size	Unit	Constant	Outside
			Dimensions				
Extra_organism			3	1	litre	☒	
Cytosol			3	1	litre	☒	Extra_organism

3.1 Compartment [Extra_organism](#)

This is a three dimensional compartment with a constant size given in litre.

3.2 Compartment [Cytosol](#)

This is a three dimensional compartment with a constant size given in litre, which is surrounded by [Extra_organism](#).

4 Species

This model contains 324 species. Section 6 provides further details and the derived rates of change of each species.

Table 3: Properties of each species.

Id	Name	Compartment	Derived Unit	Constant	Boundary Condition
EC0033	Glycine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0050_b	L-Arginine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0090	Uracil[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0063	L-Tryptophan[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0007	Oxygen[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0080	D-Fructose[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0646_b	Deoxycytidine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0210_b	Nicotinate_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0125	Adenine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0098_b	Glycerol_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0027	D-Glucose[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0132_b	L-Homocysteine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0007_b	Oxygen_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0052_b	L-Glutamine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0041_b	L-Aspartate_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0151_b	L-Valine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0057	Ascorbate[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0156_b	L-Threonine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0103	D-Ribose[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0151	L-Valine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0105_b	L-Leucine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$
EC0035	L-Alanine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	$\square$	$\square$

Id	Name	Compartment	Derived Unit	Constant	Boundary Condition
EC0011	CO2[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0057_b	Ascorbate.b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0207	Pyridoxal[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0078	sn-Glycerol 3-phosphate[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0212_b	Riboflavin_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0610	Pantetheine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0300_b	Thiamin_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0063_b	L-Tryptophan_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0025_b	H2O2_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0096_b	Choline_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0146_b	Thymine_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0029	Acetate[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0065	H+[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0207_b	Pyridoxal_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0389_b	Folic acid_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0009_b	Orthophosphate_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0065_b	H+_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0050	L-Arginine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0154_b	(S)-Lactate_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0039	L-Lysine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0210	Nicotinate[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC9483_b	Phosphatidylcholine (Mpn)_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0064	L-Phenylalanine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0199	Guanine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0058	L-Methionine[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0041	L-Aspartate[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
EC0029_b	Acetate_b[e]	Extra_organism	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
EC0052	L-Glutamine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0082_b	L-Cysteine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0062	L-Ornithine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC9471_b	Fatty Acid (Mpn)_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0067	L-Tyrosine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0309	Mannitol[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0389	Folic acid[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0129	L-Asparagine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC9483	Phosphatidylcholine (Mpn)[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0001_b	H2O_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0001	H2O[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC9324	Biomass[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC9471	Fatty Acid (Mpn)[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0035_b	L-Alanine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0300	Thiamin[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0082	L-Cysteine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0025	H2O2[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0156	L-Threonine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0033_b	Glycine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0078_b	sn-Glycerol 3-phosphate_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0125_b	Adenine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0135	D-Mannose[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0309_b	Mannitol_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0090_b	Uracil_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0062_b	L-Ornithine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0064_b	L-Phenylalanine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0103_b	D-Ribose_b[e]	Extra_organism	mol · l ⁻¹	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
EC0610_b	Pantetheine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0129_b	L-Asparagine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0646	Deoxycytidine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0067_b	L-Tyrosine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0363_b	Cytidine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0011_b	CO2_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0053	L-Serine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0126_b	L-Proline_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0023_b	L-Glutamate_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0363	Cytidine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0212	Riboflavin[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0154	(S)-Lactate[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0027_b	D-Glucose_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0135_b	D-Mannose_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0023	L-Glutamate[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0013	NH3[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0096	Choline[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0105	L-Leucine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0126	L-Proline[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0013_b	NH3_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0053_b	L-Serine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0117_b	L-Histidine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0146	Thymine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0009	Orthophosphate[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0199_b	Guanine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0039_b	L-Lysine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0058_b	L-Methionine_b[e]	Extra_organism	mol · l ⁻¹	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
EC9324_b	Biomass.b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0080_b	D-Fructose.b[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0098	Glycerol[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0317	L-Isoleucine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0132	L-Homocysteine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0117	L-Histidine[e]	Extra_organism	mol · l ⁻¹	☐	☐
EC0317_b	L-Isoleucine.b[e]	Extra_organism	mol · l ⁻¹	☐	☐
C0090	Uracil	Cytosol	mol · l ⁻¹	☐	☐
C0445	D-Glyceraldehyde	Cytosol	mol · l ⁻¹	☐	☐
C0796	L-Ribulose 5-phosphate	Cytosol	mol · l ⁻¹	☐	☐
C9537	L-Glutamyl-tRNA(Gln)	Cytosol	mol · l ⁻¹	☐	☐
C9510	Glycyl-tRNA(Gly)	Cytosol	mol · l ⁻¹	☐	☐
C0018	AMP	Cytosol	mol · l ⁻¹	☐	☐
C0125	Adenine	Cytosol	mol · l ⁻¹	☐	☐
C0016	Pyridoxal phosphate	Cytosol	mol · l ⁻¹	☐	☐
C2070	L-Xylulose 5-phosphate	Cytosol	mol · l ⁻¹	☐	☐
C0055	Thiamin diphosphate	Cytosol	mol · l ⁻¹	☐	☐
C0190	D-Xylulose 5-phosphate	Cytosol	mol · l ⁻¹	☐	☐
C9531	tRNA(Ile)	Cytosol	mol · l ⁻¹	☐	☐
C0233	dGTP	Cytosol	mol · l ⁻¹	☐	☐
C0454	Choline phosphate	Cytosol	mol · l ⁻¹	☐	☐
C0822	Pantetheine 4'-phosphate	Cytosol	mol · l ⁻¹	☐	☐
C0020	Pyruvate	Cytosol	mol · l ⁻¹	☐	☐
C0163	3-Phospho-D-glycerate	Cytosol	mol · l ⁻¹	☐	☐
C9524	tRNA(Gly)	Cytosol	mol · l ⁻¹	☐	☐
C0089	UMP	Cytosol	mol · l ⁻¹	☐	☐
C0151	L-Valine	Cytosol	mol · l ⁻¹	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0175	Adenosine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0031	GDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9319	Reduced thioredoxin	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0504	2-Deoxy-D-ribose 5-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0343	5,10-Methenyltetrahydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0035	L-Alanine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0011	CO ₂	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0078	sn-Glycerol 3-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0093	Dihydroxyacetone phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9528	L-Seryl-tRNA(Ser)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0292	dTMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9540	L-AsparaginyI-tRNA(Asn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9262	acyl carrier protein	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0029	Acetate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0325	Dihydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0065	H ⁺	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0291	dTDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0199	Guanine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0058	L-Methionine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0015	FAD	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0060	UTP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9282	L-Glutamyl-tRNA(Glu)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0284	D-Fructose 1,6-bisphosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0062	L-Ornithine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0006_b	NADP ⁺ _b	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0067	L-Tyrosine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0165	D-Ribulose 5-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9482	Cardiolipin (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0289	dGDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9535	tRNA(Leu)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9515	tRNA(Arg)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0198	dCMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9473	1-acyl-Glycerol 3-phosphate (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9485_b	5mcDNA (Mpn)_b	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0389	Folic acid	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0129	L-Asparagine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0017	S-Adenosyl-L-methionine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0002	ATP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9481	Phosphatidyl Glycerol (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9520	tRNA(Lys)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9532	L-Tyrosyl-tRNA(Tyr)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0859	Nicotinate D-ribonucleotide	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9471	Fatty Acid (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0082	L-Cysteine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0228	D-Erythrose 4-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0268	L-Citrulline	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0014	UDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9538	tRNA(Gln)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0069	Acetaldehyde	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9502	tRNA(Phe)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0049	FMN	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9484	DNA (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9475	Diacylglycerol (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0646	Deoxycytidine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9512	L-Valyl-tRNA(Val)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0446	Dihydrolipoamide	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9498	RNA (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9489	glycolipid (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9506	L-Isoleucyl-tRNA(Ile)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0212	Riboflavin	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0154	(S)-Lactate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9499	L-Alanyl-tRNA(Ala)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0352	dTTP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9480	Phosphatidyl Glycerol-3-Phosphate (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9527	tRNA(Val)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9517	L-Threonyl-tRNA(Thr)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0023	L-Glutamate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0096	Choline	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0126	L-Proline	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9508	L-Histidyl-tRNA(His)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0047	Formate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9519	L-CysteinyI-tRNA(Cys)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0008	ADP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0098	Glycerol	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9255	3-keto-L-gulonate-6-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9539	tRNA(Asn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0043	UDP-D-galactose	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0117	L-Histidine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9504	L-Leucyl-tRNA(Leu)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0485	D-Mannitol 1-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0195	1,3-Bisphospho-D-glycerate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0005	NADPH	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9534	tRNA(Trp)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0170	dADP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0033	Glycine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0063	L-Tryptophan	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9462	L-ascorbate 6 phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9522	tRNA(Met)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0007	Oxygen	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0523	dCDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9468	N-Formylmethionyl-tRNA(Met)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0026	UDPglucose	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0227	D-Mannose 6-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9557	DnaK-folded protein (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0193	10-Formyltetrahydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9500	tRNA(Ala)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0288	dAMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0099	D-Ribose 5-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0433	Deoxyadenosine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0351	dCTP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0647	Dephospho-CoA	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0790	D-Fructose 1-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9264	apoprotein [acyl carrier protein]	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0230	Sedoheptulose 7-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C2169	5-Formyltetrahydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0103	D-Ribose	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0477	2-Phospho-D-glycerate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0010	CoA	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0113	dATP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0152	Glycerone	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9530	tRNA(Tyr)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0142	Carbamoyl phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0207	Pyridoxal	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9525	tRNA(His)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0087	D-Glucose 1-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0610	Pantetheine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0077	D-Glucose 6-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0290	dGMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0123	GMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0050	L-Arginine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9558	GroEL-folded protein (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0039	L-Lysine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0210	Nicotinate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0064	L-Phenylalanine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0012	Pyrophosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0041	L-Aspartate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0052	L-Glutamine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9485	5mcDNA (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9474	phosphatidic acid (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0045	Adenosine 3',5'-bisphosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0019	S-Adenosyl-L-homocysteine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9521	tRNA(Thr)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0293	dUMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C0306	Guanosine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0503	2-Deoxy-D-ribose 1-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9320	tRNA(Glu)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0070	D-Fructose 6-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0470	D-Ribose 1-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0188	Acetyl phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9541	L-Glutaminyl-tRNA(Gln)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9483	Phosphatidylcholine (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9324	Biomass	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0001	H2O	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9518	tRNA(Asp)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0100	D-Glyceraldehyde 3-phosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0300	Thiamin	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0824	S-Acetyldihydrolipoamide	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0025	H2O2	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9472	ACP-R (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0156	L-Threonine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0250	CDPcholine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0059	Phosphoenolpyruvate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0501	sn-Glycero-3-phosphocholine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0960	dUDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0177	Thymidine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0271	Deoxyguanosine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9509	tRNA(Pro)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0053	L-Serine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9505	L-Aspartyl-tRNA(Asp)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9318	Oxidized thioredoxin	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condition
C9507	L-Arginyl-tRNA(Arg)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0122	5,10-Methylenetetrahydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9523	tRNA(Cys)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9516	L-Lysyl-tRNA(Lys)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0205	Lipoamide	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0022	Acetyl-CoA	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0045_b	Adenosine 3',5'-bisphosphate_b	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0363	Cytidine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9513	L-Prolyl-tRNA(Pro)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0046	CMP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0013	NH3	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9479	CDP-diacylglycerol (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0105	L-Leucine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9467	L-Methionyl-tRNA(Met)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0006	NADP+	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0085	Tetrahydrofolate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9526	tRNA(Ser)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0630	Deamino-NAD+	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0003	NAD+	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0094	CDP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0146	Thymine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0051	CTP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0009	Orthophosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0408	Deoxyuridine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0038	GTP	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0004	NADH	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0317	L-Isoleucine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

Id	Name	Compartment	Derived Unit	Constant	Boundary Condi- tion
C9533	L-Tryptophanyl-tRNA(Trp)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9503	L-Phenylalanyl-tRNA(Phe)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0132	L-Homocysteine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C9536	Protein (Mpn)	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0781	Thiamin monophosphate	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐
C0242	Uridine	Cytosol	$\text{mol} \cdot \text{l}^{-1}$	☐	☐

5 Reactions

This model contains 306 reactions. All reactions are listed in the following table and are subsequently described in detail. If a reaction is affected by a modifier, the identifier of this species is written above the reaction arrow.

Table 4: Overview of all reactions

Nº	Id	Name	Reaction Equation	SBO
1	RR00685	D-Glyceraldehyde-3-phosphate:NAD+ oxidoreductase(phosphorylating)	$C0003 + C0100 + C0009 \rightleftharpoons C0195 + C0065 + C0004$	
2	RR09116	SINK: L-Threonine[e]	$EC0156 \rightleftharpoons EC0156.b$	
3	RR09462	SINK: L-Leucine[e]	$EC0105 \rightleftharpoons EC0105.b$	
4	RR01850	L-Xylose 5-phosphate 3-epimerase	$C0796 \rightleftharpoons C2070$	
5	RR09050	SINK: L-Arginine[e]	$EC0050 \rightleftharpoons EC0050.b$	
6	RR04369	TRDR	$C9318 + C0065 + C0005 \rightleftharpoons C9319 + C0006$	
7	RR09428	SINK: L-Cysteine[e]	$EC0082 \rightleftharpoons EC0082.b$	
8	RR03512	AMP:pyrophosphate phosphoribosyltransferase	$C0012 + C0018 \rightleftharpoons C0125 + C0101$	
9	RR00521	sn-Glycerol-3-phosphate:oxygen oxidoreductase	$C0007 + C0078 \rightleftharpoons C0025 + C0093$	2-
10	RR01147	ATP:dGMP phosphotransferase	$C0002 + C0290 \rightleftharpoons C0008 + C0289$	
11	RR00598	5,10-Methylenetetrahydrofolate:glycine hydroxymethyltransferase	$C0122 + C0001 + C0033 \rightleftharpoons C0053 + C0085$	
12	RR03709	Adenosine:orthophosphate ribosyltransferase	$C0009 + 2 C0065 + C0175 \rightleftharpoons C0125 + C0470$	
13	RR01475	Deoxyadenosine:orthophosphate ribosyltransferase	$C0009 + C0433 \rightleftharpoons C0503 + C0125$	
14	RR01018	Sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphateglyceronetransferase	$C0100 + C0230 \rightleftharpoons C0228 + C0070$	
15	RR09769	ascorbate hydrolase	$C9462 + C0001 \rightleftharpoons C9255 + 3 C0065$	
16	RR09104	SINK: Adenine[e]	$EC0125 \rightleftharpoons EC0125.b$	

Nº	Id	Name	Reaction Equation	SBO
17	RR01592	2-Deoxy-D-ribose 1-phosphate phosphomutase	1,5- C0503 $\rightleftharpoons$ C0504	
18	RR08870	ADEt2r	EC0125 + EC0065 $\rightleftharpoons$ C0065 + C0125	
19	RR00667	Glycerol:NAD+ oxidoreductase	C0003 + C0098 $\rightleftharpoons$ C0065 + C0445 + C0004	
20	RR09796	dGTP:3-phospho-D-glycerate phosphotransferase	1- C0233 + C0163 $\rightleftharpoons$ C0195 + C0289	
21	RR00690	D-Fructose-1,6-bisphosphate glyceraldehyde-3-phosphate-lyase	D- C0284 $\rightleftharpoons$ C0100 + C0093	
22	RR03544	Acetaldehyde:NAD+ oxidoreductase (CoA-acetylating)	C0003 + C0069 + C0010 $\rightleftharpoons$ C0022 + C0065 + C0004	
23	RR09070	SINK: L-Glutamate[e]	EC0023 $\rightleftharpoons$ EC0023_b	
24	RR08739	glycerol transport via channel	C0098 $\rightleftharpoons$ EC0098	
25	RR00779	GMP:pyrophosphate phosphoribosyltransferase	C0012 + C0123 $\rightleftharpoons$ C0199 + C0101	
26	RR01146	ATP:deoxyadenosine 5'-phosphotransferase	C0002 + C0433 $\rightleftharpoons$ C0288 + C0065 + C0008	
27	RR03808	ATP:deoxyguanosine 5'-phosphotransferase	C0002 + C0271 $\rightleftharpoons$ C0065 + C0290 + C0008	
28	RR08868	CHLt2r	EC0065 + EC0096 $\rightleftharpoons$ C0096 + C0065	
29	RR09830	FAttransp	EC9471 $\rightleftharpoons$ C9471	
30	RR09052	SINK: L-Aspartate[e]	EC0041 $\rightleftharpoons$ EC0041_b	
31	RR08615	arginine/ornithine antiporter	EC0050 + C0062 $\rightleftharpoons$ C0050 + EC0062	
32	RR00452	D-Mannitol-1-phosphate:NAD+ oxidoreductase	5- C0485 + C0003 $\rightleftharpoons$ C0065 + C0070 + C0004	
33	RR00902	5,10-Methenyltetrahydrofolate hydrolase(decyclizing)	5- C0343 + C0001 $\rightleftharpoons$ C0193	
34	RR03810	Deoxyguanosine:orthophosphate ribosyltransferase	C0009 + C0271 $\rightleftharpoons$ C0503 + C0199	
35	RR09088	SINK: L-Lysine[e]	EC0039 $\rightleftharpoons$ EC0039_b	
36	RR00106	ATP:carbamate phosphotransferase	C0002 + C0011 + C0013 $\rightleftharpoons$ C0142 + C0065 + C0008	

Nº	Id	Name	Reaction Equation	SBO
37	RR01184	Guanosine:orthophosphate ribosyltransferase	$C0009 + 2 C0065 + C0306 \rightleftharpoons C0470 + C0199$	
38	RR00275	Formate:NAD ⁺ oxidoreductase	$C0047 + C0003 \rightleftharpoons C0004 + C0011$	
39	RR00157	ATP:acetate phosphotransferase	$C0029 + C0002 \rightleftharpoons C0188 + C0008$	
40	RR09421	SINK: L-Asparagine[e]	$EC0129 \rightleftharpoons EC0129_b$	
41	RR09857	phosphatidylcholine avylhydrolase (Mpn)	$C9483 + 2 C0001 \rightleftharpoons 2 C9471 + C0501$	
42	RR03873	ATP:nicotinamide-nucleotide adenylyltransferase	$C0002 + C0065 + C0859 \rightleftharpoons C0012 + C0630$	
43	RR03769	Sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphateglycolaldehyde transferase	$C0100 + C0230 \rightleftharpoons C0190 + C0099$	
44	RR03681	2-Phospho-D-glycerate 2,3-phosphomutase	$C0477 \rightleftharpoons C0163$	
45	RR00689	transketolase	$C0100 + C0070 \rightleftharpoons C0228 + C0190$	
46	RR09082	SINK: L-Alanine[e]	$EC0035 \rightleftharpoons EC0035_b$	
47	RR09054	SINK: CO ₂ [e]	$EC0011 \rightleftharpoons EC0011_b$	
48	RR09458	SINK: L-Isoleucine[e]	$EC0317 \rightleftharpoons EC0317_b$	
49	RR01151	ATP:dTMP phosphotransferase	$C0292 + C0002 \rightleftharpoons C0008 + C0291$	
50	RR01158	5,10-Methylenetetrahydrofolate:dUMP C-methyltransferase	$C0293 + C0122 \rightleftharpoons C0325 + C0292$	
51	RR09076	SINK: Glycine[e]	$EC0033 \rightleftharpoons EC0033_b$	
52	RR00910	ATP:dCMP phosphotransferase	$C0002 + C0198 \rightleftharpoons C0523 + C0008$	
53	RR08593	ATP synthase (four protons for one ATP)	$C0009 + 4 EC0065 + C0008 \rightleftharpoons C0002 + 3 C0065 + C0001$	
54	RR01236	Dihydrofolate:NAD ⁺ oxidoreductase	$C0325 + C0003 \rightleftharpoons C0389 + C0004$	
55	RR08860	cytidine transport in via proton symport, reversible	$EC0065 + EC0363 \rightleftharpoons C0065 + C0363$	
56	RR00771	5,10-Methylenetetrahydrofolate:NADP ⁺ oxidoreductase	$C0122 + C0006 \rightleftharpoons C0343 + C0065 + C0005$	
57	RR09795	dATP:3-phospho-D-glycerate phosphotransferase	1- $C0113 + C0163 \rightleftharpoons C0195 + C0170$	

Nº	Id	Name	Reaction Equation	SBO
58	RR00682	D-Ribose 1,5-phosphomutase	$C0470 \rightleftharpoons 2 C0065 + C0099$	
59	RR08672	CO2 transporter via diffusion	$EC0011 \rightleftharpoons C0011$	
60	RR09493	SINK: L-Tryptophan[e]	$EC0063 \rightleftharpoons EC0063.b$	
61	RR03649	Carbamoyl-phosphate:L-ornithine bamoyltransferase	car- $C0142 + C0062 \rightleftharpoons C0009 + C0065 + C0268$	
62	RR09789	H2O2t	$EC0025 \rightleftharpoons C0025$	
63	RR03195	L-ribulose-5-phosphate 4-epimerase	$C0796 \rightleftharpoons C0190$	
64	RR09471	SINK: L-Methionine[e]	$EC0058 \rightleftharpoons EC0058.b$	
65	RR01155	ATP:dUMP phosphotransferase	$C0002 + C0293 \rightleftharpoons C0960 + C0008$	
66	RR08971	SOURCE: H+[e]	$EC0065.b \rightleftharpoons EC0065$	
67	RR08879	adenylate kinase	$C0018 + C0002 \rightleftharpoons 2 C0008$	
68	RR08670	O2t	$EC0007 \rightleftharpoons C0007$	
69	RR00147	UDPglucose 4-epimerase	$C0026 \rightleftharpoons C0043$	
70	RR00662	sn-Glycero-3-phosphocholine phohydrolase	glycerophos- $C0501 + C0001 \rightleftharpoons C0096 + C0065 + C0078$	
71	RR03675	ATP:3-phospho-D-glycerate phosphotransferase	1- $C0002 + C0163 \rightleftharpoons C0195 + C0008$	
72	RR00590	5,6,7,8-Tetrahydrofolate:NAD+ tase	oxidoreduc- $C0003 + C0085 \rightleftharpoons C0325 + C0065 + C0004$	
73	RR00465	D-Mannose-6-phosphate ketol-isomerase	$C0227 \rightleftharpoons C0070$	
74	RR00268	ATP:CMP phosphotransferase	$C0002 + C0046 \rightleftharpoons C0094 + C0008$	
75	RR03698	ATP:dAMP phosphotransferase	$C0288 + C0002 \rightleftharpoons C0170 + C0008$	
76	RR08939	phosphoglucomutase	$C0087 \rightleftharpoons C0077$	
77	RR03717	Thymidine:orthophosphate transferase	deoxyribosyl- $C0009 + C0177 \rightleftharpoons C0503 + C0146$	
78	RR03680	GTP:3-phospho-D-glycerate phosphotransferase	1- $C0163 + C0038 \rightleftharpoons C0195 + C0031$	
79	RR01416	Deoxyuridine:orthophosphate ferase	ribosyltrans- $C0009 + C0408 \rightleftharpoons C0503 + C0090$	

Nº	Id	Name	Reaction Equation	SBO
80	RR09484	SINK: L-Serine[e]	$EC0053 \rightleftharpoons EC0053.b$	
81	RR01052	Uridine:orthophosphate ribosyltransferase	$C0009 + 2 C0065 + C0242 \rightleftharpoons C0090 + C0470$	
82	RR00404	(S)-Lactate:NAD ⁺ oxidoreductase	$C0003 + C0154 \rightleftharpoons C0020 + C0065 + C0004$	
83	RR09837	SOURCE: Fatty Acid (Mpn) [e]	$EC9471.b \rightleftharpoons EC9471$	
84	RR09229	SOURCE: Water[e]	$EC0001.b \rightleftharpoons EC0001$	
85	RR08674	H ₂ O transport via diffusion	$EC0001 \rightleftharpoons C0001$	
86	RR09086	SINK: L-Histidine[e]	$EC0117 \rightleftharpoons EC0117.b$	
87	RR09498	SINK: L-Valine[e]	$EC0151 \rightleftharpoons EC0151.b$	
88	RR09084	SINK: L-Glutamine[e]	$EC0052 \rightleftharpoons EC0052.b$	
89	RR09889	NADH oxidase (H ₂ O-forming)	$C0007 + 2 C0065 + 2 C0004 \rightleftharpoons 2 C0003 + 2 C0001$	
90	RR00617	UMP:pyrophosphate phosphoribosyltransferase	$C0012 + C0089 \rightleftharpoons C0090 + C0101$	
91	RR00464	D-Glucose-6-phosphate ketol-isomerase	$C0077 \rightleftharpoons C0070$	
92	RR09100	SINK: Orthophosphate[e]	$EC0009 \rightleftharpoons EC0009.b$	
93	RR00952	Nicotinate D-ribonucleotide:pyrophosphate phosphoribosyltransferase	$C0012 + C0859 \rightleftharpoons C0065 + C0210 + C0101$	
94	RR09496	SINK: L-Tyrosine[e]	$EC0067 \rightleftharpoons EC0067.b$	
95	RR09098	SINK: L-Phenylalanine[e]	$EC0064 \rightleftharpoons EC0064.b$	
96	RR03546	Acetyl-CoA:orthophosphate acetyltransferase	$C0022 + C0009 \rightleftharpoons C0188 + C0010$	
97	RR09124	SINK: NH ₃ [e]	$EC0013 \rightleftharpoons EC0013.b$	
98	RR00146	UTP:alpha-D-glucose-1-phosphate uridylyltransferase	$C0060 + C0065 + C0087 \rightleftharpoons C0012 + C0026$	
99	RR09342	2-oxoglutarate dehydrogenase, lipoamidedehydrogenase compone	$C0003 + C0446 \rightleftharpoons C0065 + C0205 + C0004$	
100	RR00111	ATP:UMP phosphotransferase	$C0002 + C0089 \rightleftharpoons C0014 + C0008$	
101	RR00681	D-Ribose-5-phosphate ketol-isomerase	$C0099 \rightleftharpoons C0165$	
102	RR00362	2-Phospho-D-glycerate hydro-lyase	$C0477 \rightleftharpoons C0059 + C0001$	

Nº	Id	Name	Reaction Equation	SBO
103	RR08636	URAt2r	$EC0090 + EC0065 \rightleftharpoons C0090 + C0065$	
104	RR00652	D-Glyceraldehyde-3-phosphate isomerase	ketol- $C0100 \rightleftharpoons C0093$	
105	RR03691	D-Ribulose-5-phosphate 3-epimerase	$C0165 \rightleftharpoons C0190$	
106	RR09882	SINK: Biomass[e]	$EC9324 \rightleftharpoons EC9324.b$	
107	RR09090	SINK: L-Proline[e]	$EC0126 \rightleftharpoons EC0126.b$	
108	RR01487	Acetyl-CoA:dihydrolipoamide acetyltransferase	S- $C0446 + C0022 \rightleftharpoons C0010 + C0824$	
109	RR09424	SINK: Choline[e]	$EC0096 \rightleftharpoons EC0096.b$	
110	RR00171	ATP:GMP phosphotransferase	$C0123 + C0002 \rightleftharpoons C0008 + C0031$	
111	RR09841	NH ₃ export	$C0013 \rightleftharpoons EC0013$	
112	IR09820	acyl-ACP synthase (Mpn)	$C9262 + C0002 + C9471 \longrightarrow C0012 + C0018 + C0065 + C9472$	
113	IR08663	CYSabc	$EC0082 + C0002 + C0001 \longrightarrow C0082 + C0009 + C0065 + C0008$	
114	IR03514	S-Adenosyl-L-homocysteine hydrolase	$C0019 + C0001 \longrightarrow C0132 + C0175$	
115	IR08684	L-histidine reversible transport via proton symport (R)	$C0065 + C0117 \longrightarrow EC0117 + EC0065$	
116	IR00522	ATP:glycerol 3-phosphotransferase	$C0002 + C0098 \longrightarrow C0065 + C0008 + C0078$	
117	IR09822	1-acyl-sn-glycerol-3-phosphate acyltransferase	$C9473 + C9472 \longrightarrow C9262 + C9474$	
118	IR09801	SOURCE: Pyridoxal[e]	$EC0207.b \longrightarrow EC0207$	
119	IR09815	folate transporter	$EC0389 \longrightarrow C0389$	
120	IR07938	GTP:pyruvate O ₂ -phosphotransferase (R)	$C0065 + C0059 + C0031 \longrightarrow C0020 + C0038$	
121	IR08692	ILEt2r (R)	$C0317 + C0065 \longrightarrow EC0317 + EC0065$	
122	IR09924	ASPt2 (R)	$C0065 + C0041 \longrightarrow EC0041 + EC0065$	
123	IR09910	AsparaginyI-tRNA synthetase	$C9539 + C0002 + C0129 \longrightarrow C0012 + C0018 + C9540$	

Nº	Id	Name	Reaction Equation	SBO
124	IR09934	apoACP synthesis	$ \begin{aligned} &3 \text{ C9467} + \text{C9510} + 4 \text{ C9541} + 12 \text{ C9504} + \\ &3 \text{ C9540} + 4 \text{ C9499} + 5 \text{ C9528} + 6 \text{ C9505} + \\ &\text{C9468} + \text{C9508} + 5 \text{ C9512} + 5 \text{ C9503} + 2 \text{ C9507} + \\ &199 \text{ C0001} + \text{C9517} + 199 \text{ C0002} + 8 \text{ C9506} + \\ &\text{C9513} + 11 \text{ C9516} + 11 \text{ C9282} \longrightarrow 3 \text{ C9539} + \\ &\text{C9264} + 11 \text{ C9320} + \text{C9521} + 199 \text{ C0009} + \\ &5 \text{ C9527} + 4 \text{ C9500} + 2 \text{ C9515} + 199 \text{ C0008} + \\ &5 \text{ C9526} + 4 \text{ C9522} + \text{C9509} + 8 \text{ C9531} + \\ &6 \text{ C9518} + \text{C9525} + 5 \text{ C9502} + 282 \text{ C0065} + \\ &\text{C9524} + 12 \text{ C9535} + 11 \text{ C9520} + 4 \text{ C9538} \end{aligned} $	
125	IR08686	PHEt2r (R)	$\text{C0064} + \text{C0065} \longrightarrow \text{EC0064} + \text{EC0065}$	
126	IR09898	Leucyl-tRNA synthetase	$\text{C0002} + \text{C0105} + \text{C9535} \longrightarrow \text{C0012} + \text{C0018} + \text{C9504}$	
127	IR09808	SOURCE: Cytidine[e]	$\text{EC0363.b} \longrightarrow \text{EC0363}$	
128	IR09845	L-Serine transport via ABC System	$\text{C0002} + \text{EC0053} + \text{C0001} \longrightarrow \text{C0053} + \text{C0009} + \text{C0065} + \text{C0008}$	
129	IR08746	ASPabc	$\text{EC0041} + \text{C0002} + \text{C0001} \longrightarrow \text{C0009} + \text{C0065} + \text{C0008} + \text{C0041}$	
130	IR09771	SOURCE: (L)-Ascorbate[e]	$\text{EC0057.b} \longrightarrow \text{EC0057}$	
131	IR07898	ATP:pyruvate O2-phosphotransferase (R)	$\text{C0065} + \text{C0059} + \text{C0008} \longrightarrow \text{C0020} + \text{C0002}$	
132	IR09198	SINK: Adenosine 3',5'-bisphosphate[i]	$\text{C0045} \longrightarrow \text{C0045.b}$	
133	IR09956	DnaK catalyzed protein folding (Mpn)	$ \begin{aligned} &\text{C9536} + \text{C0002} + \text{C0001} \longrightarrow \text{C0009} + \text{C0065} + \\ &\text{C0008} + \text{C9557} \end{aligned} $	
134	IR08910	fructose-1-phosphate kinase	$\text{C0790} + \text{C0002} \longrightarrow \text{C0284} + \text{C0065} + \text{C0008}$	
135	IR04342	RNDR4	$\text{C0014} + \text{C9319} \longrightarrow \text{C9318} + \text{C0960} + \text{C0001}$	
136	IR03894	ATP:pantetheine-4'-phosphate adenylyltransferase	$\text{C0002} + \text{C0065} + \text{C0822} \longrightarrow \text{C0647} + \text{C0012}$	
137	IR04690	CTP:pyruvate O2-phosphotransferase (R)	$\text{C0065} + \text{C0059} + \text{C0094} \longrightarrow \text{C0051} + \text{C0020}$	
138	IR03497	ATP:pyridoxal 5'-phosphotransferase	$\text{C0002} + \text{C0207} \longrightarrow \text{C0065} + \text{C0008} + \text{C0016}$	

Nº	Id	Name	Reaction Equation	SBO
139	IR09829	cardiolipin synthase (Mpn)	$C9479 + C9481 \longrightarrow C9482 + C0065 + C0046$	
140	IR04340	RNDR1	$C9319 + C0008 \longrightarrow C9318 + C0170 + C0001$	
141	IR09590	EX_pi(e)	$C0009 \longrightarrow EC0009$	
142	IR00932	Pyruvate:lipoamide oxidoreductase(decarboxylating and acceptor- acetylating)	$C0020 + C0065 + C0205 \longrightarrow C0011 + C0824$	
143	IR09954	protein degradation (Mpn)	$C9536 + 688 C0002 + 1032 C0001 \longrightarrow 22 C0317 + 14 C0126 + 32 C0039 + 688 C0008 + 10 C0067 + 21 C0156 + 6 C0117 + 18 C0041 + 13 C0050 + 3 C0082 + 688 C0065 + 29 C0105 + 3 C0063 + 15 C0064 + 688 C0009 + 27 C0035 + 25 C0151 + 17 C0052 + 22 C0023 + 21 C0053 + 6 C0058 + 20 C0129 + 21 C0033$	
144	IR08822	PROt2r (R)	$C0126 + C0065 \longrightarrow EC0126 + EC0065$	
145	IR09079	SOURCE: Guanine[e]	$EC0199.b \longrightarrow EC0199$	
146	IR08755	L-lysine transport via ABC system	$EC0039 + C0002 + C0001 \longrightarrow C0009 + C0039 + C0065 + C0008$	
147	IR08750	GLUt2r (R)	$C0065 + C0023 \longrightarrow EC0023 + EC0065$	
148	IR08661	L-asparagine reversible transport via proton symport (R)	$C0065 + C0129 \longrightarrow EC0065 + EC0129$	
149	IR09109	SOURCE: D-Ribose[e]	$EC0103.b \longrightarrow EC0103$	
150	IR09906	Lysyl-tRNA synthetase	$C0002 + C0039 + C9520 \longrightarrow C0012 + C0018 + C9516$	
151	IR09798	SOURCE: Folic Acid[e]	$EC0389.b \longrightarrow EC0389$	
152	IR08786	MANpts	$C0059 + EC0135 \longrightarrow C0020 + C0227$	
153	IR03510	Adenosine 3',5'-bisphosphate 3'(2')- phosphohydrolase	$C0045 + C0001 \longrightarrow C0018 + C0009$	
154	IR00114	ATP:FMN adenylyltransferase	$C0002 + C0065 + C0049 \longrightarrow C0012 + C0015$	

Nº	Id	Name	Reaction Equation	SBO
155	IR09895	Tyrosyl-tRNA synthetase	$C0002 + C0067 + C9530 \longrightarrow C0012 + C0018 + C9532$	
156	IR09803	SOURCE: Thiamine[e]	$EC0300.b \longrightarrow EC0300$	
157	IR08807	DCYTt2	$EC0065 + EC0646 \longrightarrow C0646 + C0065$	
158	IR08707	glycine reversible transport via proton sym- port (R)	$C0065 + C0033 \longrightarrow EC0033 + EC0065$	
159	IR00649	ATP:glycerone phosphotransferase	$C0152 + C0002 \longrightarrow C0065 + C0093 + C0008$	
160	IR08754	L-histidine transport via ABC system	$EC0117 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0117$	
161	IR09905	Isoleucyl-tRNA synthetase	$C9531 + C0317 + C0002 \longrightarrow C0012 + C0018 + C9506$	
162	IR09773	Ascorbate PTS	$2 C0065 + C0059 + EC0057 \longrightarrow C0020 + C9462$	
163	IR08829	D-fructose transport via PEP:Pyr PTS	$C0059 + EC0080 \longrightarrow C0790 + C0020$	
164	IR09863	DNA methylierung (Mpn)	$C0017 + C9484 \longrightarrow C9485 + C0019$	
165	IR00648	Glycerone phosphate phosphohydrolase (al- kaline optimum)	$C0093 + C0001 \longrightarrow C0152 + C0009$	
166	IR09847	ACP phosphodiesterase	$C9262 + C0001 \longrightarrow C9264 + C0065 + C0822$	
167	IR09884	RNA synthesis (Mpn)	$18 C0051 + 25 C0060 + 69 C0002 + 40 C0001 + 28 C0038 \longrightarrow 100 C0012 + 40 C0009 + 40 C0065 + C9498 + 40 C0008$	
168	IR09860	Phosphatidylcholine ABC transporter (Mpn)	$C0002 + EC9483 + C0001 \longrightarrow C9483 + C0009 + C0065 + C0008$	
169	IR08768	THRabc	$EC0156 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0156$	
170	IR09958	TYRabc	$EC0067 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0067 + C0008$	
171	IR00656	ATP:choline phosphotransferase	$C0096 + C0002 \longrightarrow C0454 + C0065 + C0008$	
172	IR09899	Glycyl-tRNA synthetase	$C0002 + C9524 + C0033 \longrightarrow C0012 + C0018 + C9510$	

Nº	Id	Name	Reaction Equation	SBO
173	IR09902	Valyl-tRNA synthetase	$C0002 + C9527 + C0151 \longrightarrow C0012 + C0018 + C9512$	
174	IR09871	glycolipid synthesis (Mpn)	$3 C0043 + 3 C0026 + C9475 \longrightarrow 6 C0014 + 6 C0065 + C9489$	
175	IR09886	Lactate export (Mpn)	$C0065 + C0154 \longrightarrow EC0154 + EC0065$	
176	IR09957	TRPabc	$C0002 + EC0063 + C0001 \longrightarrow C0063 + C0009 + C0065 + C0008$	
177	IR09891	Alanyl-tRNA synthetase	$C0002 + C9500 + C0035 \longrightarrow C0012 + C0018 + C9499$	
178	IR08844	L-tryptophan reversible transport via proton symport (R)	$C0063 + C0065 \longrightarrow EC0063 + EC0065$	
179	IR00911	ATP:deoxycytidine 5'-phosphotransferase	$C0002 + C0646 \longrightarrow C0065 + C0008 + C0198$	
180	IR09865	homocysteine export (Mpn)	$C0132 \longrightarrow EC0132$	
181	IR08970	SOURCE: Glucose[e]	$EC0027.b \longrightarrow EC0027$	
182	IR09913	glutaminyl-tRNA synthetase (Glu)	$C9537 + C0002 + C0052 + C0001 \longrightarrow C0009 + C9541 + C0065 + C0008 + C0023$	
183	IR09893	Methionyl-tRNA synthetase	$C0002 + C0058 + C9522 \longrightarrow C0012 + C0018 + C9467$	
184	IR04284	Glutamyl-tRNA synthetase	$C9320 + C0002 + C0023 \longrightarrow C0012 + C0018 + C9282$	
185	IR08984	atp consumption proxy	$C0002 + C0001 \longrightarrow C0009 + C0065 + C0008$	
186	IR00674	ATP:D-ribose-5-phosphate pyrophospho-transferase	$C0002 + C0099 \longrightarrow C0018 + C0065 + C0101$	
187	IR08771	LEUabc	$C0002 + EC0105 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0105$	
188	IR08659	L-asparagine transport via ABC system	$C0002 + EC0129 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0129$	
189	IR09827	phosphatidylglycerophosphate synthase	$C9479 + C0078 \longrightarrow C9480 + C0065 + C0046$	

Nº	Id	Name	Reaction Equation	SBO
190	IR08824	phosphate transport via ABC system	$EC0009 + C0002 + C0001 \longrightarrow 2 C0009 + C0065 + C0008$	
191	IR00596	Formate:tetrahydrofolate ligase (ADP-forming)	$C0047 + C0002 + C0085 \longrightarrow C0193 + C0009 + C0008$	
192	IR08635	ALAt2r (R)	$C0065 + C0035 \longrightarrow EC0035 + EC0065$	
193	IR09961	GroEL catalyzed protein folding (Mpn)	$C9536 + 7 C0002 + 7 C0001 \longrightarrow 7 C0009 + C9558 + 7 C0065 + 7 C0008$	
194	IR08738	L-glutamine transport via ABC system	$EC0052 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0052$	
195	IR09821	glycerol-3-phosphate O-acyltransferase (Mpn)	$C9472 + C0078 \longrightarrow C9262 + C9473$	
196	IR09926	GLNt2 (R)	$C0065 + C0052 \longrightarrow EC0052 + EC0065$	
197	IR09807	SOURCE: Uracil[e]	$EC0090_b \longrightarrow EC0090$	
198	IR08208	S-Aminomethyldihydrolipoylprotein:(6S)-tetrahydrofolateaminomethyltransferase (ammonia-forming) (R)	$C0343 + C0065 + C0001 \longrightarrow C2169$	
199	IR09095	SOURCE: Oxygen[e]	$EC0007_b \longrightarrow EC0007$	
200	IR09792	pyruvate kinase 3	$C0523 + C0065 + C0059 \longrightarrow C0351 + C0020$	
201	IR01710	ATP:pantothenate 4'-phosphotransferase	$C0002 + C0610 \longrightarrow C0065 + C0008 + C0822$	
202	IR03511	Deamino-NAD+:ammonia ligase (AMP-forming)	$C0002 + C0630 + C0013 \longrightarrow C0003 + C0012 + C0018$	
203	IR09908	Phenylalanyl-tRNA synthetase	$C0064 + C0002 + C9502 \longrightarrow C0012 + C0018 + C9503$	
204	IR05416	dGTP:pyruvate O2-phosphotransferase (R)	$C0065 + C0059 + C0289 \longrightarrow C0020 + C0233$	
205	IR09271	SINK: NADP+[i]	$C0006 \longrightarrow C0006_b$	
206	IR01054	Cytidine aminohydrolase	$C0363 + C0001 \longrightarrow C0242 + C0013$	
207	IR09818	riboflavin transporter	$EC0212 \longrightarrow C0212$	
208	IR09824	phosphatidate phosphatase	$C9474 + C0001 \longrightarrow C0009 + C9475$	
209	IR09925	CYSt2r (R)	$C0082 + C0065 \longrightarrow EC0082 + EC0065$	

Nº	Id	Name	Reaction Equation	SBO
210	IR04341	RNDR2	$C9319 + C0031 \longrightarrow C9318 + C0289 + C0001$	
211	IR09816	pantetheine transporter	$EC0610 \longrightarrow C0610$	
212	IR08826	MNLpts	$EC0309 + C0059 \longrightarrow C0485 + C0020$	
213	IR08835	SERt2r (R)	$C0053 + C0065 \longrightarrow EC0065 + EC0053$	
214	IR09802	SOURCE: Riboflavin[e]	$EC0212_b \longrightarrow EC0212$	
215	IR07876	ATP:L-methione S-adenosyltransferase (R)	$C0002 + C0058 + C0001 \longrightarrow C0017 + C0012 + C0009$	
216	IR08658	Nicotinic acid uptake	$EC0210 \longrightarrow C0210$	
217	IR09073	SOURCE: Glycerol[e]	$EC0098_b \longrightarrow EC0098$	
218	IR09866	SINK: L-homocysteine[e]	$EC0132 \longrightarrow EC0132_b$	
219	IR00074	ATP:NAD ⁺ 2'-phosphotransferase	$C0003 + C0002 \longrightarrow C0065 + C0008 + C0006$	
220	IR04740	UTP:pyruvate O ₂ -phosphotransferase (R)	$C0014 + C0065 + C0059 \longrightarrow C0060 + C0020$	
221	IR09881	doubling energy mpn	$25\ C0002 + C9324 + 25\ C0001 \longrightarrow EC9324 + 25\ C0009 + 25\ C0065 + 25\ C0008$	
222	IR00676	ATP:D-ribose 5-phosphotransferase	$C0103 + C0002 \longrightarrow C0065 + C0099 + C0008$	
223	IR09799	SOURCE: Pantetheine[e]	$EC0610_b \longrightarrow EC0610$	
224	IR09805	SOURCE: Mannitol[e]	$EC0309_b \longrightarrow EC0309$	
225	IR09959	PHEabc	$EC0064 + C0002 + C0001 \longrightarrow C0064 + C0009 + C0065 + C0008$	
226	IR08769	L-alanine transport via ABC system	$EC0035 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0035 + C0008$	
227	IR09828	phosphatidylglycerophosphatase	$C9480 + C0001 \longrightarrow C9481 + C0009$	
228	IR09911	Glutamyl-tRNA synthetase (GIntRNA)	$C0002 + C0023 + C9538 \longrightarrow C0012 + C0018 + C9537$	
229	IR08767	ILEabc	$EC0317 + C0002 + C0001 \longrightarrow C0317 + C0009 + C0065 + C0008$	
230	IR08789	L-methionine transport via ABC system	$EC0058 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0058 + C0008$	
231	IR00341	ATP:thiamin-phosphate phosphotransferase	$C0002 + C0781 \longrightarrow C0008 + C0055$	

Nº	Id	Name	Reaction Equation	SBO
232	IR09854	SOURCE: Thymine[e]	EC0146.b $\longrightarrow$ EC0146	
233	IR00615	ATP:uridine 5'-phosphotransferase	C0002 + C0242 $\longrightarrow$ C0065 + C0008 + C0089	
234	IR09885	RNA degradation (Mpn)	C9498 + 100 C0001 $\longrightarrow$ 29 C0018 + 28 C0123 + 100 C0065 + 18 C0046 + 25 C0089	
235	IR09907	Seryl-tRNA synthetase	C0053 + C0002 + C9526 $\longrightarrow$ C0012 + C0018 + C9528	
236	IR03716	Thymidylate 5'-phosphohydrolase	C0292 + C0001 $\longrightarrow$ C0009 + C0177	
237	IR09075	SOURCE: sn-Glycerol 3-phosphate[e]	EC0078.b $\longrightarrow$ EC0078	
238	IR09093	SOURCE: D-Mannose[e]	EC0135.b $\longrightarrow$ EC0135	
239	IR09909	Protein synthesis (Mpn)	3 C9519 + 5 C9467 + 21 C9510 + 17 C9541 + 29 C9504 + 3 C9533 + 20 C9540 + 27 C9499 + 21 C9528 + 18 C9505 + C9468 + 6 C9508 + 25 C9512 + 15 C9503 + 13 C9507 + 482 C0001 + 21 C9517 + 826 C0002 + 22 C9506 + 14 C9513 + 32 C9516 + 22 C9282 + 10 C9532 $\longrightarrow$ 20 C9539 + C0047 + 826 C0008 + 21 C9526 + 10 C9530 + 18 C9518 + 6 C9525 + C9536 + 826 C0065 + 21 C9524 + 29 C9535 + 17 C9538 + 3 C9534 + 22 C9320 + 21 C9521 + 25 C9527 + 826 C0009 + 13 C9515 + 27 C9500 + 14 C9509 + 6 C9522 + 22 C9531 + 3 C9523 + 15 C9502 + 32 C9520	
240	IR01174	ATP:thiamine phosphotransferase	C0002 + C0300 $\longrightarrow$ C0781 + C0065 + C0008	
241	IR08747	GLUabc	C0002 + EC0023 + C0001 $\longrightarrow$ C0009 + C0065 + C0008 + C0023	
242	IR00688	2-Deoxy-D-ribose-5-phosphate acetaldehyde-lyase	C0504 $\longrightarrow$ C0100 + C0069	
243	IR09861	DNA synthesis (Mpn)	30 C0113 + 30 C0352 + 20 C0351 + 140 C0002 + 20 C0233 + 140 C0001 $\longrightarrow$ 100 C0012 + 140 C0009 + 140 C0065 + C9484 + 140 C0008	

Nº	Id	Name	Reaction Equation	SBO
244	IR04343	RNDR3	$C9319 + C0094 \longrightarrow C0523 + C9318 + C0001$	
245	IR09859	SOURCE: Phosphatidylcholine (Mpn)	$EC9483_b \longrightarrow EC9483$	
246	IR00293	ATP:riboflavin 5'-phosphotransferase	$C0212 + C0002 \longrightarrow C0065 + C0049 + C0008$	
247	IR09955	biomass production (Mpn)	$167760 C0077 + 6837 C0126 +$ $100 C0005 + 1981 C0175 + 2489 C0156 +$ $148168 C9474 + 9318 C0041 + 3913 C0050 +$ $16049 C9536 + 67 C0082 + 100 C0016 +$ $100 C0055 + 5122 C0064 + 1864 C0063 +$ $41537 C0009 + 9824 C0035 + 100 C2169 +$ $2793 C0151 + 1681 C9484 + 294 C0052 +$ $18651 C0023 + 100 C0010 + 1370 C0058 +$ $2015 C9498 + 100 C0006 + 54 C0129 +$ $63702 C9489 + 100 C0017 + 1858 C0317 +$ $1741 C0039 + 100 C0015 + 1366 C0067 +$ $2422 C0117 + 3 C9262 + 1657 C0306 +$ $2541 C0242 + 21313 C0105 + 100 C0250 +$ $503 C0363 + 3202 C0053 + 1242 C0177 +$ $9220 C0033 \longrightarrow 1000000 C9324$	
248	IR08779	LYSt2r (R)	$C0039 + C0065 \longrightarrow EC0039 + EC0065$	
249	IR08770	VALabc	$C0002 + EC0151 + C0001 \longrightarrow C0009 + C0065 +$ $C0151 + C0008$	
250	IR09826	phosphatidate cytidyltransferase	$C0051 + C0065 + C9474 \longrightarrow C0012 + C9479$	
251	IR09903	Cysteinyl-tRNA synthetase	$C9523 + C0082 + C0002 \longrightarrow C0012 + C0018 +$ $C9519$	
252	IR08690	VALt2r (R)	$C0065 + C0151 \longrightarrow EC0065 + EC0151$	
253	IR08753	ARGabc	$C0002 + EC0050 + C0001 \longrightarrow C0050 + C0009 +$ $C0065 + C0008$	
254	IR00001	Pyrophosphate phosphohydrolase	$C0012 + C0001 \longrightarrow 2 C0009 + C0065$	
255	IR03714	ATP:thymidine 5'-phosphotransferase	$C0002 + C0177 \longrightarrow C0292 + C0065 + C0008$	

Nº	Id	Name	Reaction Equation	SBO
256	IR09904	Arginyl-tRNA synthetase	$C0050 + C0002 + C9515 \longrightarrow C0012 + C0018 + C9507$	
257	IR08996	SINK: Acetate[e]	$EC0029 \longrightarrow EC0029_b$	
258	IR00909	2'-Deoxycytidine 5'-monophosphate phosphohydrolase	$C0198 + C0001 \longrightarrow C0009 + C0646$	
259	IR08828	GLCpts	$C0059 + EC0027 \longrightarrow C0077 + C0020$	
260	IR09912	glutaminyl-tRNA synthetase (Asp)	$C9537 + C0002 + C0129 + C0001 \longrightarrow C0009 + C9541 + C0065 + C0008 + C0041$	
261	IR09793	pyruvate kinase 4	$C0065 + C0059 + C0291 \longrightarrow C0352 + C0020$	
262	IR09892	Histidyl-tRNA synthetase	$C9525 + C0002 + C0117 \longrightarrow C0012 + C0018 + C9508$	
263	IR01159	dUMP + H2O $\rightleftharpoons$ Deoxyuridine + Orthophosphate	$C0293 + C0001 \longrightarrow C0009 + C0408$	
264	IR08849	GLYC3Pabc	$C0002 + EC0078 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0078$	
265	IR00075	ATP:NADH 2'-phosphotransferase	$C0002 + C0004 \longrightarrow C0065 + C0005 + C0008$	
266	IR09960	glycine transport via ABC system	$EC0033 + C0002 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0033$	
267	IR08645	GUA _t 2	$EC0199 + EC0065 \longrightarrow C0065 + C0199$	
268	IR00614	Uridine 5'-monophosphate phosphohydrolase	$C0089 + C0001 \longrightarrow C0009 + C0242$	
269	IR08848	L-tyrosine reversible transport via proton symport (R)	$C0065 + C0067 \longrightarrow EC0067 + EC0065$	
270	IR09864	SINK: 5mcDNA (Mpn)	$C9485 \longrightarrow C9485_b$	
271	IR04278	ACPS1	$C9264 + C0010 \longrightarrow C0045 + C9262 + C0065$	
272	IR00450	ATP:D-fructose-6-phosphate 1-phosphotransferase	$C0002 + C0070 \longrightarrow C0284 + C0065 + C0008$	
273	IR09896	Prolyl-tRNA synthetase	$C0126 + C0002 + C9509 \longrightarrow C0012 + C0018 + C9513$	

Nº	Id	Name	Reaction Equation	SBO
274	IR09923	ARGt2r (R)	$C0050 + C0065 \longrightarrow EC0050 + EC0065$	
275	IR07969	dATP:pyruvate O2-phosphotransferase (R)	$C0065 + C0170 + C0059 \longrightarrow C0113 + C0020$	
276	IR09840	SINK: H2O2[e]	$EC0025 \longrightarrow EC0025_b$	
277	IR09855	Thymine t2	$EC0146 + EC0065 \longrightarrow C0065 + C0146$	
278	IR09894	Aspartyl-tRNA synthetase	$C9518 + C0002 + C0041 \longrightarrow C0012 + C9505 + C0018$	
279	IR09809	SOURCE: dCytidine[e]	$EC0646_b \longrightarrow EC0646$	
280	IR08688	L-leucine reversible transport via proton sym- port (R)	$C0065 + C0105 \longrightarrow EC0065 + EC0105$	
281	IR09505	SINK: (S)-Lactate[e]	$EC0154 \longrightarrow EC0154_b$	
282	IR00267	Cytidine-5'-monophosphate phosphohydro- lase	$C0046 + C0001 \longrightarrow C0009 + C0363$	
283	IR09900	Tryptophanyl-tRNA synthetase	$C0063 + C9534 + C0002 \longrightarrow C0012 + C0018 + C9533$	
284	IR00777	Guanosine 5'-monophosphate phosphohy- drolase	$C0123 + C0001 \longrightarrow C0009 + C0306$	
285	IR00269	ATP:cytidine 5'-phosphotransferase	$C0002 + C0363 \longrightarrow C0065 + C0046 + C0008$	
286	IR08836	thiamine transport via ABC system	$C0002 + EC0300 + C0001 \longrightarrow C0009 + C0065 + C0008 + C0300$	
287	IR04231	3-keto-L-gulonate 6-phosphate decarboxy- lase	$C9255 + C0065 \longrightarrow C2070 + C0011$	
288	IR09888	Acetate export (Mpn)	$C0029 + C0065 \longrightarrow EC0065 + EC0029$	
289	IR09800	SOURCE: Nicotinate[e]	$EC0210_b \longrightarrow EC0210$	
290	IR03832	5-Formyltetrahydrofolate cyclo-ligase (ADP- forming)	$C0002 + C2169 \longrightarrow C0343 + C0009 + 2\ C0065 + C0008$	
291	IR01064	CTP:choline-phosphate cytidyltransferase	$C0051 + C0454 + C0065 \longrightarrow C0012 + C0250$	
292	IR08823	L-proline transport via ABC system	$EC0126 + C0002 + C0001 \longrightarrow C0126 + C0009 + C0065 + C0008$	
293	IR09479	SINK: L-Ornithine[e]	$EC0062 \longrightarrow EC0062_b$	

Nº	Id	Name	Reaction Equation	SBO
294	IR09901	Threonyl-tRNA synthetase	$C9521 + C0002 + C0156 \longrightarrow C0012 + C9517 + C0018$	
295	IR09811	Ribose transporter	$EC0103 \longrightarrow C0103$	
296	IR09927	MET2t (R)	$C0058 + C0065 \longrightarrow EC0058 + EC0065$	
297	IR09817	pyridoxal transporter	$EC0207 \longrightarrow C0207$	
298	IR08842	THRT2r (R)	$C0065 + C0156 \longrightarrow EC0156 + EC0065$	
299	IR01418	Deoxycytidine aminohydrolase	$C0646 + C0001 \longrightarrow C0408 + C0013$	
300	IR01156	ATP:deoxyuridine 5'-phosphotransferase	$C0002 + C0408 \longrightarrow C0293 + C0065 + C0008$	
301	IR00094	ATP:dephospho-CoA 3'-phosphotransferase	$C0647 + C0002 \longrightarrow C0065 + C0008 + C0010$	
302	IR09797	methionyl-tRNA formyltransferase	$C0193 + C9467 \longrightarrow C9468 + C0085$	
303	IR01486	D-Fructose 1-phosphate D-glyceraldehyde-3-phosphate-lyase	$C0790 \longrightarrow C0093 + C0445$	
304	IR09059	SOURCE: D-Fructose[e]	$EC0080.b \longrightarrow EC0080$	
305	IR03505	Adenosine 5'-monophosphate phosphohydrolase	$C0018 + C0001 \longrightarrow C0009 + C0175$	
306	IR00296	L-Arginine iminohydrolase	$C0050 + C0001 \longrightarrow C0065 + C0268 + C0013$	

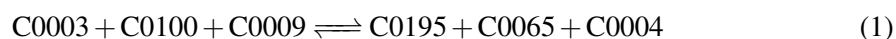
5.1 Reaction RR00685

This is a reversible reaction of three reactants forming three products.

Name D-Glyceraldehyde-3-phosphate:NAD+ oxidoreductase(phosphorylating)

Notes GENE ASSOCIATION: MPN430

Reaction equation



Reactants

Table 5: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0100	D-Glyceraldehyde 3-phosphate	
C0009	Orthophosphate	

Products

Table 6: Properties of each product.

Id	Name	SBO
C0195	1,3-Bisphospho-D-glycerate	
C0065	H+	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_1 = \text{FLUX_VALUE} \quad (2)$$

Table 7: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.2 Reaction RR09116

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Threonine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 8: Properties of each reactant.

Id	Name	SBO
EC0156	L-Threonine[e]	

Product

Table 9: Properties of each product.

Id	Name	SBO
EC0156_b	L-Threonine_b[e]	

Kinetic Law

Derived unit not available

$$v_2 = \text{FLUX_VALUE} \quad (4)$$

Table 10: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.3 Reaction RR09462

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Leucine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 11: Properties of each reactant.

Id	Name	SBO
EC0105	L-Leucine[e]	

Product

Table 12: Properties of each product.

Id	Name	SBO
EC0105_b	L-Leucine_b[e]	

Kinetic Law

Derived unit not available

$$v_3 = \text{FLUX_VALUE} \quad (6)$$

Table 13: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.4 Reaction RR01850

This is a reversible reaction of one reactant forming one product.

Name L-Xylose 5-phosphate 3-epimerase

Notes GENE ASSOCIATION: MPN492

Reaction equation



Reactant

Table 14: Properties of each reactant.

Id	Name	SBO
C0796	L-Ribulose 5-phosphate	

Product

Table 15: Properties of each product.

Id	Name	SBO
C2070	L-Xylulose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_4 = \text{FLUX_VALUE} \quad (8)$$

Table 16: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.5 Reaction RR09050

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Arginine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 17: Properties of each reactant.

Id	Name	SBO
EC0050	L-Arginine[e]	

Product

Table 18: Properties of each product.

Id	Name	SBO
EC0050_b	L-Arginine_b[e]	

Kinetic Law

Derived unit not available

$$v_5 = \text{FLUX_VALUE} \quad (10)$$

Table 19: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

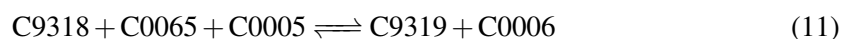
5.6 Reaction RR04369

This is a reversible reaction of three reactants forming two products.

Name TRDR

Notes GENE ASSOCIATION: MPN240

Reaction equation



Reactants

Table 20: Properties of each reactant.

Id	Name	SBO
C9318	Oxidized thioredoxin	
C0065	H+	
C0005	NADPH	

Products

Table 21: Properties of each product.

Id	Name	SBO
C9319	Reduced thioredoxin	
C0006	NADP+	

Kinetic Law

Derived unit not available

$$v_6 = \text{FLUX_VALUE} \quad (12)$$

Table 22: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.7 Reaction RR09428

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Cysteine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 23: Properties of each reactant.

Id	Name	SBO
EC0082	L-Cysteine[e]	

Product

Table 24: Properties of each product.

Id	Name	SBO
EC0082_b	L-Cysteine_b[e]	

Kinetic Law

Derived unit not available

$$v_7 = \text{FLUX_VALUE} \quad (14)$$

Table 25: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.8 Reaction RR03512

This is a reversible reaction of two reactants forming two products.

Name AMP:pyrophosphate phosphoribosyltransferase

Notes GENE ASSOCIATION: MPN395

Reaction equation



Reactants

Table 26: Properties of each reactant.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	

Products

Table 27: Properties of each product.

Id	Name	SBO
C0125	Adenine	
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	

Kinetic Law

Derived unit not available

$$v_8 = \text{FLUX_VALUE} \quad (16)$$

Table 28: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.9 Reaction RR00521

This is a reversible reaction of two reactants forming two products.

Name sn-Glycerol-3-phosphate:oxygen 2-oxidoreductase

Notes GENE ASSOCIATION: MPN051

Reaction equation



Reactants

Table 29: Properties of each reactant.

Id	Name	SBO
C0007	Oxygen	
C0078	sn-Glycerol 3-phosphate	

Products

Table 30: Properties of each product.

Id	Name	SBO
C0025	H2O2	
C0093	Dihydroxyacetone phosphate	

Kinetic Law

Derived unit not available

$$v_9 = \text{FLUX_VALUE} \quad (18)$$

Table 31: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.10 Reaction RR01147

This is a reversible reaction of two reactants forming two products.

Name ATP:dGMP phosphotransferase

Notes GENE ASSOCIATION: MPN246

Reaction equation



Reactants

Table 32: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0290	dGMP	

Products

Table 33: Properties of each product.

Id	Name	SBO
C0008	ADP	
C0289	dGDP	

Kinetic Law

Derived unit not available

$$v_{10} = \text{FLUX_VALUE} \quad (20)$$

Table 34: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

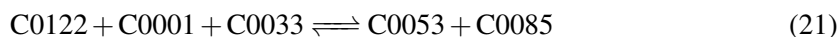
5.11 Reaction RR00598

This is a reversible reaction of three reactants forming two products.

Name 5,10-Methylenetetrahydrofolate:glycine hydroxymethyltransferase

Notes GENE ASSOCIATION: MPN576

Reaction equation



Reactants

Table 35: Properties of each reactant.

Id	Name	SBO
C0122	5,10-Methylenetetrahydrofolate	
C0001	H2O	
C0033	Glycine	

Products

Table 36: Properties of each product.

Id	Name	SBO
C0053	L-Serine	
C0085	Tetrahydrofolate	

Kinetic Law

Derived unit not available

$$v_{11} = \text{FLUX_VALUE} \quad (22)$$

Table 37: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

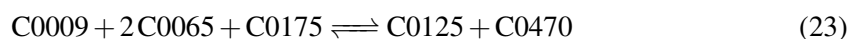
5.12 Reaction RR03709

This is a reversible reaction of three reactants forming two products.

Name Adenosine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN062

Reaction equation



Reactants

Table 38: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0175	Adenosine	

Products

Table 39: Properties of each product.

Id	Name	SBO
C0125	Adenine	
C0470	D-Ribose 1-phosphate	

Kinetic Law

Derived unit not available

$$v_{12} = \text{FLUX_VALUE} \quad (24)$$

Table 40: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		✓
	UPPER_BOUND		999999.0		✓
	OBJECTIVE- _COEFFICIENT		0.0		✓

5.13 Reaction RR01475

This is a reversible reaction of two reactants forming two products.

Name Deoxyadenosine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN062

Reaction equation



Reactants

Table 41: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0433	Deoxyadenosine	

Products

Table 42: Properties of each product.

Id	Name	SBO
C0503	2-Deoxy-D-ribose 1-phosphate	
C0125	Adenine	

Kinetic Law

Derived unit not available

$$v_{13} = \text{FLUX_VALUE} \quad (26)$$

Table 43: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.14 Reaction RR01018

This is a reversible reaction of two reactants forming two products.

Name Sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphateglyceronetransferase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 44: Properties of each reactant.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	
C0230	Sedoheptulose 7-phosphate	

Products

Table 45: Properties of each product.

Id	Name	SBO
C0228	D-Erythrose 4-phosphate	
C0070	D-Fructose 6-phosphate	

Kinetic Law

Derived unit not available

$$v_{14} = \text{FLUX_VALUE} \quad (28)$$

Table 46: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.15 Reaction RR09769

This is a reversible reaction of two reactants forming two products.

Name ascorbate hydrolase

Notes GENE ASSOCIATION: MPN497

Reaction equation



Reactants

Table 47: Properties of each reactant.

Id	Name	SBO
C9462	L-ascorbate 6 phosphate	
C0001	H2O	

Products

Table 48: Properties of each product.

Id	Name	SBO
C9255	3-keto-L-gulonate-6-phosphate	
C0065	H+	

Kinetic Law

Derived unit not available

$$v_{15} = \text{FLUX_VALUE} \quad (30)$$

Table 49: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.16 Reaction RR09104

This is a reversible reaction of one reactant forming one product.

Name SINK: Adenine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 50: Properties of each reactant.

Id	Name	SBO
EC0125	Adenine[e]	

Product

Table 51: Properties of each product.

Id	Name	SBO
EC0125_b	Adenine_b[e]	

Kinetic Law

Derived unit not available

$$v_{16} = \text{FLUX_VALUE} \quad (32)$$

Table 52: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.17 Reaction RR01592

This is a reversible reaction of one reactant forming one product.

Name 2-Deoxy-D-ribose 1-phosphate 1,5-phosphomutase

Notes GENE ASSOCIATION: MPN066

Reaction equation



Reactant

Table 53: Properties of each reactant.

Id	Name	SBO
C0503	2-Deoxy-D-ribose 1-phosphate	

Product

Table 54: Properties of each product.

Id	Name	SBO
C0504	2-Deoxy-D-ribose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{17} = \text{FLUX_VALUE} \quad (34)$$

Table 55: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.18 Reaction RR08870

This is a reversible reaction of two reactants forming two products.

Name ADEt2r

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 56: Properties of each reactant.

Id	Name	SBO
EC0125	Adenine[e]	
EC0065	H+[e]	

Products

Table 57: Properties of each product.

Id	Name	SBO
C0065	H+	
C0125	Adenine	

Kinetic Law

Derived unit not available

$$v_{18} = \text{FLUX_VALUE} \quad (36)$$

Table 58: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

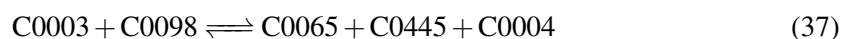
5.19 Reaction RR00667

This is a reversible reaction of two reactants forming three products.

Name Glycerol:NAD+ oxidoreductase

Notes GENE ASSOCIATION: MPN564

Reaction equation



Reactants

Table 59: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0098	Glycerol	

Products

Table 60: Properties of each product.

Id	Name	SBO
C0065	H+	
C0445	D-Glyceraldehyde	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{19} = \text{FLUX_VALUE} \quad (38)$$

Table 61: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.20 Reaction RR09796

This is a reversible reaction of two reactants forming two products.

Name dGTP:3-phospho-D-glycerate 1-phosphotransferase

Notes GENE ASSOCIATION: MPN429

Reaction equation



Reactants

Table 62: Properties of each reactant.

Id	Name	SBO
C0233	dGTP	
C0163	3-Phospho-D-glycerate	

Products

Table 63: Properties of each product.

Id	Name	SBO
C0195	1,3-Bisphospho-D-glycerate	
C0289	dGDP	

Kinetic Law

Derived unit not available

$$v_{20} = \text{FLUX_VALUE} \quad (40)$$

Table 64: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.21 Reaction RR00690

This is a reversible reaction of one reactant forming two products.

Name D-Fructose-1,6-bisphosphate D-glyceraldehyde-3-phosphate-lyase

Notes GENE ASSOCIATION: MPN025

Reaction equation



Reactant

Table 65: Properties of each reactant.

Id	Name	SBO
C0284	D-Fructose 1,6-bisphosphate	

Products

Table 66: Properties of each product.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	
C0093	Dihydroxyacetone phosphate	

Kinetic Law

Derived unit not available

$$v_{21} = \text{FLUX_VALUE} \quad (42)$$

Table 67: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

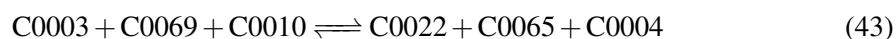
5.22 Reaction RR03544

This is a reversible reaction of three reactants forming three products.

Name Acetaldehyde:NAD+ oxidoreductase (CoA-acetylating)

Notes GENE ASSOCIATION: MPN564

Reaction equation



Reactants

Table 68: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0069	Acetaldehyde	
C0010	CoA	

Products

Table 69: Properties of each product.

Id	Name	SBO
C0022	Acetyl-CoA	
C0065	H+	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{22} = \text{FLUX_VALUE} \quad (44)$$

Table 70: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.23 Reaction RR09070

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Glutamate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 71: Properties of each reactant.

Id	Name	SBO
EC0023	L-Glutamate[e]	

Product

Table 72: Properties of each product.

Id	Name	SBO
EC0023_b	L-Glutamate_b[e]	

Kinetic Law

Derived unit not available

$$v_{23} = \text{FLUX_VALUE} \quad (46)$$

Table 73: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.24 Reaction RR08739

This is a reversible reaction of one reactant forming one product.

Name glycerol transport via channel

Notes GENE ASSOCIATION: MPN043

Reaction equation



Reactant

Table 74: Properties of each reactant.

Id	Name	SBO
C0098	Glycerol	

Product

Table 75: Properties of each product.

Id	Name	SBO
EC0098	Glycerol[e]	

Kinetic Law

Derived unit not available

$$v_{24} = \text{FLUX_VALUE} \quad (48)$$

Table 76: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.25 Reaction RR00779

This is a reversible reaction of two reactants forming two products.

Name GMP:pyrophosphate phosphoribosyltransferase

Notes GENE ASSOCIATION: MPN672

Reaction equation



Reactants

Table 77: Properties of each reactant.

Id	Name	SBO
C0012	Pyrophosphate	
C0123	GMP	

Products

Table 78: Properties of each product.

Id	Name	SBO
C0199	Guanine	
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	

Kinetic Law

Derived unit not available

$$v_{25} = \text{FLUX_VALUE} \quad (50)$$

Table 79: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

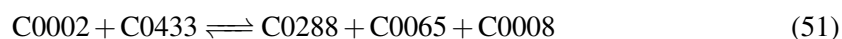
5.26 Reaction RR01146

This is a reversible reaction of two reactants forming three products.

Name ATP:deoxyadenosine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN386

Reaction equation



Reactants

Table 80: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0433	Deoxyadenosine	

Products

Table 81: Properties of each product.

Id	Name	SBO
C0288	dAMP	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{26} = \text{FLUX_VALUE} \quad (52)$$

Table 82: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		✓
	UPPER_BOUND		999999.0		✓
	OBJECTIVE- _COEFFICIENT		0.0		✓

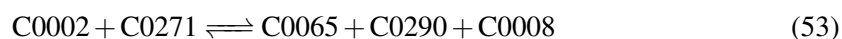
5.27 Reaction RR03808

This is a reversible reaction of two reactants forming three products.

Name ATP:deoxyguanosine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN386

Reaction equation



Reactants

Table 83: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0271	Deoxyguanosine	

Products

Table 84: Properties of each product.

Id	Name	SBO
C0065	H+	
C0290	dGMP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{27} = \text{FLUX_VALUE} \quad (54)$$

Table 85: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.28 Reaction RR08868

This is a reversible reaction of two reactants forming two products.

Name CHLt2r

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 86: Properties of each reactant.

Id	Name	SBO
EC0065	H+[e]	
EC0096	Choline[e]	

Products

Table 87: Properties of each product.

Id	Name	SBO
C0096	Choline	
C0065	H+	

Kinetic Law

Derived unit not available

$$v_{28} = \text{FLUX_VALUE} \quad (56)$$

Table 88: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.29 Reaction RR09830

This is a reversible reaction of one reactant forming one product.

Name FAtransp

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 89: Properties of each reactant.

Id	Name	SBO
EC9471	Fatty Acid (Mpn)[e]	

Product

Table 90: Properties of each product.

Id	Name	SBO
C9471	Fatty Acid (Mpn)	

Kinetic Law

Derived unit not available

$$v_{29} = \text{FLUX_VALUE} \quad (58)$$

Table 91: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.30 Reaction RR09052

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Aspartate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 92: Properties of each reactant.

Id	Name	SBO
EC0041	L-Aspartate[e]	

Product

Table 93: Properties of each product.

Id	Name	SBO
EC0041_b	L-Aspartate_b[e]	

Kinetic Law

Derived unit not available

$$v_{30} = \text{FLUX_VALUE} \quad (60)$$

Table 94: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.31 Reaction RR08615

This is a reversible reaction of two reactants forming two products.

Name arginine/ornithine antiporter

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 95: Properties of each reactant.

Id	Name	SBO
EC0050	L-Arginine[e]	
C0062	L-Ornithine	

Products

Table 96: Properties of each product.

Id	Name	SBO
C0050	L-Arginine	
EC0062	L-Ornithine[e]	

Kinetic Law

Derived unit not available

$$v_{31} = \text{FLUX_VALUE} \quad (62)$$

Table 97: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

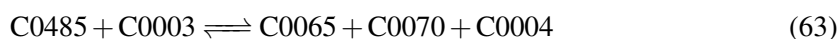
5.32 Reaction RR00452

This is a reversible reaction of two reactants forming three products.

Name D-Mannitol-1-phosphate:NAD+ 5-oxidoreductase

Notes GENE ASSOCIATION: MPN652

Reaction equation



Reactants

Table 98: Properties of each reactant.

Id	Name	SBO
C0485	D-Mannitol 1-phosphate	
C0003	NAD+	

Products

Table 99: Properties of each product.

Id	Name	SBO
C0065	H+	
C0070	D-Fructose 6-phosphate	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{32} = \text{FLUX_VALUE} \quad (64)$$

Table 100: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		✓
	UPPER_BOUND		999999.0		✓
	OBJECTIVE- _COEFFICIENT		0.0		✓

5.33 Reaction RR00902

This is a reversible reaction of two reactants forming one product.

Name 5,10-Methenyltetrahydrofolate 5-hydrolase(decyclizing)

Notes GENE ASSOCIATION: MPN017

Reaction equation



Reactants

Table 101: Properties of each reactant.

Id	Name	SBO
C0343	5,10-Methenyltetrahydrofolate	
C0001	H2O	

Product

Table 102: Properties of each product.

Id	Name	SBO
C0193	10-Formyltetrahydrofolate	

Kinetic Law

Derived unit not available

$$v_{33} = \text{FLUX_VALUE} \quad (66)$$

Table 103: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.34 Reaction RR03810

This is a reversible reaction of two reactants forming two products.

Name Deoxyguanosine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN062

Reaction equation



Reactants

Table 104: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0271	Deoxyguanosine	

Products

Table 105: Properties of each product.

Id	Name	SBO
C0503	2-Deoxy-D-ribose 1-phosphate	
C0199	Guanine	

Kinetic Law

Derived unit not available

$$v_{34} = \text{FLUX_VALUE} \quad (68)$$

Table 106: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.35 Reaction RR09088

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Lysine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 107: Properties of each reactant.

Id	Name	SBO
EC0039	L-Lysine[e]	

Product

Table 108: Properties of each product.

Id	Name	SBO
EC0039_b	L-Lysine_b[e]	

Kinetic Law

Derived unit not available

$$v_{35} = \text{FLUX_VALUE} \quad (70)$$

Table 109: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

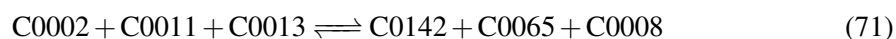
5.36 Reaction RR00106

This is a reversible reaction of three reactants forming three products.

Name ATP:carbamate phosphotransferase

Notes GENE ASSOCIATION: MPN307

Reaction equation



Reactants

Table 110: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0011	CO2	
C0013	NH3	

Products

Table 111: Properties of each product.

Id	Name	SBO
C0142	Carbamoyl phosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{36} = \text{FLUX_VALUE} \quad (72)$$

Table 112: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

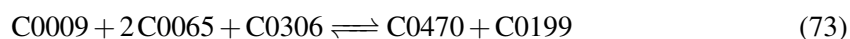
5.37 Reaction RR01184

This is a reversible reaction of three reactants forming two products.

Name Guanosine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN062

Reaction equation



Reactants

Table 113: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0306	Guanosine	

Products

Table 114: Properties of each product.

Id	Name	SBO
C0470	D-Ribose 1-phosphate	
C0199	Guanine	

Kinetic Law

Derived unit not available

$$v_{37} = \text{FLUX_VALUE} \quad (74)$$

Table 115: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.38 Reaction RR00275

This is a reversible reaction of two reactants forming two products.

Name Formate:NAD+ oxidoreductase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 116: Properties of each reactant.

Id	Name	SBO
C0047	Formate	
C0003	NAD+	

Products

Table 117: Properties of each product.

Id	Name	SBO
C0004	NADH	
C0011	CO2	

Kinetic Law

Derived unit not available

$$v_{38} = \text{FLUX_VALUE} \quad (76)$$

Table 118: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.39 Reaction RR00157

This is a reversible reaction of two reactants forming two products.

Name ATP:acetate phosphotransferase

Notes GENE ASSOCIATION: MPN533

Reaction equation



Reactants

Table 119: Properties of each reactant.

Id	Name	SBO
C0029	Acetate	
C0002	ATP	

Products

Table 120: Properties of each product.

Id	Name	SBO
C0188	Acetyl phosphate	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{39} = \text{FLUX_VALUE} \quad (78)$$

Table 121: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.40 Reaction RR09421

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Asparagine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 122: Properties of each reactant.

Id	Name	SBO
EC0129	L-Asparagine[e]	

Product

Table 123: Properties of each product.

Id	Name	SBO
EC0129_b	L-Asparagine_b[e]	

Kinetic Law

Derived unit not available

$$v_{40} = \text{FLUX_VALUE} \quad (80)$$

Table 124: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.41 Reaction RR09857

This is a reversible reaction of two reactants forming two products.

Name phosphatidylcholine avylhydrolase (Mpn)

Notes GENE ASSOCIATION: MPN445

Reaction equation



Reactants

Table 125: Properties of each reactant.

Id	Name	SBO
C9483	Phosphatidylcholine (Mpn)	
C0001	H2O	

Products

Table 126: Properties of each product.

Id	Name	SBO
C9471	Fatty Acid (Mpn)	
C0501	sn-Glycero-3-phosphocholine	

Kinetic Law

Derived unit not available

$$v_{41} = \text{FLUX_VALUE} \quad (82)$$

Table 127: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

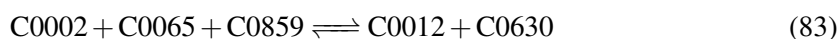
5.42 Reaction RR03873

This is a reversible reaction of three reactants forming two products.

Name ATP:nicotinamide-nucleotide adenylyltransferase

Notes GENE ASSOCIATION: MPN336

Reaction equation



Reactants

Table 128: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0065	H+	
C0859	Nicotinate D-ribonucleotide	

Products

Table 129: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0630	Deamino-NAD+	

Kinetic Law

Derived unit not available

$$v_{42} = \text{FLUX_VALUE} \quad (84)$$

Table 130: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		✓
	UPPER_BOUND		999999.0		✓
	OBJECTIVE- _COEFFICIENT		0.0		✓

5.43 Reaction RR03769

This is a reversible reaction of two reactants forming two products.

Name Sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphateglycolaldehyde transferase

Notes GENE ASSOCIATION: MPN082

Reaction equation



Reactants

Table 131: Properties of each reactant.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	
C0230	Sedoheptulose 7-phosphate	

Products

Table 132: Properties of each product.

Id	Name	SBO
C0190	D-Xylulose 5-phosphate	
C0099	D-Ribose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{43} = \text{FLUX_VALUE} \quad (86)$$

Table 133: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.44 Reaction RR03681

This is a reversible reaction of one reactant forming one product.

Name 2-Phospho-D-glycerate 2,3-phosphomutase

Notes GENE ASSOCIATION: MPN628

Reaction equation



Reactant

Table 134: Properties of each reactant.

Id	Name	SBO
C0477	2-Phospho-D-glycerate	

Product

Table 135: Properties of each product.

Id	Name	SBO
C0163	3-Phospho-D-glycerate	

Kinetic Law

Derived unit not available

$$v_{44} = \text{FLUX_VALUE} \quad (88)$$

Table 136: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.45 Reaction RR00689

This is a reversible reaction of two reactants forming two products.

Name transketolase

Notes GENE ASSOCIATION: MPN082

Reaction equation



Reactants

Table 137: Properties of each reactant.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	
C0070	D-Fructose 6-phosphate	

Products

Table 138: Properties of each product.

Id	Name	SBO
C0228	D-Erythrose 4-phosphate	
C0190	D-Xylulose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{45} = \text{FLUX_VALUE} \quad (90)$$

Table 139: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.46 Reaction RR09082

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Alanine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 140: Properties of each reactant.

Id	Name	SBO
EC0035	L-Alanine[e]	

Product

Table 141: Properties of each product.

Id	Name	SBO
EC0035_b	L-Alanine_b[e]	

Kinetic Law

Derived unit not available

$$v_{46} = \text{FLUX_VALUE} \quad (92)$$

Table 142: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.47 Reaction RR09054

This is a reversible reaction of one reactant forming one product.

Name SINK: CO2[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 143: Properties of each reactant.

Id	Name	SBO
EC0011	CO2[e]	

Product

Table 144: Properties of each product.

Id	Name	SBO
EC0011_b	CO2_b[e]	

Kinetic Law

Derived unit not available

$$v_{47} = \text{FLUX_VALUE} \quad (94)$$

Table 145: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.48 Reaction RR09458

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Isoleucine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 146: Properties of each reactant.

Id	Name	SBO
EC0317	L-Isoleucine[e]	

Product

Table 147: Properties of each product.

Id	Name	SBO
EC0317_b	L-Isoleucine_b[e]	

Kinetic Law

Derived unit not available

$$v_{48} = \text{FLUX_VALUE} \quad (96)$$

Table 148: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.49 Reaction RR01151

This is a reversible reaction of two reactants forming two products.

Name ATP:dTMP phosphotransferase

Notes GENE ASSOCIATION: MPN006

Reaction equation



Reactants

Table 149: Properties of each reactant.

Id	Name	SBO
C0292	dTMP	
C0002	ATP	

Products

Table 150: Properties of each product.

Id	Name	SBO
C0008	ADP	
C0291	dTDP	

Kinetic Law

Derived unit not available

$$v_{49} = \text{FLUX_VALUE} \quad (98)$$

Table 151: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.50 Reaction RR01158

This is a reversible reaction of two reactants forming two products.

Name 5,10-Methylenetetrahydrofolate:dUMP C-methyltransferase

Notes GENE ASSOCIATION: MPN320

Reaction equation



Reactants

Table 152: Properties of each reactant.

Id	Name	SBO
C0293	dUMP	
C0122	5,10-Methylenetetrahydrofolate	

Products

Table 153: Properties of each product.

Id	Name	SBO
C0325	Dihydrofolate	
C0292	dTMP	

Kinetic Law

Derived unit not available

$$v_{50} = \text{FLUX_VALUE} \quad (100)$$

Table 154: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.51 Reaction RR09076

This is a reversible reaction of one reactant forming one product.

Name SINK: Glycine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 155: Properties of each reactant.

Id	Name	SBO
EC0033	Glycine[e]	

Product

Table 156: Properties of each product.

Id	Name	SBO
EC0033_b	Glycine_b[e]	

Kinetic Law

Derived unit not available

$$v_{51} = \text{FLUX_VALUE} \quad (102)$$

Table 157: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.52 Reaction RR00910

This is a reversible reaction of two reactants forming two products.

Name ATP:dCMP phosphotransferase

Notes GENE ASSOCIATION: MPN476

Reaction equation



Reactants

Table 158: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0198	dCMP	

Products

Table 159: Properties of each product.

Id	Name	SBO
C0523	dCDP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{52} = \text{FLUX_VALUE} \quad (104)$$

Table 160: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

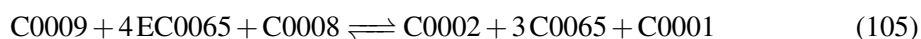
5.53 Reaction RR08593

This is a reversible reaction of three reactants forming three products.

Name ATP synthase (four protons for one ATP)

Notes GENE ASSOCIATION: MPN597 and MPN598 and MPN599 and MPN600 and MPN601 and MPN602 and MPN603 and MPN604

Reaction equation



Reactants

Table 161: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
EC0065	H+[e]	
C0008	ADP	

Products

Table 162: Properties of each product.

Id	Name	SBO
C0002	ATP	
C0065	H+	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{53} = \text{FLUX_VALUE} \quad (106)$$

Table 163: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		-999999.0		☒
	UPPER_BOUND		999999.0		☒

Id	Name	SBO	Value	Unit	Constant
OBJECTIVE- _COEFFICIENT			0.0		☒

5.54 Reaction RR01236

This is a reversible reaction of two reactants forming two products.

Name Dihydrofolate:NAD+ oxidoreductase

Notes GENE ASSOCIATION: MPN321 or MPN300

Reaction equation



Reactants

Table 164: Properties of each reactant.

Id	Name	SBO
C0325	Dihydrofolate	
C0003	NAD+	

Products

Table 165: Properties of each product.

Id	Name	SBO
C0389	Folic acid	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{54} = \text{FLUX_VALUE} \quad (108)$$

Table 166: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.55 Reaction RR08860

This is a reversible reaction of two reactants forming two products.

Name cytidine transport in via proton symport, reversible

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 167: Properties of each reactant.

Id	Name	SBO
EC0065	H+[e]	
EC0363	Cytidine[e]	

Products

Table 168: Properties of each product.

Id	Name	SBO
C0065	H+	
C0363	Cytidine	

Kinetic Law

Derived unit not available

$$v_{55} = \text{FLUX_VALUE} \quad (110)$$

Table 169: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

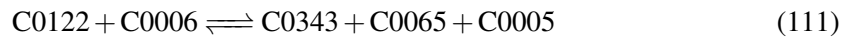
5.56 Reaction RR00771

This is a reversible reaction of two reactants forming three products.

Name 5,10-Methylenetetrahydrofolate:NADP+ oxidoreductase

Notes GENE ASSOCIATION: MPN017

Reaction equation



Reactants

Table 170: Properties of each reactant.

Id	Name	SBO
C0122	5,10-Methylenetetrahydrofolate	
C0006	NADP+	

Products

Table 171: Properties of each product.

Id	Name	SBO
C0343	5,10-Methenyltetrahydrofolate	
C0065	H+	
C0005	NADPH	

Kinetic Law

Derived unit not available

$$v_{56} = \text{FLUX_VALUE} \quad (112)$$

Table 172: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.57 Reaction RR09795

This is a reversible reaction of two reactants forming two products.

Name dATP:3-phospho-D-glycerate 1-phosphotransferase

Notes GENE ASSOCIATION: MPN429

Reaction equation



Reactants

Table 173: Properties of each reactant.

Id	Name	SBO
C0113	dATP	
C0163	3-Phospho-D-glycerate	

Products

Table 174: Properties of each product.

Id	Name	SBO
C0195	1,3-Bisphospho-D-glycerate	
C0170	dADP	

Kinetic Law

Derived unit not available

$$v_{57} = \text{FLUX_VALUE} \quad (114)$$

Table 175: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.58 Reaction RR00682

This is a reversible reaction of one reactant forming two products.

Name D-Ribose 1,5-phosphomutase

Notes GENE ASSOCIATION: MPN066

Reaction equation



Reactant

Table 176: Properties of each reactant.

Id	Name	SBO
C0470	D-Ribose 1-phosphate	

Products

Table 177: Properties of each product.

Id	Name	SBO
C0065	H+	
C0099	D-Ribose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{58} = \text{FLUX_VALUE} \quad (116)$$

Table 178: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.59 Reaction RR08672

This is a reversible reaction of one reactant forming one product.

Name CO2 transporter via diffusion

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 179: Properties of each reactant.

Id	Name	SBO
EC0011	CO2[e]	

Product

Table 180: Properties of each product.

Id	Name	SBO
C0011	CO2	

Kinetic Law

Derived unit not available

$$v_{59} = \text{FLUX_VALUE} \quad (118)$$

Table 181: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.60 Reaction RR09493

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Tryptophan[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 182: Properties of each reactant.

Id	Name	SBO
EC0063	L-Tryptophan[e]	

Product

Table 183: Properties of each product.

Id	Name	SBO
EC0063_b	L-Tryptophan_b[e]	

Kinetic Law

Derived unit not available

$$v_{60} = \text{FLUX_VALUE} \quad (120)$$

Table 184: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

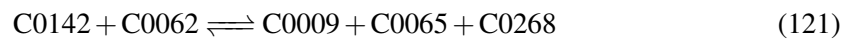
5.61 Reaction RR03649

This is a reversible reaction of two reactants forming three products.

Name Carbamoyl-phosphate:L-ornithine carbamoyltransferase

Notes GENE ASSOCIATION: MPN306

Reaction equation



Reactants

Table 185: Properties of each reactant.

Id	Name	SBO
C0142	Carbamoyl phosphate	
C0062	L-Ornithine	

Products

Table 186: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0268	L-Citrulline	

Kinetic Law

Derived unit not available

$$v_{61} = \text{FLUX_VALUE} \quad (122)$$

Table 187: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.62 Reaction RR09789

This is a reversible reaction of one reactant forming one product.

Name H2O2t

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 188: Properties of each reactant.

Id	Name	SBO
EC0025	H2O2[e]	

Product

Table 189: Properties of each product.

Id	Name	SBO
C0025	H2O2	

Kinetic Law

Derived unit not available

$$v_{62} = \text{FLUX_VALUE} \quad (124)$$

Table 190: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.63 Reaction RR03195

This is a reversible reaction of one reactant forming one product.

Name L-ribulose-5-phosphate 4-epimerase

Notes GENE ASSOCIATION: MPN498

Reaction equation



Reactant

Table 191: Properties of each reactant.

Id	Name	SBO
C0796	L-Ribulose 5-phosphate	

Product

Table 192: Properties of each product.

Id	Name	SBO
C0190	D-Xylulose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{63} = \text{FLUX_VALUE} \quad (126)$$

Table 193: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.64 Reaction RR09471

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Methionine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 194: Properties of each reactant.

Id	Name	SBO
EC0058	L-Methionine[e]	

Product

Table 195: Properties of each product.

Id	Name	SBO
EC0058_b	L-Methionine_b[e]	

Kinetic Law

Derived unit not available

$$v_{64} = \text{FLUX_VALUE} \quad (128)$$

Table 196: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.65 Reaction RR01155

This is a reversible reaction of two reactants forming two products.

Name ATP:dUMP phosphotransferase

Notes GENE ASSOCIATION: MPN006

Reaction equation



Reactants

Table 197: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0293	dUMP	

Products

Table 198: Properties of each product.

Id	Name	SBO
C0960	dUDP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{65} = \text{FLUX_VALUE} \quad (130)$$

Table 199: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.66 Reaction RR08971

This is a reversible reaction of one reactant forming one product.

Name SOURCE: H+[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 200: Properties of each reactant.

Id	Name	SBO
EC0065_b	H+_b[e]	

Product

Table 201: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{66} = \text{FLUX_VALUE} \quad (132)$$

Table 202: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.67 Reaction RR08879

This is a reversible reaction of two reactants forming one product.

Name adenylate kinase

Notes GENE ASSOCIATION: MPN185

Reaction equation



Reactants

Table 203: Properties of each reactant.

Id	Name	SBO
C0018	AMP	
C0002	ATP	

Product

Table 204: Properties of each product.

Id	Name	SBO
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{67} = \text{FLUX_VALUE} \quad (134)$$

Table 205: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.68 Reaction RR08670

This is a reversible reaction of one reactant forming one product.

Name O2t

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 206: Properties of each reactant.

Id	Name	SBO
EC0007	Oxygen[e]	

Product

Table 207: Properties of each product.

Id	Name	SBO
C0007	Oxygen	

Kinetic Law

Derived unit not available

$$v_{68} = \text{FLUX_VALUE} \quad (136)$$

Table 208: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.69 Reaction RR00147

This is a reversible reaction of one reactant forming one product.

Name UDPglucose 4-epimerase

Notes GENE ASSOCIATION: MPN257

Reaction equation



Reactant

Table 209: Properties of each reactant.

Id	Name	SBO
C0026	UDPglucose	

Product

Table 210: Properties of each product.

Id	Name	SBO
C0043	UDP-D-galactose	

Kinetic Law

Derived unit not available

$$v_{69} = \text{FLUX_VALUE} \quad (138)$$

Table 211: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

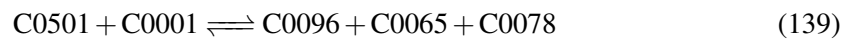
5.70 Reaction RR00662

This is a reversible reaction of two reactants forming three products.

Name sn-Glycero-3-phosphocholine glycerophosphohydrolase

Notes GENE ASSOCIATION: MPN420

Reaction equation



Reactants

Table 212: Properties of each reactant.

Id	Name	SBO
C0501	sn-Glycero-3-phosphocholine	
C0001	H2O	

Products

Table 213: Properties of each product.

Id	Name	SBO
C0096	Choline	
C0065	H+	
C0078	sn-Glycerol 3-phosphate	

Kinetic Law

Derived unit not available

$$v_{70} = \text{FLUX_VALUE} \quad (140)$$

Table 214: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.71 Reaction RR03675

This is a reversible reaction of two reactants forming two products.

Name ATP:3-phospho-D-glycerate 1-phosphotransferase

Notes GENE ASSOCIATION: MPN429

Reaction equation



Reactants

Table 215: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0163	3-Phospho-D-glycerate	

Products

Table 216: Properties of each product.

Id	Name	SBO
C0195	1,3-Bisphospho-D-glycerate	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{71} = \text{FLUX_VALUE} \quad (142)$$

Table 217: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

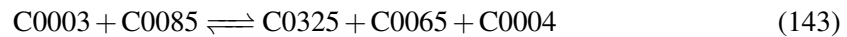
5.72 Reaction RR00590

This is a reversible reaction of two reactants forming three products.

Name 5,6,7,8-Tetrahydrofolate:NAD+ oxidoreductase

Notes GENE ASSOCIATION: MPN321 or MPN300

Reaction equation



Reactants

Table 218: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0085	Tetrahydrofolate	

Products

Table 219: Properties of each product.

Id	Name	SBO
C0325	Dihydrofolate	
C0065	H+	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{72} = \text{FLUX_VALUE} \quad (144)$$

Table 220: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.73 Reaction RR00465

This is a reversible reaction of one reactant forming one product.

Name D-Mannose-6-phosphate ketol-isomerase

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 221: Properties of each reactant.

Id	Name	SBO
C0227	D-Mannose 6-phosphate	

Product

Table 222: Properties of each product.

Id	Name	SBO
C0070	D-Fructose 6-phosphate	

Kinetic Law

Derived unit not available

$$v_{73} = \text{FLUX_VALUE} \quad (146)$$

Table 223: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.74 Reaction RR00268

This is a reversible reaction of two reactants forming two products.

Name ATP:CMP phosphotransferase

Notes GENE ASSOCIATION: MPN476

Reaction equation



Reactants

Table 224: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0046	CMP	

Products

Table 225: Properties of each product.

Id	Name	SBO
C0094	CDP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{74} = \text{FLUX_VALUE} \quad (148)$$

Table 226: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.75 Reaction RR03698

This is a reversible reaction of two reactants forming two products.

Name ATP:dAMP phosphotransferase

Notes GENE ASSOCIATION: MPN185

Reaction equation



Reactants

Table 227: Properties of each reactant.

Id	Name	SBO
C0288	dAMP	
C0002	ATP	

Products

Table 228: Properties of each product.

Id	Name	SBO
C0170	dADP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{75} = \text{FLUX_VALUE} \quad (150)$$

Table 229: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.76 Reaction RR08939

This is a reversible reaction of one reactant forming one product.

Name phosphoglucomutase

Notes GENE ASSOCIATION: MPN066

Reaction equation



Reactant

Table 230: Properties of each reactant.

Id	Name	SBO
C0087	D-Glucose 1-phosphate	

Product

Table 231: Properties of each product.

Id	Name	SBO
C0077	D-Glucose 6-phosphate	

Kinetic Law

Derived unit not available

$$v_{76} = \text{FLUX_VALUE} \quad (152)$$

Table 232: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.77 Reaction RR03717

This is a reversible reaction of two reactants forming two products.

Name Thymidine:orthophosphate deoxyribosyltransferase

Notes GENE ASSOCIATION: MPN064

Reaction equation



Reactants

Table 233: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0177	Thymidine	

Products

Table 234: Properties of each product.

Id	Name	SBO
C0503	2-Deoxy-D-ribose 1-phosphate	
C0146	Thymine	

Kinetic Law

Derived unit not available

$$v_{77} = \text{FLUX_VALUE} \quad (154)$$

Table 235: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.78 Reaction RR03680

This is a reversible reaction of two reactants forming two products.

Name GTP:3-phospho-D-glycerate 1-phosphotransferase

Notes GENE ASSOCIATION: MPN429

Reaction equation



Reactants

Table 236: Properties of each reactant.

Id	Name	SBO
C0163	3-Phospho-D-glycerate	
C0038	GTP	

Products

Table 237: Properties of each product.

Id	Name	SBO
C0195	1,3-Bisphospho-D-glycerate	
C0031	GDP	

Kinetic Law

Derived unit not available

$$v_{78} = \text{FLUX_VALUE} \quad (156)$$

Table 238: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.79 Reaction RR01416

This is a reversible reaction of two reactants forming two products.

Name Deoxyuridine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN064

Reaction equation



Reactants

Table 239: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0408	Deoxyuridine	

Products

Table 240: Properties of each product.

Id	Name	SBO
C0503	2-Deoxy-D-ribose 1-phosphate	
C0090	Uracil	

Kinetic Law

Derived unit not available

$$v_{79} = \text{FLUX_VALUE} \quad (158)$$

Table 241: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.80 Reaction RR09484

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Serine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 242: Properties of each reactant.

Id	Name	SBO
EC0053	L-Serine[e]	

Product

Table 243: Properties of each product.

Id	Name	SBO
EC0053_b	L-Serine_b[e]	

Kinetic Law

Derived unit not available

$$v_{80} = \text{FLUX_VALUE} \quad (160)$$

Table 244: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

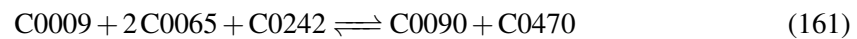
5.81 Reaction RR01052

This is a reversible reaction of three reactants forming two products.

Name Uridine:orthophosphate ribosyltransferase

Notes GENE ASSOCIATION: MPN064

Reaction equation



Reactants

Table 245: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0242	Uridine	

Products

Table 246: Properties of each product.

Id	Name	SBO
C0090	Uracil	
C0470	D-Ribose 1-phosphate	

Kinetic Law

Derived unit not available

$$v_{81} = \text{FLUX_VALUE} \quad (162)$$

Table 247: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

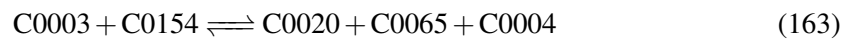
5.82 Reaction RR00404

This is a reversible reaction of two reactants forming three products.

Name (S)-Lactate:NAD+ oxidoreductase

Notes GENE ASSOCIATION: MPN674

Reaction equation



Reactants

Table 248: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0154	(S)-Lactate	

Products

Table 249: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C0065	H+	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{82} = \text{FLUX_VALUE} \quad (164)$$

Table 250: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.83 Reaction RR09837

This is a reversible reaction of one reactant forming one product.

Name SOURCE: Fatty Acid (Mpn) [e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 251: Properties of each reactant.

Id	Name	SBO
EC9471_b	Fatty Acid (Mpn)_b[e]	

Product

Table 252: Properties of each product.

Id	Name	SBO
EC9471	Fatty Acid (Mpn)[e]	

Kinetic Law

Derived unit not available

$$v_{83} = \text{FLUX_VALUE} \quad (166)$$

Table 253: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.84 Reaction RR09229

This is a reversible reaction of one reactant forming one product.

Name SOURCE: Water[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 254: Properties of each reactant.

Id	Name	SBO
EC0001_b	H2O_b[e]	

Product

Table 255: Properties of each product.

Id	Name	SBO
EC0001	H2O[e]	

Kinetic Law

Derived unit not available

$$v_{84} = \text{FLUX_VALUE} \quad (168)$$

Table 256: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.85 Reaction RR08674

This is a reversible reaction of one reactant forming one product.

Name H2O transport via diffusion

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 257: Properties of each reactant.

Id	Name	SBO
EC0001	H2O[e]	

Product

Table 258: Properties of each product.

Id	Name	SBO
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{85} = \text{FLUX_VALUE} \quad (170)$$

Table 259: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.86 Reaction RR09086

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Histidine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 260: Properties of each reactant.

Id	Name	SBO
EC0117	L-Histidine[e]	

Product

Table 261: Properties of each product.

Id	Name	SBO
EC0117_b	L-Histidine_b[e]	

Kinetic Law

Derived unit not available

$$v_{86} = \text{FLUX_VALUE} \quad (172)$$

Table 262: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.87 Reaction RR09498

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Valine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 263: Properties of each reactant.

Id	Name	SBO
EC0151	L-Valine[e]	

Product

Table 264: Properties of each product.

Id	Name	SBO
EC0151_b	L-Valine_b[e]	

Kinetic Law

Derived unit not available

$$v_{87} = \text{FLUX_VALUE} \quad (174)$$

Table 265: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.88 Reaction RR09084

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Glutamine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 266: Properties of each reactant.

Id	Name	SBO
EC0052	L-Glutamine[e]	

Product

Table 267: Properties of each product.

Id	Name	SBO
EC0052_b	L-Glutamine_b[e]	

Kinetic Law

Derived unit not available

$$v_{88} = \text{FLUX_VALUE} \quad (176)$$

Table 268: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.89 Reaction RR09889

This is a reversible reaction of three reactants forming two products.

Name NADH oxidase (H₂O-forming)

Notes GENE ASSOCIATION: MPN394

Reaction equation



Reactants

Table 269: Properties of each reactant.

Id	Name	SBO
C0007	Oxygen	
C0065	H ⁺	
C0004	NADH	

Products

Table 270: Properties of each product.

Id	Name	SBO
C0003	NAD ⁺	
C0001	H ₂ O	

Kinetic Law

Derived unit not available

$$v_{89} = \text{FLUX_VALUE} \quad (178)$$

Table 271: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.90 Reaction RR00617

This is a reversible reaction of two reactants forming two products.

Name UMP:pyrophosphate phosphoribosyltransferase

Notes GENE ASSOCIATION: MPN033

Reaction equation



Reactants

Table 272: Properties of each reactant.

Id	Name	SBO
C0012	Pyrophosphate	
C0089	UMP	

Products

Table 273: Properties of each product.

Id	Name	SBO
C0090	Uracil	
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	

Kinetic Law

Derived unit not available

$$v_{90} = \text{FLUX_VALUE} \quad (180)$$

Table 274: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.91 Reaction RR00464

This is a reversible reaction of one reactant forming one product.

Name D-Glucose-6-phosphate ketol-isomerase

Notes GENE ASSOCIATION: MPN250

Reaction equation



Reactant

Table 275: Properties of each reactant.

Id	Name	SBO
C0077	D-Glucose 6-phosphate	

Product

Table 276: Properties of each product.

Id	Name	SBO
C0070	D-Fructose 6-phosphate	

Kinetic Law

Derived unit not available

$$v_{91} = \text{FLUX_VALUE} \quad (182)$$

Table 277: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.92 Reaction RR09100

This is a reversible reaction of one reactant forming one product.

Name SINK: Orthophosphate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 278: Properties of each reactant.

Id	Name	SBO
EC0009	Orthophosphate[e]	

Product

Table 279: Properties of each product.

Id	Name	SBO
EC0009_b	Orthophosphate_b[e]	

Kinetic Law

Derived unit not available

$$v_{92} = \text{FLUX_VALUE} \quad (184)$$

Table 280: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

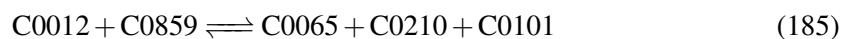
5.93 Reaction RR00952

This is a reversible reaction of two reactants forming three products.

Name Nicotinate D-ribonucleotide:pyrophosphate phosphoribosyltransferase

Notes GENE ASSOCIATION: MPN047

Reaction equation



Reactants

Table 281: Properties of each reactant.

Id	Name	SBO
C0012	Pyrophosphate	
C0859	Nicotinate D-ribonucleotide	

Products

Table 282: Properties of each product.

Id	Name	SBO
C0065	H+	
C0210	Nicotinate	
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	

Kinetic Law

Derived unit not available

$$v_{93} = \text{FLUX_VALUE} \quad (186)$$

Table 283: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.94 Reaction RR09496

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Tyrosine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 284: Properties of each reactant.

Id	Name	SBO
EC0067	L-Tyrosine[e]	

Product

Table 285: Properties of each product.

Id	Name	SBO
EC0067_b	L-Tyrosine_b[e]	

Kinetic Law

Derived unit not available

$$v_{94} = \text{FLUX_VALUE} \quad (188)$$

Table 286: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.95 Reaction RR09098

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Phenylalanine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 287: Properties of each reactant.

Id	Name	SBO
EC0064	L-Phenylalanine[e]	

Product

Table 288: Properties of each product.

Id	Name	SBO
EC0064_b	L-Phenylalanine_b[e]	

Kinetic Law

Derived unit not available

$$v_{95} = \text{FLUX_VALUE} \quad (190)$$

Table 289: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.96 Reaction RR03546

This is a reversible reaction of two reactants forming two products.

Name Acetyl-CoA:orthophosphate acetyltransferase

Notes GENE ASSOCIATION: MPN428

Reaction equation



Reactants

Table 290: Properties of each reactant.

Id	Name	SBO
C0022	Acetyl-CoA	
C0009	Orthophosphate	

Products

Table 291: Properties of each product.

Id	Name	SBO
C0188	Acetyl phosphate	
C0010	CoA	

Kinetic Law

Derived unit not available

$$v_{96} = \text{FLUX_VALUE} \quad (192)$$

Table 292: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.97 Reaction RR09124

This is a reversible reaction of one reactant forming one product.

Name SINK: NH3[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 293: Properties of each reactant.

Id	Name	SBO
EC0013	NH3[e]	

Product

Table 294: Properties of each product.

Id	Name	SBO
EC0013_b	NH3_b[e]	

Kinetic Law

Derived unit not available

$$v_{97} = \text{FLUX_VALUE} \quad (194)$$

Table 295: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

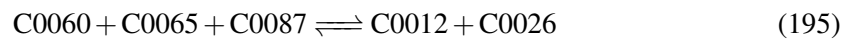
5.98 Reaction RR00146

This is a reversible reaction of three reactants forming two products.

Name UTP:alpha-D-glucose-1-phosphate uridylyltransferase

Notes GENE ASSOCIATION: MPN667

Reaction equation



Reactants

Table 296: Properties of each reactant.

Id	Name	SBO
C0060	UTP	
C0065	H+	
C0087	D-Glucose 1-phosphate	

Products

Table 297: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0026	UDPglucose	

Kinetic Law

Derived unit not available

$$v_{98} = \text{FLUX_VALUE} \quad (196)$$

Table 298: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

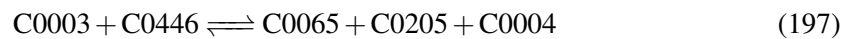
5.99 Reaction RR09342

This is a reversible reaction of two reactants forming three products.

Name 2-oxoglutarate dehydrogenase, lipoamidedehydrogenase compone

Notes GENE ASSOCIATION: MPN390

Reaction equation



Reactants

Table 299: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0446	Dihydrolipoamide	

Products

Table 300: Properties of each product.

Id	Name	SBO
C0065	H+	
C0205	Lipoamide	
C0004	NADH	

Kinetic Law

Derived unit not available

$$v_{99} = \text{FLUX_VALUE} \quad (198)$$

Table 301: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.100 Reaction RR00111

This is a reversible reaction of two reactants forming two products.

Name ATP:UMP phosphotransferase

Notes GENE ASSOCIATION: MPN632

Reaction equation



Reactants

Table 302: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0089	UMP	

Products

Table 303: Properties of each product.

Id	Name	SBO
C0014	UDP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{100} = \text{FLUX_VALUE} \quad (200)$$

Table 304: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.101 Reaction RR00681

This is a reversible reaction of one reactant forming one product.

Name D-Ribose-5-phosphate ketol-isomerase

Notes GENE ASSOCIATION: MPN595

Reaction equation



Reactant

Table 305: Properties of each reactant.

Id	Name	SBO
C0099	D-Ribose 5-phosphate	

Product

Table 306: Properties of each product.

Id	Name	SBO
C0165	D-Ribulose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{101} = \text{FLUX_VALUE} \quad (202)$$

Table 307: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.102 Reaction RR00362

This is a reversible reaction of one reactant forming two products.

Name 2-Phospho-D-glycerate hydro-lyase

Notes GENE ASSOCIATION: MPN606

Reaction equation



Reactant

Table 308: Properties of each reactant.

Id	Name	SBO
C0477	2-Phospho-D-glycerate	

Products

Table 309: Properties of each product.

Id	Name	SBO
C0059	Phosphoenolpyruvate	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{102} = \text{FLUX_VALUE} \quad (204)$$

Table 310: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

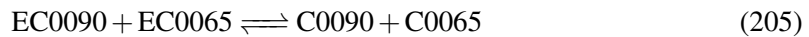
5.103 Reaction RR08636

This is a reversible reaction of two reactants forming two products.

Name URAt2r

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 311: Properties of each reactant.

Id	Name	SBO
EC0090	Uracil[e]	
EC0065	H+[e]	

Products

Table 312: Properties of each product.

Id	Name	SBO
C0090	Uracil	
C0065	H+	

Kinetic Law

Derived unit not available

$$v_{103} = \text{FLUX_VALUE} \quad (206)$$

Table 313: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.104 Reaction RR00652

This is a reversible reaction of one reactant forming one product.

Name D-Glyceraldehyde-3-phosphate ketol-isomerase

Notes GENE ASSOCIATION: MPN629

Reaction equation



Reactant

Table 314: Properties of each reactant.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	

Product

Table 315: Properties of each product.

Id	Name	SBO
C0093	Dihydroxyacetone phosphate	

Kinetic Law

Derived unit not available

$$v_{104} = \text{FLUX_VALUE} \quad (208)$$

Table 316: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.105 Reaction RR03691

This is a reversible reaction of one reactant forming one product.

Name D-Ribulose-5-phosphate 3-epimerase

Notes GENE ASSOCIATION: MPN251

Reaction equation



Reactant

Table 317: Properties of each reactant.

Id	Name	SBO
C0165	D-Ribulose 5-phosphate	

Product

Table 318: Properties of each product.

Id	Name	SBO
C0190	D-Xylulose 5-phosphate	

Kinetic Law

Derived unit not available

$$v_{105} = \text{FLUX_VALUE} \quad (210)$$

Table 319: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.106 Reaction RR09882

This is a reversible reaction of one reactant forming one product.

Name SINK: Biomass[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 320: Properties of each reactant.

Id	Name	SBO
EC9324	Biomass[e]	

Product

Table 321: Properties of each product.

Id	Name	SBO
EC9324_b	Biomass_b[e]	

Kinetic Law

Derived unit not available

$$v_{106} = \text{FLUX_VALUE} \quad (212)$$

Table 322: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			1.0		☒

5.107 Reaction RR09090

This is a reversible reaction of one reactant forming one product.

Name SINK: L-Proline[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 323: Properties of each reactant.

Id	Name	SBO
EC0126	L-Proline[e]	

Product

Table 324: Properties of each product.

Id	Name	SBO
EC0126_b	L-Proline_b[e]	

Kinetic Law

Derived unit not available

$$v_{107} = \text{FLUX_VALUE} \quad (214)$$

Table 325: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			-999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

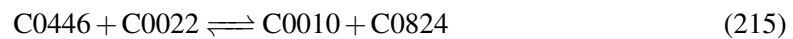
5.108 Reaction RR01487

This is a reversible reaction of two reactants forming two products.

Name Acetyl-CoA: dihydrolipoamide S-acetyltransferase

Notes GENE ASSOCIATION: MPN391

Reaction equation



Reactants

Table 326: Properties of each reactant.

Id	Name	SBO
C0446	Dihydrolipoamide	
C0022	Acetyl-CoA	

Products

Table 327: Properties of each product.

Id	Name	SBO
C0010	CoA	
C0824	S-Acetyldihydrolipoamide	

Kinetic Law

Derived unit not available

$$v_{108} = \text{FLUX_VALUE} \quad (216)$$

Table 328: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.109 Reaction RR09424

This is a reversible reaction of one reactant forming one product.

Name SINK: Choline[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 329: Properties of each reactant.

Id	Name	SBO
EC0096	Choline[e]	

Product

Table 330: Properties of each product.

Id	Name	SBO
EC0096_b	Choline_b[e]	

Kinetic Law

Derived unit not available

$$v_{109} = \text{FLUX_VALUE} \quad (218)$$

Table 331: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.110 Reaction RR00171

This is a reversible reaction of two reactants forming two products.

Name ATP:GMP phosphotransferase

Notes GENE ASSOCIATION: MPN246

Reaction equation



Reactants

Table 332: Properties of each reactant.

Id	Name	SBO
C0123	GMP	
C0002	ATP	

Products

Table 333: Properties of each product.

Id	Name	SBO
C0008	ADP	
C0031	GDP	

Kinetic Law

Derived unit not available

$$v_{110} = \text{FLUX_VALUE} \quad (220)$$

Table 334: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.111 Reaction RR09841

This is a reversible reaction of one reactant forming one product.

Name NH3 export

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 335: Properties of each reactant.

Id	Name	SBO
C0013	NH3	

Product

Table 336: Properties of each product.

Id	Name	SBO
EC0013	NH3[e]	

Kinetic Law

Derived unit not available

$$v_{111} = \text{FLUX_VALUE} \quad (222)$$

Table 337: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			−999999.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

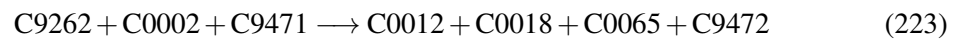
5.112 Reaction IR09820

This is an irreversible reaction of three reactants forming four products.

Name acyl-ACP synthase (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 338: Properties of each reactant.

Id	Name	SBO
C9262	acyl carrier protein	
C0002	ATP	
C9471	Fatty Acid (Mpn)	

Products

Table 339: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C0065	H+	
C9472	ACP-R (Mpn)	

Kinetic Law

Derived unit not available

$$v_{112} = \text{FLUX_VALUE} \quad (224)$$

Table 340: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

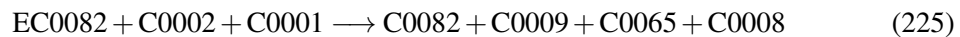
5.113 Reaction IR08663

This is an irreversible reaction of three reactants forming four products.

Name CYSabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 341: Properties of each reactant.

Id	Name	SBO
EC0082	L-Cysteine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 342: Properties of each product.

Id	Name	SBO
C0082	L-Cysteine	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{113} = \text{FLUX_VALUE} \quad (226)$$

Table 343: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

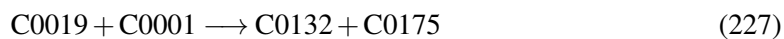
5.114 Reaction IR03514

This is an irreversible reaction of two reactants forming two products.

Name S-Adenosyl-L-homocysteine hydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 344: Properties of each reactant.

Id	Name	SBO
C0019	S-Adenosyl-L-homocysteine	
C0001	H2O	

Products

Table 345: Properties of each product.

Id	Name	SBO
C0132	L-Homocysteine	
C0175	Adenosine	

Kinetic Law

Derived unit not available

$$v_{114} = \text{FLUX_VALUE} \quad (228)$$

Table 346: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.115 Reaction IR08684

This is an irreversible reaction of two reactants forming two products.

Name L-histidine reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 347: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0117	L-Histidine	

Products

Table 348: Properties of each product.

Id	Name	SBO
EC0117	L-Histidine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{115} = \text{FLUX_VALUE} \quad (230)$$

Table 349: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

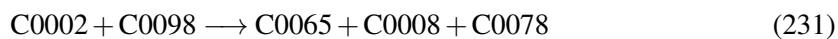
5.116 Reaction IR00522

This is an irreversible reaction of two reactants forming three products.

Name ATP:glycerol 3-phosphotransferase

Notes GENE ASSOCIATION: MPN050

Reaction equation



Reactants

Table 350: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0098	Glycerol	

Products

Table 351: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0078	sn-Glycerol 3-phosphate	

Kinetic Law

Derived unit not available

$$v_{116} = \text{FLUX_VALUE} \quad (232)$$

Table 352: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

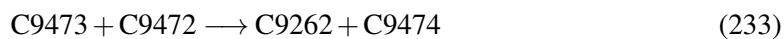
5.117 Reaction IR09822

This is an irreversible reaction of two reactants forming two products.

Name 1-acyl-sn-glycerol-3-phosphate acyltransferase

Notes GENE ASSOCIATION: MPN299

Reaction equation



Reactants

Table 353: Properties of each reactant.

Id	Name	SBO
C9473	1-acyl-Glycerol 3-phosphate (Mpn)	
C9472	ACP-R (Mpn)	

Products

Table 354: Properties of each product.

Id	Name	SBO
C9262	acyl carrier protein	
C9474	phosphatidic acid (Mpn)	

Kinetic Law

Derived unit not available

$$v_{117} = \text{FLUX_VALUE} \quad (234)$$

Table 355: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.118 Reaction IR09801

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Pyridoxal[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 356: Properties of each reactant.

Id	Name	SBO
EC0207_b	Pyridoxal_b[e]	

Product

Table 357: Properties of each product.

Id	Name	SBO
EC0207	Pyridoxal[e]	

Kinetic Law

Derived unit not available

$$v_{118} = \text{FLUX_VALUE} \quad (236)$$

Table 358: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.119 Reaction IR09815

This is an irreversible reaction of one reactant forming one product.

Name folate transporter

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 359: Properties of each reactant.

Id	Name	SBO
EC0389	Folic acid[e]	

Product

Table 360: Properties of each product.

Id	Name	SBO
C0389	Folic acid	

Kinetic Law

Derived unit not available

$$v_{119} = \text{FLUX_VALUE} \quad (238)$$

Table 361: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

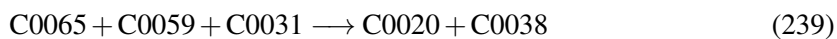
5.120 Reaction IR07938

This is an irreversible reaction of three reactants forming two products.

Name GTP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 362: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
C0031	GDP	

Products

Table 363: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C0038	GTP	

Kinetic Law

Derived unit not available

$$v_{120} = \text{FLUX_VALUE} \quad (240)$$

Table 364: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.121 Reaction IR08692

This is an irreversible reaction of two reactants forming two products.

Name ILEt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 365: Properties of each reactant.

Id	Name	SBO
C0317	L-Isoleucine	
C0065	H+	

Products

Table 366: Properties of each product.

Id	Name	SBO
EC0317	L-Isoleucine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{121} = \text{FLUX_VALUE} \quad (242)$$

Table 367: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.122 Reaction IR09924

This is an irreversible reaction of two reactants forming two products.

Name ASPT2 (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 368: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0041	L-Aspartate	

Products

Table 369: Properties of each product.

Id	Name	SBO
EC0041	L-Aspartate[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{122} = \text{FLUX_VALUE} \quad (244)$$

Table 370: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

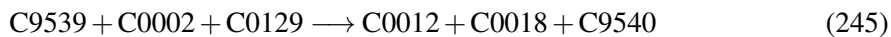
5.123 Reaction IR09910

This is an irreversible reaction of three reactants forming three products.

Name AsparaginyI-tRNA synthetase

Notes GENE ASSOCIATION: MPN252

Reaction equation



Reactants

Table 371: Properties of each reactant.

Id	Name	SBO
C9539	tRNA(Asn)	
C0002	ATP	
C0129	L-Asparagine	

Products

Table 372: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	

Id	Name	SBO
C9540	L-AsparaginyI-tRNA(Asn)	

Kinetic Law

Derived unit not available

$$v_{123} = \text{FLUX_VALUE} \quad (246)$$

Table 373: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.124 Reaction IR09934

This is an irreversible reaction of 20 reactants forming 21 products.

Name apoACP synthesis

Notes GENE ASSOCIATION:

Reaction equation

$$3 \text{ C9467} + \text{C9510} + 4 \text{ C9541} + 12 \text{ C9504} + 3 \text{ C9540} + 4 \text{ C9499} + 5 \text{ C9528} + 6 \text{ C9505} + \text{C9468} + \text{C9508} + 5 \text{ C9512} + \dots \quad (247)$$

Reactants

Table 374: Properties of each reactant.

Id	Name	SBO
C9467	L-Methionyl-tRNA(Met)	
C9510	Glycyl-tRNA(Gly)	
C9541	L-GlutaminyI-tRNA(Gln)	
C9504	L-Leucyl-tRNA(Leu)	
C9540	L-AsparaginyI-tRNA(Asn)	
C9499	L-AlanyI-tRNA(Ala)	
C9528	L-Seryl-tRNA(Ser)	

Id	Name	SBO
C9505	L-Aspartyl-tRNA(Asp)	
C9468	N-Formylmethionyl-tRNA(Met)	
C9508	L-Histidyl-tRNA(His)	
C9512	L-Valyl-tRNA(Val)	
C9503	L-Phenylalanyl-tRNA(Phe)	
C9507	L-Arginyl-tRNA(Arg)	
C0001	H ₂ O	
C9517	L-Threonyl-tRNA(Thr)	
C0002	ATP	
C9506	L-Isoleucyl-tRNA(Ile)	
C9513	L-Prolyl-tRNA(Pro)	
C9516	L-Lysyl-tRNA(Lys)	
C9282	L-Glutamyl-tRNA(Glu)	

Products

Table 375: Properties of each product.

Id	Name	SBO
C9539	tRNA(Asn)	
C9264	apoprotein [acyl carrier protein]	
C9320	tRNA(Glu)	
C9521	tRNA(Thr)	
C0009	Orthophosphate	
C9527	tRNA(Val)	
C9500	tRNA(Ala)	
C9515	tRNA(Arg)	
C0008	ADP	
C9526	tRNA(Ser)	
C9522	tRNA(Met)	
C9509	tRNA(Pro)	
C9531	tRNA(Ile)	
C9518	tRNA(Asp)	
C9525	tRNA(His)	
C9502	tRNA(Phe)	
C0065	H ⁺	
C9524	tRNA(Gly)	
C9535	tRNA(Leu)	
C9520	tRNA(Lys)	
C9538	tRNA(Gln)	

Kinetic Law

Derived unit not available

$$v_{124} = \text{FLUX_VALUE} \quad (248)$$

Table 376: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.125 Reaction IR08686

This is an irreversible reaction of two reactants forming two products.

Name PHEt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 377: Properties of each reactant.

Id	Name	SBO
C0064	L-Phenylalanine	
C0065	H+	

Products

Table 378: Properties of each product.

Id	Name	SBO
EC0064	L-Phenylalanine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{125} = \text{FLUX_VALUE} \quad (250)$$

Table 379: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

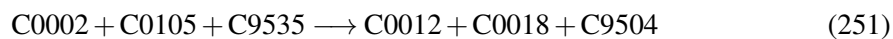
5.126 Reaction IR09898

This is an irreversible reaction of three reactants forming three products.

Name Leucyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN384

Reaction equation



Reactants

Table 380: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0105	L-Leucine	
C9535	tRNA(Leu)	

Products

Table 381: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	

Id	Name	SBO
C9504	L-Leucyl-tRNA(Leu)	

Kinetic Law

Derived unit not available

$$v_{126} = \text{FLUX_VALUE} \quad (252)$$

Table 382: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.127 Reaction IR09808

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Cytidine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 383: Properties of each reactant.

Id	Name	SBO
EC0363_b	Cytidine_b[e]	

Product

Table 384: Properties of each product.

Id	Name	SBO
EC0363	Cytidine[e]	

Kinetic Law**Derived unit** not available

$$v_{127} = \text{FLUX_VALUE} \quad (254)$$

Table 385: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.128 Reaction IR09845

This is an irreversible reaction of three reactants forming four products.

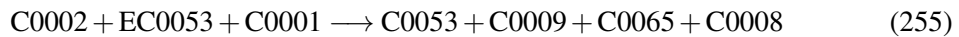
Name L-Serine transport via ABC System**Notes** GENE ASSOCIATION:**Reaction equation****Reactants**

Table 386: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0053	L-Serine[e]	
C0001	H2O	

Products

Table 387: Properties of each product.

Id	Name	SBO
C0053	L-Serine	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{128} = \text{FLUX_VALUE} \quad (256)$$

Table 388: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

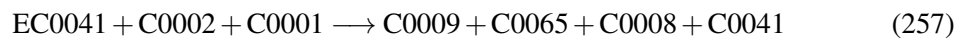
5.129 Reaction IR08746

This is an irreversible reaction of three reactants forming four products.

Name ASPabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 389: Properties of each reactant.

Id	Name	SBO
EC0041	L-Aspartate[e]	
C0002	ATP	
C0001	H2O	

Products

Table 390: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0041	L-Aspartate	

Kinetic Law

Derived unit not available

$$v_{129} = \text{FLUX_VALUE} \quad (258)$$

Table 391: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.130 Reaction IR09771

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: (L)-Ascorbate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 392: Properties of each reactant.

Id	Name	SBO
EC0057_b	Ascorbate_b[e]	

Product

Table 393: Properties of each product.

Id	Name	SBO
EC0057	Ascorbate[e]	

Kinetic Law

Derived unit not available

$$v_{130} = \text{FLUX_VALUE} \quad (260)$$

Table 394: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			0.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

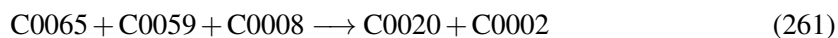
5.131 Reaction IR07898

This is an irreversible reaction of three reactants forming two products.

Name ATP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 395: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
C0008	ADP	

Products

Table 396: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C0002	ATP	

Kinetic Law

Derived unit not available

$$v_{131} = \text{FLUX_VALUE} \quad (262)$$

Table 397: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.132 Reaction IR09198

This is an irreversible reaction of one reactant forming one product.

Name SINK: Adenosine 3',5'-bisphosphate[i]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 398: Properties of each reactant.

Id	Name	SBO
C0045	Adenosine 3',5'-bisphosphate	

Product

Table 399: Properties of each product.

Id	Name	SBO
C0045_b	Adenosine 3',5'-bisphosphate_b	

Kinetic Law

Derived unit not available

$$v_{132} = \text{FLUX_VALUE} \quad (264)$$

Table 400: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.133 Reaction IR09956

This is an irreversible reaction of three reactants forming four products.

Name DnaK catalyzed protein folding (Mpn)

Notes GENE ASSOCIATION: MPN434 and MPN002 and MPN120

Reaction equation



Reactants

Table 401: Properties of each reactant.

Id	Name	SBO
C9536	Protein (Mpn)	
C0002	ATP	
C0001	H2O	

Products

Table 402: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C9557	DnaK-folded protein (Mpn)	

Kinetic Law

Derived unit not available

$$v_{133} = \text{FLUX_VALUE} \quad (266)$$

Table 403: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

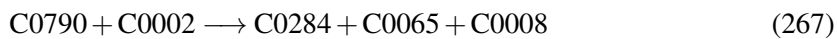
5.134 Reaction IR08910

This is an irreversible reaction of two reactants forming three products.

Name fructose-1-phosphate kinase

Notes GENE ASSOCIATION: MPN079

Reaction equation



Reactants

Table 404: Properties of each reactant.

Id	Name	SBO
C0790	D-Fructose 1-phosphate	

Id	Name	SBO
C0002	ATP	

Products

Table 405: Properties of each product.

Id	Name	SBO
C0284	D-Fructose 1,6-bisphosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{134} = \text{FLUX_VALUE} \quad (268)$$

Table 406: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.135 Reaction [IR04342](#)

This is an irreversible reaction of two reactants forming three products.

Name RNDR4

Notes GENE ASSOCIATION: MPN322 and MPN323 and MPN324

Reaction equation



Reactants

Table 407: Properties of each reactant.

Id	Name	SBO
C0014	UDP	
C9319	Reduced thioredoxin	

Products

Table 408: Properties of each product.

Id	Name	SBO
C9318	Oxidized thioredoxin	
C0960	dUDP	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{135} = \text{FLUX_VALUE} \quad (270)$$

Table 409: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.136 Reaction IR03894

This is an irreversible reaction of three reactants forming two products.

Name ATP:pantetheine-4'-phosphate adenylyltransferase

Notes GENE ASSOCIATION: MPN336

Reaction equation



Reactants

Table 410: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0065	H+	
C0822	Pantetheine 4'-phosphate	

Products

Table 411: Properties of each product.

Id	Name	SBO
C0647	Dephospho-CoA	
C0012	Pyrophosphate	

Kinetic Law

Derived unit not available

$$v_{136} = \text{FLUX_VALUE} \quad (272)$$

Table 412: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

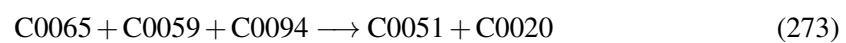
5.137 Reaction IR04690

This is an irreversible reaction of three reactants forming two products.

Name CTP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 413: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
C0094	CDP	

Products

Table 414: Properties of each product.

Id	Name	SBO
C0051	CTP	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{137} = \text{FLUX_VALUE} \quad (274)$$

Table 415: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.138 Reaction IR03497

This is an irreversible reaction of two reactants forming three products.

Name ATP:pyridoxal 5'-phosphotransferase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 416: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0207	Pyridoxal	

Products

Table 417: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0016	Pyridoxal phosphate	

Kinetic Law

Derived unit not available

$$v_{138} = \text{FLUX_VALUE} \quad (276)$$

Table 418: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

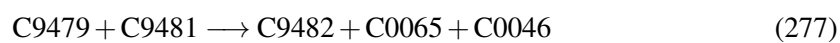
5.139 Reaction IR09829

This is an irreversible reaction of two reactants forming three products.

Name cardiolipin synthase (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 419: Properties of each reactant.

Id	Name	SBO
C9479	CDP-diacylglycerol (Mpn)	
C9481	Phosphatidyl Glycerol (Mpn)	

Products

Table 420: Properties of each product.

Id	Name	SBO
C9482	Cardiolipin (Mpn)	
C0065	H+	
C0046	CMP	

Kinetic Law

Derived unit not available

$$v_{139} = \text{FLUX_VALUE} \quad (278)$$

Table 421: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

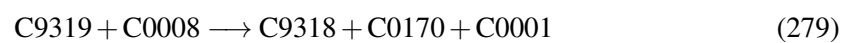
5.140 Reaction IR04340

This is an irreversible reaction of two reactants forming three products.

Name RNDR1

Notes GENE ASSOCIATION: MPN322 and MPN323 and MPN324

Reaction equation



Reactants

Table 422: Properties of each reactant.

Id	Name	SBO
C9319	Reduced thioredoxin	
C0008	ADP	

Products

Table 423: Properties of each product.

Id	Name	SBO
C9318	Oxidized thioredoxin	
C0170	dADP	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{140} = \text{FLUX_VALUE} \quad (280)$$

Table 424: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.141 Reaction IR09590

This is an irreversible reaction of one reactant forming one product.

Name EX_pi(e)

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 425: Properties of each reactant.

Id	Name	SBO
C0009	Orthophosphate	

Product

Table 426: Properties of each product.

Id	Name	SBO
EC0009	Orthophosphate[e]	

Kinetic Law

Derived unit not available

$$v_{141} = \text{FLUX_VALUE} \quad (282)$$

Table 427: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

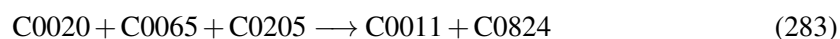
5.142 Reaction IR00932

This is an irreversible reaction of three reactants forming two products.

Name Pyruvate:lipoamide 2-oxidoreductase(decarboxylating andacceptor-acetylating)

Notes GENE ASSOCIATION: MPN392 and MPN393

Reaction equation



Reactants

Table 428: Properties of each reactant.

Id	Name	SBO
C0020	Pyruvate	
C0065	H+	
C0205	Lipoamide	

Products

Table 429: Properties of each product.

Id	Name	SBO
C0011	CO ₂	
C0824	S-Acetyldihydrolipoamide	

Kinetic Law

Derived unit not available

$$v_{142} = \text{FLUX_VALUE} \quad (284)$$

Table 430: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

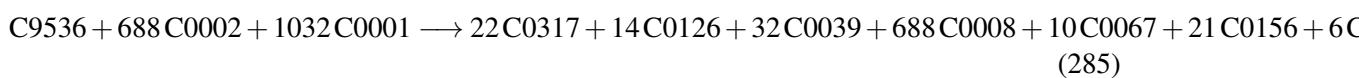
5.143 Reaction IR09954

This is an irreversible reaction of three reactants forming 23 products.

Name protein degradation (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 431: Properties of each reactant.

Id	Name	SBO
C9536	Protein (Mpn)	
C0002	ATP	
C0001	H2O	

Products

Table 432: Properties of each product.

Id	Name	SBO
C0317	L-Isoleucine	
C0126	L-Proline	
C0039	L-Lysine	
C0008	ADP	
C0067	L-Tyrosine	
C0156	L-Threonine	
C0117	L-Histidine	
C0041	L-Aspartate	
C0050	L-Arginine	
C0082	L-Cysteine	
C0065	H+	
C0105	L-Leucine	
C0063	L-Tryptophan	
C0064	L-Phenylalanine	
C0009	Orthophosphate	
C0035	L-Alanine	
C0151	L-Valine	
C0052	L-Glutamine	
C0023	L-Glutamate	
C0053	L-Serine	
C0058	L-Methionine	
C0129	L-Asparagine	
C0033	Glycine	

Kinetic Law

Derived unit not available

$$v_{143} = \text{FLUX_VALUE} \quad (286)$$

Table 433: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			$3.489 \cdot 10^{-4}$		☒
UPPER_BOUND			999999.000		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

5.144 Reaction IR08822

This is an irreversible reaction of two reactants forming two products.

Name PROt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 434: Properties of each reactant.

Id	Name	SBO
C0126	L-Proline	
C0065	H+	

Products

Table 435: Properties of each product.

Id	Name	SBO
EC0126	L-Proline[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{144} = \text{FLUX_VALUE} \quad (288)$$

Table 436: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.145 Reaction IR09079

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Guanine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 437: Properties of each reactant.

Id	Name	SBO
EC0199_b	Guanine_b[e]	

Product

Table 438: Properties of each product.

Id	Name	SBO
EC0199	Guanine[e]	

Kinetic Law

Derived unit not available

$$v_{145} = \text{FLUX_VALUE} \quad (290)$$

Table 439: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

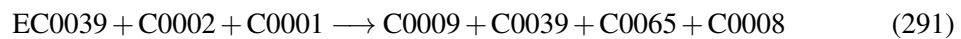
5.146 Reaction IR08755

This is an irreversible reaction of three reactants forming four products.

Name L-lysine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 440: Properties of each reactant.

Id	Name	SBO
EC0039	L-Lysine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 441: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0039	L-Lysine	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{146} = \text{FLUX_VALUE} \quad (292)$$

Table 442: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.147 Reaction IR08750

This is an irreversible reaction of two reactants forming two products.

Name GLUt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 443: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0023	L-Glutamate	

Products

Table 444: Properties of each product.

Id	Name	SBO
EC0023	L-Glutamate[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{147} = \text{FLUX_VALUE} \quad (294)$$

Table 445: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.148 Reaction IR08661

This is an irreversible reaction of two reactants forming two products.

Name L-asparagine reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 446: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0129	L-Asparagine	

Products

Table 447: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	
EC0129	L-Asparagine[e]	

Kinetic Law

Derived unit not available

$$v_{148} = \text{FLUX_VALUE} \quad (296)$$

Table 448: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.149 Reaction IR09109

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: D-Ribose[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 449: Properties of each reactant.

Id	Name	SBO
EC0103_b	D-Ribose_b[e]	

Product

Table 450: Properties of each product.

Id	Name	SBO
EC0103	D-Ribose[e]	

Kinetic Law

Derived unit not available

$$v_{149} = \text{FLUX_VALUE} \quad (298)$$

Table 451: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.000		☒
UPPER_BOUND			0.128		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

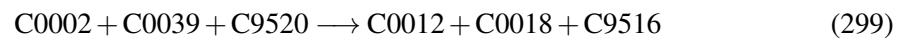
5.150 Reaction IR09906

This is an irreversible reaction of three reactants forming three products.

Name Lysyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN277

Reaction equation



Reactants

Table 452: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0039	L-Lysine	
C9520	tRNA(Lys)	

Products

Table 453: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9516	L-Lysyl-tRNA(Lys)	

Kinetic Law

Derived unit not available

$$v_{150} = \text{FLUX_VALUE} \quad (300)$$

Table 454: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.151 Reaction IR09798

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Folic Acid[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 455: Properties of each reactant.

Id	Name	SBO
EC0389_b	Folic acid.b[e]	

Product

Table 456: Properties of each product.

Id	Name	SBO
EC0389	Folic acid[e]	

Kinetic Law

Derived unit not available

$$v_{151} = \text{FLUX_VALUE} \quad (302)$$

Table 457: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.152 Reaction IR08786

This is an irreversible reaction of two reactants forming two products.

Name MANpts

Notes GENE ASSOCIATION: MPN078

Reaction equation



Reactants

Table 458: Properties of each reactant.

Id	Name	SBO
C0059	Phosphoenolpyruvate	
EC0135	D-Mannose[e]	

Products

Table 459: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C0227	D-Mannose 6-phosphate	

Kinetic Law

Derived unit not available

$$v_{152} = \text{FLUX_VALUE} \quad (304)$$

Table 460: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.153 Reaction IR03510

This is an irreversible reaction of two reactants forming two products.

Name Adenosine 3',5'-bisphosphate 3'(2')-phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 461: Properties of each reactant.

Id	Name	SBO
C0045	Adenosine 3',5'-bisphosphate	
C0001	H2O	

Products

Table 462: Properties of each product.

Id	Name	SBO
C0018	AMP	
C0009	Orthophosphate	

Kinetic Law

Derived unit not available

$$v_{153} = \text{FLUX_VALUE} \quad (306)$$

Table 463: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

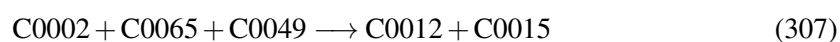
5.154 Reaction IR00114

This is an irreversible reaction of three reactants forming two products.

Name ATP:FMN adenylyltransferase

Notes GENE ASSOCIATION: MPN158

Reaction equation



Reactants

Table 464: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0065	H+	
C0049	FMN	

Products

Table 465: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0015	FAD	

Kinetic Law

Derived unit not available

$$v_{154} = \text{FLUX_VALUE} \quad (308)$$

Table 466: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

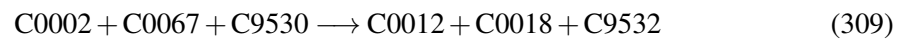
5.155 Reaction IR09895

This is an irreversible reaction of three reactants forming three products.

Name Tyrosyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN669

Reaction equation



Reactants

Table 467: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0067	L-Tyrosine	
C9530	tRNA(Tyr)	

Products

Table 468: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9532	L-Tyrosyl-tRNA(Tyr)	

Kinetic Law

Derived unit not available

$$v_{155} = \text{FLUX_VALUE} \quad (310)$$

Table 469: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.156 Reaction IR09803

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Thiamine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 470: Properties of each reactant.

Id	Name	SBO
EC0300_b	Thiamin_b[e]	

Product

Table 471: Properties of each product.

Id	Name	SBO
EC0300	Thiamin[e]	

Kinetic Law

Derived unit not available

$$v_{156} = \text{FLUX_VALUE} \quad (312)$$

Table 472: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

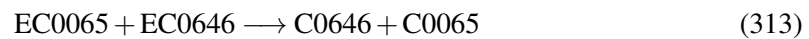
5.157 Reaction IR08807

This is an irreversible reaction of two reactants forming two products.

Name DCYTt2

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 473: Properties of each reactant.

Id	Name	SBO
EC0065	H+[e]	
EC0646	Deoxycytidine[e]	

Products

Table 474: Properties of each product.

Id	Name	SBO
C0646	Deoxycytidine	
C0065	H+	

Kinetic Law

Derived unit not available

$$v_{157} = \text{FLUX_VALUE} \quad (314)$$

Table 475: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.158 Reaction IR08707

This is an irreversible reaction of two reactants forming two products.

Name glycine reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 476: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0033	Glycine	

Products

Table 477: Properties of each product.

Id	Name	SBO
EC0033	Glycine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{158} = \text{FLUX_VALUE} \quad (316)$$

Table 478: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

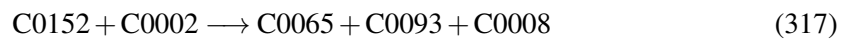
5.159 Reaction IR00649

This is an irreversible reaction of two reactants forming three products.

Name ATP:glycerone phosphotransferase

Notes GENE ASSOCIATION: MPN547

Reaction equation



Reactants

Table 479: Properties of each reactant.

Id	Name	SBO
C0152	Glycerone	
C0002	ATP	

Products

Table 480: Properties of each product.

Id	Name	SBO
C0065	H+	
C0093	Dihydroxyacetone phosphate	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{159} = \text{FLUX_VALUE} \quad (318)$$

Table 481: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

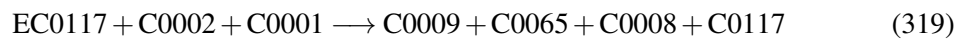
5.160 Reaction IR08754

This is an irreversible reaction of three reactants forming four products.

Name L-histidine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 482: Properties of each reactant.

Id	Name	SBO
EC0117	L-Histidine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 483: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0117	L-Histidine	

Kinetic Law

Derived unit not available

$$v_{160} = \text{FLUX_VALUE} \quad (320)$$

Table 484: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

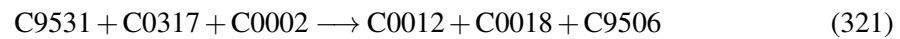
5.161 Reaction IR09905

This is an irreversible reaction of three reactants forming three products.

Name Isoleucyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN520

Reaction equation



Reactants

Table 485: Properties of each reactant.

Id	Name	SBO
C9531	tRNA(Ile)	
C0317	L-Isoleucine	
C0002	ATP	

Products

Table 486: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9506	L-Isoleucyl-tRNA(Ile)	

Kinetic Law

Derived unit not available

$$v_{161} = \text{FLUX_VALUE} \quad (322)$$

Table 487: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.162 Reaction IR09773

This is an irreversible reaction of three reactants forming two products.

Name Ascorbate PTS

Notes GENE ASSOCIATION: MPN494 and MPN495 and MPN496

Reaction equation



Reactants

Table 488: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
EC0057	Ascorbate[e]	

Products

Table 489: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C9462	L-ascorbate 6 phosphate	

Kinetic Law

Derived unit not available

$$v_{162} = \text{FLUX_VALUE} \quad (324)$$

Table 490: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.163 Reaction IR08829

This is an irreversible reaction of two reactants forming two products.

Name D-fructose transport via PEP:Pyr PTS

Notes GENE ASSOCIATION: MPN078

Reaction equation



Reactants

Table 491: Properties of each reactant.

Id	Name	SBO
C0059	Phosphoenolpyruvate	
EC0080	D-Fructose[e]	

Products

Table 492: Properties of each product.

Id	Name	SBO
C0790	D-Fructose 1-phosphate	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{163} = \text{FLUX_VALUE} \quad (326)$$

Table 493: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.164 Reaction IR09863

This is an irreversible reaction of two reactants forming two products.

Name DNA methylierung (Mpn)

Notes GENE ASSOCIATION: MPN108

Reaction equation



Reactants

Table 494: Properties of each reactant.

Id	Name	SBO
C0017	S-Adenosyl-L-methionine	
C9484	DNA (Mpn)	

Products

Table 495: Properties of each product.

Id	Name	SBO
C9485	5mcDNA (Mpn)	
C0019	S-Adenosyl-L-homocysteine	

Kinetic Law

Derived unit not available

$$v_{164} = \text{FLUX_VALUE} \quad (328)$$

Table 496: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.165 Reaction IR00648

This is an irreversible reaction of two reactants forming two products.

Name Glycerone phosphate phosphohydrolase (alkaline optimum)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 497: Properties of each reactant.

Id	Name	SBO
C0093	Dihydroxyacetone phosphate	
C0001	H2O	

Products

Table 498: Properties of each product.

Id	Name	SBO
C0152	Glycerone	
C0009	Orthophosphate	

Kinetic Law

Derived unit not available

$$v_{165} = \text{FLUX_VALUE} \quad (330)$$

Table 499: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.01		☒
UPPER_BOUND			999999.00		☒
OBJECTIVE- _COEFFICIENT			0.00		☒

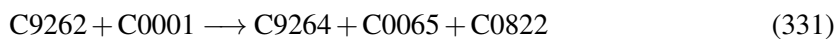
5.166 Reaction IR09847

This is an irreversible reaction of two reactants forming three products.

Name ACP phosphodiesterase

Notes GENE ASSOCIATION: MPN479

Reaction equation



Reactants

Table 500: Properties of each reactant.

Id	Name	SBO
C9262	acyl carrier protein	
C0001	H2O	

Products

Table 501: Properties of each product.

Id	Name	SBO
C9264	apoprotein [acyl carrier protein]	
C0065	H+	
C0822	Pantetheine 4'-phosphate	

Kinetic Law

Derived unit not available

$$v_{166} = \text{FLUX_VALUE} \quad (332)$$

Table 502: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

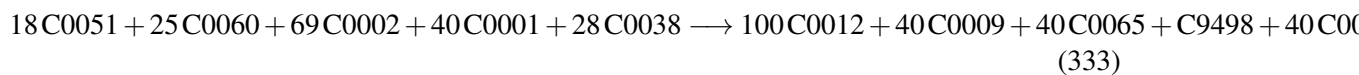
5.167 Reaction IR09884

This is an irreversible reaction of five reactants forming five products.

Name RNA synthesis (Mpn)

Notes GENE ASSOCIATION: MPN516

Reaction equation



Reactants

Table 503: Properties of each reactant.

Id	Name	SBO
C0051	CTP	
C0060	UTP	
C0002	ATP	
C0001	H2O	
C0038	GTP	

Products

Table 504: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0009	Orthophosphate	
C0065	H+	
C9498	RNA (Mpn)	
C0008	ADP	

Kinetic Law**Derived unit** not available

$$v_{167} = \text{FLUX_VALUE} \quad (334)$$

Table 505: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.168 Reaction IR09860

This is an irreversible reaction of three reactants forming four products.

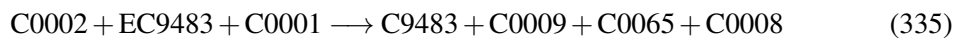
Name Phosphatidylcholine ABC transporter (Mpn)**Notes** GENE ASSOCIATION:**Reaction equation****Reactants**

Table 506: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC9483	Phosphatidylcholine (Mpn)[e]	

Id	Name	SBO
C0001	H2O	

Products

Table 507: Properties of each product.

Id	Name	SBO
C9483	Phosphatidylcholine (Mpn)	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{168} = \text{FLUX_VALUE} \quad (336)$$

Table 508: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

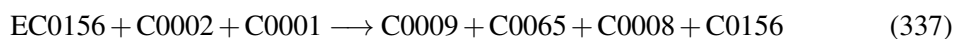
5.169 Reaction IR08768

This is an irreversible reaction of three reactants forming four products.

Name THRabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 509: Properties of each reactant.

Id	Name	SBO
EC0156	L-Threonine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 510: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0156	L-Threonine	

Kinetic Law

Derived unit not available

$$v_{169} = \text{FLUX_VALUE} \quad (338)$$

Table 511: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

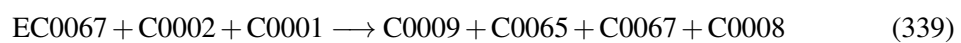
5.170 Reaction IR09958

This is an irreversible reaction of three reactants forming four products.

Name TYRabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 512: Properties of each reactant.

Id	Name	SBO
EC0067	L-Tyrosine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 513: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0067	L-Tyrosine	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{170} = \text{FLUX_VALUE} \quad (340)$$

Table 514: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

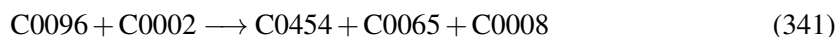
5.171 Reaction IR00656

This is an irreversible reaction of two reactants forming three products.

Name ATP:choline phosphotransferase

Notes GENE ASSOCIATION: MPN532

Reaction equation



Reactants

Table 515: Properties of each reactant.

Id	Name	SBO
C0096	Choline	
C0002	ATP	

Products

Table 516: Properties of each product.

Id	Name	SBO
C0454	Choline phosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{171} = \text{FLUX_VALUE} \quad (342)$$

Table 517: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

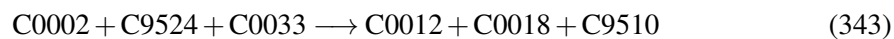
5.172 Reaction IR09899

This is an irreversible reaction of three reactants forming three products.

Name Glycyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN354

Reaction equation



Reactants

Table 518: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C9524	tRNA(Gly)	
C0033	Glycine	

Products

Table 519: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9510	Glycyl-tRNA(Gly)	

Kinetic Law

Derived unit not available

$$v_{172} = \text{FLUX_VALUE} \quad (344)$$

Table 520: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

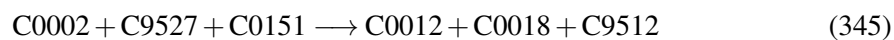
5.173 Reaction IR09902

This is an irreversible reaction of three reactants forming three products.

Name Valyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN480

Reaction equation



Reactants

Table 521: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C9527	tRNA(Val)	
C0151	L-Valine	

Products

Table 522: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9512	L-Valyl-tRNA(Val)	

Kinetic Law

Derived unit not available

$$v_{173} = \text{FLUX_VALUE} \quad (346)$$

Table 523: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.174 Reaction IR09871

This is an irreversible reaction of three reactants forming three products.

Name glycolipid synthesis (Mpn)

Notes GENE ASSOCIATION: MPN483

Reaction equation



Reactants

Table 524: Properties of each reactant.

Id	Name	SBO
C0043	UDP-D-galactose	
C0026	UDPglucose	
C9475	Diacylglycerol (Mpn)	

Products

Table 525: Properties of each product.

Id	Name	SBO
C0014	UDP	
C0065	H+	
C9489	glycolipid (Mpn)	

Kinetic Law

Derived unit not available

$$v_{174} = \text{FLUX_VALUE} \quad (348)$$

Table 526: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.175 Reaction IR09886

This is an irreversible reaction of two reactants forming two products.

Name Lactate export (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 527: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0154	(S)-Lactate	

Products

Table 528: Properties of each product.

Id	Name	SBO
EC0154	(S)-Lactate[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{175} = \text{FLUX_VALUE} \quad (350)$$

Table 529: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

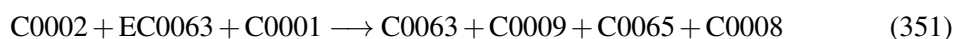
5.176 Reaction IR09957

This is an irreversible reaction of three reactants forming four products.

Name TRPabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 530: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0063	L-Tryptophan[e]	
C0001	H2O	

Products

Table 531: Properties of each product.

Id	Name	SBO
C0063	L-Tryptophan	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{176} = \text{FLUX_VALUE} \quad (352)$$

Table 532: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓

Id	Name	SBO	Value	Unit	Constant
OBJECTIVE- _COEFFICIENT			0.0		☒

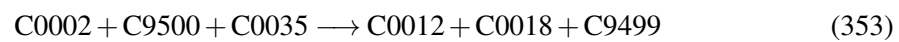
5.177 Reaction IR09891

This is an irreversible reaction of three reactants forming three products.

Name Alanyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN418

Reaction equation



Reactants

Table 533: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C9500	tRNA(Ala)	
C0035	L-Alanine	

Products

Table 534: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9499	L-Alanyl-tRNA(Ala)	

Kinetic Law

Derived unit not available

$$v_{177} = \text{FLUX_VALUE} \quad (354)$$

Table 535: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.178 Reaction IR08844

This is an irreversible reaction of two reactants forming two products.

Name L-tryptophan reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 536: Properties of each reactant.

Id	Name	SBO
C0063	L-Tryptophan	
C0065	H+	

Products

Table 537: Properties of each product.

Id	Name	SBO
EC0063	L-Tryptophan[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{178} = \text{FLUX_VALUE} \quad (356)$$

Table 538: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

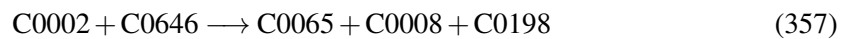
5.179 Reaction IR00911

This is an irreversible reaction of two reactants forming three products.

Name ATP:deoxycytidine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN386

Reaction equation



Reactants

Table 539: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0646	Deoxycytidine	

Products

Table 540: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0198	dCMP	

Kinetic Law

Derived unit not available

$$v_{179} = \text{FLUX_VALUE} \quad (358)$$

Table 541: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.180 Reaction IR09865

This is an irreversible reaction of one reactant forming one product.

Name homocysteine export (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 542: Properties of each reactant.

Id	Name	SBO
C0132	L-Homocysteine	

Product

Table 543: Properties of each product.

Id	Name	SBO
EC0132	L-Homocysteine[e]	

Kinetic Law

Derived unit not available

$$v_{180} = \text{FLUX_VALUE} \quad (360)$$

Table 544: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.181 Reaction IR08970

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Glucose[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 545: Properties of each reactant.

Id	Name	SBO
EC0027_b	D-Glucose_b[e]	

Product

Table 546: Properties of each product.

Id	Name	SBO
EC0027	D-Glucose[e]	

Kinetic Law

Derived unit not available

$$v_{181} = \text{FLUX_VALUE} \quad (362)$$

Table 547: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.000		☒
UPPER_BOUND			5.108		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

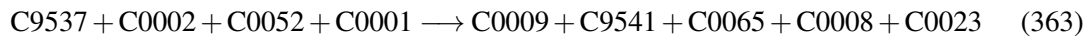
5.182 Reaction IR09913

This is an irreversible reaction of four reactants forming five products.

Name glutaminyl-tRNA synthetase (Glu)

Notes GENE ASSOCIATION: MPN236 and MPN237 and MPN238

Reaction equation



Reactants

Table 548: Properties of each reactant.

Id	Name	SBO
C9537	L-Glutamyl-tRNA(Gln)	
C0002	ATP	
C0052	L-Glutamine	
C0001	H2O	

Products

Table 549: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C9541	L-Glutaminyl-tRNA(Gln)	
C0065	H+	
C0008	ADP	
C0023	L-Glutamate	

Kinetic Law

Derived unit not available

$$v_{182} = \text{FLUX_VALUE} \quad (364)$$

Table 550: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

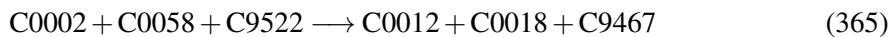
5.183 Reaction IR09893

This is an irreversible reaction of three reactants forming three products.

Name Methionyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN023

Reaction equation



Reactants

Table 551: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0058	L-Methionine	
C9522	tRNA(Met)	

Products

Table 552: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	

Id	Name	SBO
C9467	L-Methionyl-tRNA(Met)	

Kinetic Law

Derived unit not available

$$v_{183} = \text{FLUX_VALUE} \quad (366)$$

Table 553: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

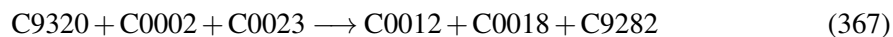
5.184 Reaction IR04284

This is an irreversible reaction of three reactants forming three products.

Name Glutamyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN678

Reaction equation



Reactants

Table 554: Properties of each reactant.

Id	Name	SBO
C9320	tRNA(Glu)	
C0002	ATP	
C0023	L-Glutamate	

Products

Table 555: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9282	L-Glutamyl-tRNA(Glu)	

Kinetic Law

Derived unit not available

$$v_{184} = \text{FLUX_VALUE} \quad (368)$$

Table 556: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

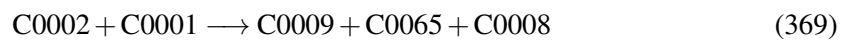
5.185 Reaction IR08984

This is an irreversible reaction of two reactants forming three products.

Name atp consumption proxy

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 557: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0001	H2O	

Products

Table 558: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{185} = \text{FLUX_VALUE} \quad (370)$$

Table 559: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			13.103		☒
UPPER_BOUND			999999.000		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

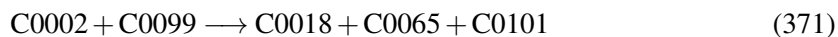
5.186 Reaction IR00674

This is an irreversible reaction of two reactants forming three products.

Name ATP:D-ribose-5-phosphate pyrophosphotransferase

Notes GENE ASSOCIATION: MPN073

Reaction equation



Reactants

Table 560: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0099	D-Ribose 5-phosphate	

Products

Table 561: Properties of each product.

Id	Name	SBO
C0018	AMP	
C0065	H+	
C0101	5-Phospho-alpha-D-ribose 1-diphosphate	

Kinetic Law

Derived unit not available

$$v_{186} = \text{FLUX_VALUE} \quad (372)$$

Table 562: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

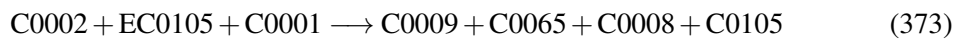
5.187 Reaction IR08771

This is an irreversible reaction of three reactants forming four products.

Name LEUabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 563: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0105	L-Leucine[e]	

Id	Name	SBO
C0001	H2O	

Products

Table 564: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0105	L-Leucine	

Kinetic Law

Derived unit not available

$$v_{187} = \text{FLUX_VALUE} \quad (374)$$

Table 565: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

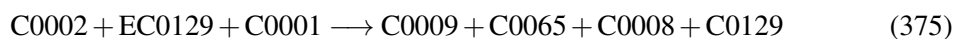
5.188 Reaction IR08659

This is an irreversible reaction of three reactants forming four products.

Name L-asparagine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 566: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0129	L-Asparagine[e]	
C0001	H2O	

Products

Table 567: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0129	L-Asparagine	

Kinetic Law

Derived unit not available

$$v_{188} = \text{FLUX_VALUE} \quad (376)$$

Table 568: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

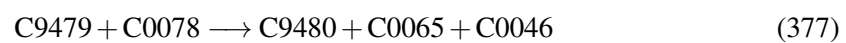
5.189 Reaction IR09827

This is an irreversible reaction of two reactants forming three products.

Name phosphatidylglycerophosphate synthase

Notes GENE ASSOCIATION: MPN253

Reaction equation



Reactants

Table 569: Properties of each reactant.

Id	Name	SBO
C9479	CDP-diacylglycerol (Mpn)	
C0078	sn-Glycerol 3-phosphate	

Products

Table 570: Properties of each product.

Id	Name	SBO
C9480	Phosphatidyl Glycerol-3-Phosphate (Mpn)	
C0065	H+	
C0046	CMP	

Kinetic Law

Derived unit not available

$$v_{189} = \text{FLUX_VALUE} \quad (378)$$

Table 571: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.190 Reaction IR08824

This is an irreversible reaction of three reactants forming three products.

Name phosphate transport via ABC system

Notes GENE ASSOCIATION: MPN609 and MPN610 and MPN611

Reaction equation



Reactants

Table 572: Properties of each reactant.

Id	Name	SBO
EC0009	Orthophosphate[e]	
C0002	ATP	
C0001	H2O	

Products

Table 573: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{190} = \text{FLUX_VALUE} \quad (380)$$

Table 574: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

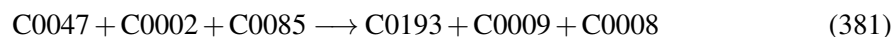
5.191 Reaction IR00596

This is an irreversible reaction of three reactants forming three products.

Name Formate:tetrahydrofolate ligase (ADP-forming)

Notes GENE ASSOCIATION: MPN017

Reaction equation



Reactants

Table 575: Properties of each reactant.

Id	Name	SBO
C0047	Formate	
C0002	ATP	
C0085	Tetrahydrofolate	

Products

Table 576: Properties of each product.

Id	Name	SBO
C0193	10-Formyltetrahydrofolate	
C0009	Orthophosphate	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{191} = \text{FLUX_VALUE} \quad (382)$$

Table 577: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.192 Reaction IR08635

This is an irreversible reaction of two reactants forming two products.

Name ALAt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 578: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0035	L-Alanine	

Products

Table 579: Properties of each product.

Id	Name	SBO
EC0035	L-Alanine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{192} = \text{FLUX_VALUE} \quad (384)$$

Table 580: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

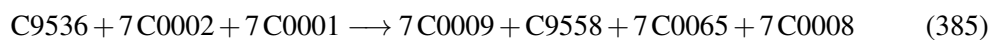
5.193 Reaction IR09961

This is an irreversible reaction of three reactants forming four products.

Name GroEL catalyzed protein folding (Mpn)

Notes GENE ASSOCIATION: MPN573 and MPN574

Reaction equation



Reactants

Table 581: Properties of each reactant.

Id	Name	SBO
C9536	Protein (Mpn)	
C0002	ATP	
C0001	H2O	

Products

Table 582: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C9558	GroEL-folded protein (Mpn)	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{193} = \text{FLUX_VALUE} \quad (386)$$

Table 583: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

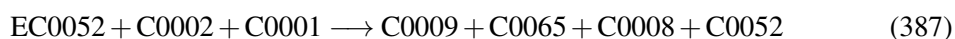
5.194 Reaction IR08738

This is an irreversible reaction of three reactants forming four products.

Name L-glutamine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 584: Properties of each reactant.

Id	Name	SBO
EC0052	L-Glutamine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 585: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0052	L-Glutamine	

Kinetic Law

Derived unit not available

$$v_{194} = \text{FLUX_VALUE} \quad (388)$$

Table 586: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓

Id	Name	SBO	Value	Unit	Constant
OBJECTIVE- _COEFFICIENT			0.0		☒

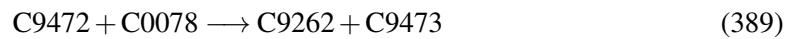
5.195 Reaction IR09821

This is an irreversible reaction of two reactants forming two products.

Name glycerol-3-phosphate O-acyltransferase (Mpn)

Notes GENE ASSOCIATION: MPN350 and MPN546

Reaction equation



Reactants

Table 587: Properties of each reactant.

Id	Name	SBO
C9472	ACP-R (Mpn)	
C0078	sn-Glycerol 3-phosphate	

Products

Table 588: Properties of each product.

Id	Name	SBO
C9262	acyl carrier protein	
C9473	1-acyl-Glycerol 3-phosphate (Mpn)	

Kinetic Law

Derived unit not available

$$v_{195} = \text{FLUX_VALUE} \quad (390)$$

Table 589: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.196 Reaction IR09926

This is an irreversible reaction of two reactants forming two products.

Name GLNt2 (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 590: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0052	L-Glutamine	

Products

Table 591: Properties of each product.

Id	Name	SBO
EC0052	L-Glutamine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{196} = \text{FLUX_VALUE} \quad (392)$$

Table 592: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.197 Reaction IR09807

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Uracil[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 593: Properties of each reactant.

Id	Name	SBO
EC0090_b	Uracil_b[e]	

Product

Table 594: Properties of each product.

Id	Name	SBO
EC0090	Uracil[e]	

Kinetic Law

Derived unit not available

$$v_{197} = \text{FLUX_VALUE} \quad (394)$$

Table 595: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

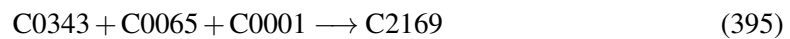
5.198 Reaction IR08208

This is an irreversible reaction of three reactants forming one product.

Name S-Aminomethyldihydrolipoylprotein:(6S)-tetrahydrofolateaminomethyltransferase (ammonia-forming) (R)

Notes GENE ASSOCIATION: MPN576

Reaction equation



Reactants

Table 596: Properties of each reactant.

Id	Name	SBO
C0343	5,10-Methenyltetrahydrofolate	
C0065	H+	
C0001	H2O	

Product

Table 597: Properties of each product.

Id	Name	SBO
C2169	5-Formyltetrahydrofolate	

Kinetic Law

Derived unit not available

$$v_{198} = \text{FLUX_VALUE} \quad (396)$$

Table 598: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.01		☒
UPPER_BOUND			999999.00		☒
OBJECTIVE- _COEFFICIENT			0.00		☒

5.199 Reaction IR09095

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Oxygen[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 599: Properties of each reactant.

Id	Name	SBO
EC0007_b	Oxygen.b[e]	

Product

Table 600: Properties of each product.

Id	Name	SBO
EC0007	Oxygen[e]	

Kinetic Law

Derived unit not available

$$v_{199} = \text{FLUX_VALUE} \quad (398)$$

Table 601: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

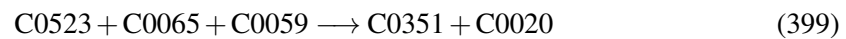
5.200 Reaction IR09792

This is an irreversible reaction of three reactants forming two products.

Name pyruvate kinase 3

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 602: Properties of each reactant.

Id	Name	SBO
C0523	dCDP	
C0065	H+	
C0059	Phosphoenolpyruvate	

Products

Table 603: Properties of each product.

Id	Name	SBO
C0351	dCTP	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{200} = \text{FLUX_VALUE} \quad (400)$$

Table 604: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

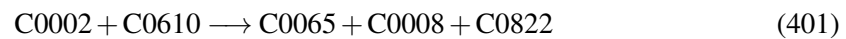
5.201 Reaction IR01710

This is an irreversible reaction of two reactants forming three products.

Name ATP:pantothenate 4'-phosphotransferase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 605: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0610	Pantetheine	

Products

Table 606: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0822	Pantetheine 4'-phosphate	

Kinetic Law

Derived unit not available

$$v_{201} = \text{FLUX_VALUE} \quad (402)$$

Table 607: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

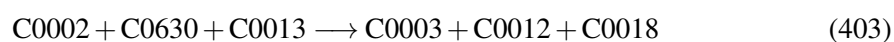
5.202 Reaction IR03511

This is an irreversible reaction of three reactants forming three products.

Name Deamino-NAD+:ammonia ligase (AMP-forming)

Notes GENE ASSOCIATION: MPN562

Reaction equation



Reactants

Table 608: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0630	Deamino-NAD+	
C0013	NH3	

Products

Table 609: Properties of each product.

Id	Name	SBO
C0003	NAD+	
C0012	Pyrophosphate	
C0018	AMP	

Kinetic Law

Derived unit not available

$$v_{202} = \text{FLUX_VALUE} \quad (404)$$

Table 610: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

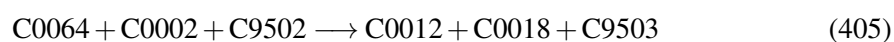
5.203 Reaction IR09908

This is an irreversible reaction of three reactants forming three products.

Name Phenylalanyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN105 and MPN106

Reaction equation



Reactants

Table 611: Properties of each reactant.

Id	Name	SBO
C0064	L-Phenylalanine	
C0002	ATP	
C9502	tRNA(Phe)	

Products

Table 612: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9503	L-Phenylalanyl-tRNA(Phe)	

Kinetic Law

Derived unit not available

$$v_{203} = \text{FLUX_VALUE} \quad (406)$$

Table 613: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

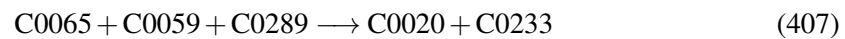
5.204 Reaction IR05416

This is an irreversible reaction of three reactants forming two products.

Name dGTP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 614: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
C0289	dGDP	

Products

Table 615: Properties of each product.

Id	Name	SBO
C0020	Pyruvate	
C0233	dGTP	

Kinetic Law

Derived unit not available

$$v_{204} = \text{FLUX_VALUE} \quad (408)$$

Table 616: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.205 Reaction IR09271

This is an irreversible reaction of one reactant forming one product.

Name SINK: NADP+[i]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 617: Properties of each reactant.

Id	Name	SBO
C0006	NADP+	

Product

Table 618: Properties of each product.

Id	Name	SBO
C0006_b	NADP+_b	

Kinetic Law

Derived unit not available

$$v_{205} = \text{FLUX_VALUE} \quad (410)$$

Table 619: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.206 Reaction IR01054

This is an irreversible reaction of two reactants forming two products.

Name Cytidine aminohydrolase

Notes GENE ASSOCIATION: MPN065

Reaction equation



Reactants

Table 620: Properties of each reactant.

Id	Name	SBO
C0363	Cytidine	
C0001	H2O	

Products

Table 621: Properties of each product.

Id	Name	SBO
C0242	Uridine	
C0013	NH3	

Kinetic Law

Derived unit not available

$$v_{206} = \text{FLUX_VALUE} \quad (412)$$

Table 622: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.207 Reaction IR09818

This is an irreversible reaction of one reactant forming one product.

Name riboflavin transporter

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 623: Properties of each reactant.

Id	Name	SBO
EC0212	Riboflavin[e]	

Product

Table 624: Properties of each product.

Id	Name	SBO
C0212	Riboflavin	

Kinetic Law

Derived unit not available

$$v_{207} = \text{FLUX_VALUE} \quad (414)$$

Table 625: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

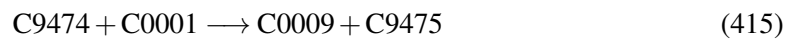
5.208 Reaction IR09824

This is an irreversible reaction of two reactants forming two products.

Name phosphatidate phosphatase

Notes GENE ASSOCIATION: MPN455

Reaction equation



Reactants

Table 626: Properties of each reactant.

Id	Name	SBO
C9474	phosphatidic acid (Mpn)	
C0001	H2O	

Products

Table 627: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C9475	Diacylglycerol (Mpn)	

Kinetic Law

Derived unit not available

$$v_{208} = \text{FLUX_VALUE} \quad (416)$$

Table 628: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.209 Reaction IR09925

This is an irreversible reaction of two reactants forming two products.

Name CYSt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 629: Properties of each reactant.

Id	Name	SBO
C0082	L-Cysteine	
C0065	H+	

Products

Table 630: Properties of each product.

Id	Name	SBO
EC0082	L-Cysteine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{209} = \text{FLUX_VALUE} \quad (418)$$

Table 631: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

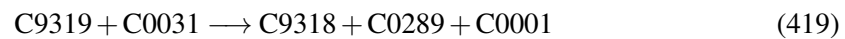
5.210 Reaction IR04341

This is an irreversible reaction of two reactants forming three products.

Name RNDR2

Notes GENE ASSOCIATION: MPN322 and MPN323 and MPN324

Reaction equation



Reactants

Table 632: Properties of each reactant.

Id	Name	SBO
C9319	Reduced thioredoxin	
C0031	GDP	

Products

Table 633: Properties of each product.

Id	Name	SBO
C9318	Oxidized thioredoxin	
C0289	dGDP	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{210} = \text{FLUX.VALUE} \quad (420)$$

Table 634: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.211 Reaction IR09816

This is an irreversible reaction of one reactant forming one product.

Name pantetheine transporter

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 635: Properties of each reactant.

Id	Name	SBO
EC0610	Pantetheine[e]	

Product

Table 636: Properties of each product.

Id	Name	SBO
C0610	Pantetheine	

Kinetic Law

Derived unit not available

$$v_{211} = \text{FLUX_VALUE} \quad (422)$$

Table 637: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.212 Reaction IR08826

This is an irreversible reaction of two reactants forming two products.

Name MNLpts

Notes GENE ASSOCIATION: MPN651 and MPN653

Reaction equation



Reactants

Table 638: Properties of each reactant.

Id	Name	SBO
EC0309	Mannitol[e]	
C0059	Phosphoenolpyruvate	

Products

Table 639: Properties of each product.

Id	Name	SBO
C0485	D-Mannitol 1-phosphate	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{212} = \text{FLUX_VALUE} \quad (424)$$

Table 640: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.213 Reaction IR08835

This is an irreversible reaction of two reactants forming two products.

Name SERT2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 641: Properties of each reactant.

Id	Name	SBO
C0053	L-Serine	
C0065	H+	

Products

Table 642: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	
EC0053	L-Serine[e]	

Kinetic Law

Derived unit not available

$$v_{213} = \text{FLUX_VALUE} \quad (426)$$

Table 643: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.214 Reaction IR09802

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Riboflavin[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 644: Properties of each reactant.

Id	Name	SBO
EC0212_b	Riboflavin_b[e]	

Product

Table 645: Properties of each product.

Id	Name	SBO
EC0212	Riboflavin[e]	

Kinetic Law

Derived unit not available

$$v_{214} = \text{FLUX_VALUE} \quad (428)$$

Table 646: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

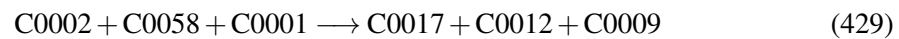
5.215 Reaction IR07876

This is an irreversible reaction of three reactants forming three products.

Name ATP:L-methione S-adenosyltransferase (R)

Notes GENE ASSOCIATION: MPN060

Reaction equation



Reactants

Table 647: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0058	L-Methionine	
C0001	H2O	

Products

Table 648: Properties of each product.

Id	Name	SBO
C0017	S-Adenosyl-L-methionine	
C0012	Pyrophosphate	
C0009	Orthophosphate	

Kinetic Law

Derived unit not available

$$v_{215} = \text{FLUX_VALUE} \quad (430)$$

Table 649: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.216 Reaction IR08658

This is an irreversible reaction of one reactant forming one product.

Name Nicotinic acid uptake

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 650: Properties of each reactant.

Id	Name	SBO
EC0210	Nicotinate[e]	

Product

Table 651: Properties of each product.

Id	Name	SBO
C0210	Nicotinate	

Kinetic Law

Derived unit not available

$$v_{216} = \text{FLUX_VALUE} \quad (432)$$

Table 652: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.217 Reaction IR09073

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Glycerol[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 653: Properties of each reactant.

Id	Name	SBO
EC0098_b	Glycerol_b[e]	

Product

Table 654: Properties of each product.

Id	Name	SBO
EC0098	Glycerol[e]	

Kinetic Law

Derived unit not available

$$v_{217} = \text{FLUX_VALUE} \quad (434)$$

Table 655: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.000		☒
UPPER_BOUND			0.128		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

5.218 Reaction IR09866

This is an irreversible reaction of one reactant forming one product.

Name SINK: L-homocysteine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 656: Properties of each reactant.

Id	Name	SBO
EC0132	L-Homocysteine[e]	

Product

Table 657: Properties of each product.

Id	Name	SBO
EC0132_b	L-Homocysteine_b[e]	

Kinetic Law

Derived unit not available

$$v_{218} = \text{FLUX_VALUE} \quad (436)$$

Table 658: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

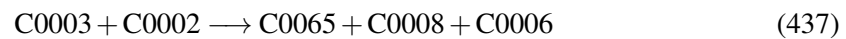
5.219 Reaction IR00074

This is an irreversible reaction of two reactants forming three products.

Name ATP:NAD+ 2'-phosphotransferase

Notes GENE ASSOCIATION: MPN267

Reaction equation



Reactants

Table 659: Properties of each reactant.

Id	Name	SBO
C0003	NAD+	
C0002	ATP	

Products

Table 660: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0006	NADP+	

Kinetic Law

Derived unit not available

$$v_{219} = \text{FLUX_VALUE} \quad (438)$$

Table 661: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

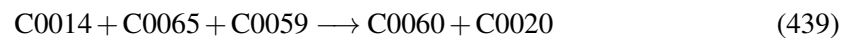
5.220 Reaction IR04740

This is an irreversible reaction of three reactants forming two products.

Name UTP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 662: Properties of each reactant.

Id	Name	SBO
C0014	UDP	
C0065	H+	
C0059	Phosphoenolpyruvate	

Products

Table 663: Properties of each product.

Id	Name	SBO
C0060	UTP	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{220} = \text{FLUX_VALUE} \quad (440)$$

Table 664: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

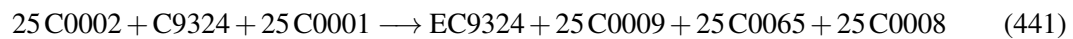
5.221 Reaction IR09881

This is an irreversible reaction of three reactants forming four products.

Name doubling energy mpn

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 665: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C9324	Biomass	
C0001	H2O	

Products

Table 666: Properties of each product.

Id	Name	SBO
EC9324	Biomass[e]	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{221} = \text{FLUX_VALUE} \quad (442)$$

Table 667: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

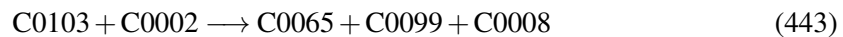
5.222 Reaction IR00676

This is an irreversible reaction of two reactants forming three products.

Name ATP:D-ribose 5-phosphotransferase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 668: Properties of each reactant.

Id	Name	SBO
C0103	D-Ribose	
C0002	ATP	

Products

Table 669: Properties of each product.

Id	Name	SBO
C0065	H+	
C0099	D-Ribose 5-phosphate	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{222} = \text{FLUX_VALUE} \quad (444)$$

Table 670: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.223 Reaction IR09799

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Pantetheine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 671: Properties of each reactant.

Id	Name	SBO
EC0610_b	Pantetheine_b[e]	

Product

Table 672: Properties of each product.

Id	Name	SBO
EC0610	Pantetheine[e]	

Kinetic Law

Derived unit not available

$$v_{223} = \text{FLUX_VALUE} \quad (446)$$

Table 673: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.224 Reaction IR09805

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Mannitol[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 674: Properties of each reactant.

Id	Name	SBO
EC0309_b	Mannitol_b[e]	

Product

Table 675: Properties of each product.

Id	Name	SBO
EC0309	Mannitol[e]	

Kinetic Law

Derived unit not available

$$v_{224} = \text{FLUX_VALUE} \quad (448)$$

Table 676: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			0.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

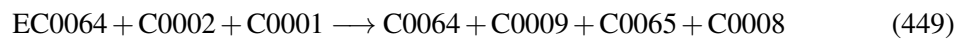
5.225 Reaction IR09959

This is an irreversible reaction of three reactants forming four products.

Name PHEabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 677: Properties of each reactant.

Id	Name	SBO
EC0064	L-Phenylalanine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 678: Properties of each product.

Id	Name	SBO
C0064	L-Phenylalanine	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{225} = \text{FLUX_VALUE} \quad (450)$$

Table 679: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

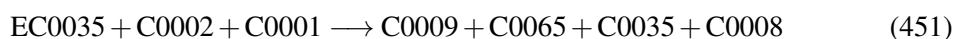
5.226 Reaction IR08769

This is an irreversible reaction of three reactants forming four products.

Name L-alanine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 680: Properties of each reactant.

Id	Name	SBO
EC0035	L-Alanine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 681: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0035	L-Alanine	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{226} = \text{FLUX_VALUE} \quad (452)$$

Table 682: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.227 Reaction IR09828

This is an irreversible reaction of two reactants forming two products.

Name phosphatidylglycerophosphatase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 683: Properties of each reactant.

Id	Name	SBO
C9480	Phosphatidyl Glycerol-3-Phosphate (Mpn)	
C0001	H2O	

Products

Table 684: Properties of each product.

Id	Name	SBO
C9481	Phosphatidyl Glycerol (Mpn)	
C0009	Orthophosphate	

Kinetic Law

Derived unit not available

$$v_{227} = \text{FLUX_VALUE} \quad (454)$$

Table 685: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

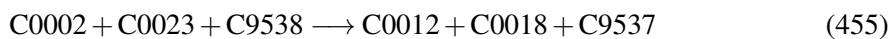
5.228 Reaction IR09911

This is an irreversible reaction of three reactants forming three products.

Name Glutamyl-tRNA synthetase (GIntRNA)

Notes GENE ASSOCIATION: MPN678

Reaction equation



Reactants

Table 686: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0023	L-Glutamate	
C9538	tRNA(Gln)	

Products

Table 687: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	

Id	Name	SBO
C9537	L-Glutamyl-tRNA(Gln)	

Kinetic Law

Derived unit not available

$$v_{228} = \text{FLUX_VALUE} \quad (456)$$

Table 688: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

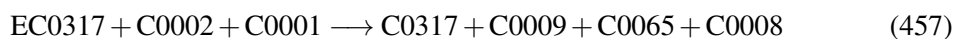
5.229 Reaction IR08767

This is an irreversible reaction of three reactants forming four products.

Name ILEabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 689: Properties of each reactant.

Id	Name	SBO
EC0317	L-Isoleucine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 690: Properties of each product.

Id	Name	SBO
C0317	L-Isoleucine	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{229} = \text{FLUX_VALUE} \quad (458)$$

Table 691: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

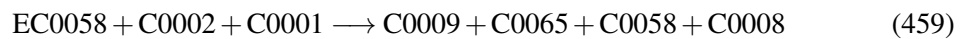
5.230 Reaction IR08789

This is an irreversible reaction of three reactants forming four products.

Name L-methionine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 692: Properties of each reactant.

Id	Name	SBO
EC0058	L-Methionine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 693: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0058	L-Methionine	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{230} = \text{FLUX_VALUE} \quad (460)$$

Table 694: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.231 Reaction IR00341

This is an irreversible reaction of two reactants forming two products.

Name ATP:thiamin-phosphate phosphotransferase

Notes GENE ASSOCIATION: MPN550

Reaction equation



Reactants

Table 695: Properties of each reactant.

Id	Name	SBO
C0002	ATP	

Id	Name	SBO
C0781	Thiamin monophosphate	

Products

Table 696: Properties of each product.

Id	Name	SBO
C0008	ADP	
C0055	Thiamin diphosphate	

Kinetic Law

Derived unit not available

$$v_{231} = \text{FLUX_VALUE} \quad (462)$$

Table 697: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.232 Reaction IR09854

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Thymine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 698: Properties of each reactant.

Id	Name	SBO
EC0146_b	Thymine_b[e]	

Product

Table 699: Properties of each product.

Id	Name	SBO
EC0146	Thymine[e]	

Kinetic Law

Derived unit not available

$$v_{232} = \text{FLUX_VALUE} \quad (464)$$

Table 700: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

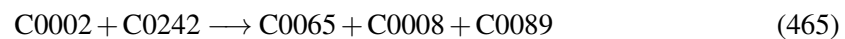
5.233 Reaction IR00615

This is an irreversible reaction of two reactants forming three products.

Name ATP:uridine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN561

Reaction equation



Reactants

Table 701: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0242	Uridine	

Products

Table 702: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0089	UMP	

Kinetic Law

Derived unit not available

$$v_{233} = \text{FLUX_VALUE} \quad (466)$$

Table 703: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

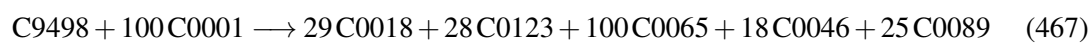
5.234 Reaction IR09885

This is an irreversible reaction of two reactants forming five products.

Name RNA degradation (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 704: Properties of each reactant.

Id	Name	SBO
C9498	RNA (Mpn)	
C0001	H2O	

Products

Table 705: Properties of each product.

Id	Name	SBO
C0018	AMP	
C0123	GMP	
C0065	H+	
C0046	CMP	
C0089	UMP	

Kinetic Law

Derived unit not available

$$v_{234} = \text{FLUX_VALUE} \quad (468)$$

Table 706: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.008		☒
UPPER_BOUND			999999.000		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

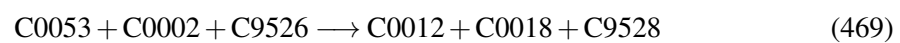
5.235 Reaction IR09907

This is an irreversible reaction of three reactants forming three products.

Name Seryl-tRNA synthetase

Notes GENE ASSOCIATION: MPN005

Reaction equation



Reactants

Table 707: Properties of each reactant.

Id	Name	SBO
C0053	L-Serine	
C0002	ATP	
C9526	tRNA(Ser)	

Products

Table 708: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9528	L-Seryl-tRNA(Ser)	

Kinetic Law

Derived unit not available

$$v_{235} = \text{FLUX_VALUE} \quad (470)$$

Table 709: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.236 Reaction IR03716

This is an irreversible reaction of two reactants forming two products.

Name Thymidylate 5'-phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 710: Properties of each reactant.

Id	Name	SBO
C0292	dTMP	
C0001	H2O	

Products

Table 711: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0177	Thymidine	

Kinetic Law

Derived unit not available

$$v_{236} = \text{FLUX_VALUE} \quad (472)$$

Table 712: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.237 Reaction IR09075

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: sn-Glycerol 3-phosphate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 713: Properties of each reactant.

Id	Name	SBO
EC0078_b	sn-Glycerol 3-phosphate_b[e]	

Product

Table 714: Properties of each product.

Id	Name	SBO
EC0078	sn-Glycerol 3-phosphate[e]	

Kinetic Law

Derived unit not available

$$v_{237} = \text{FLUX_VALUE} \quad (474)$$

Table 715: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.000		☒
UPPER_BOUND			0.128		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

5.238 Reaction IR09093

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: D-Mannose[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 716: Properties of each reactant.

Id	Name	SBO
EC0135_b	D-Mannose_b[e]	

Product

Table 717: Properties of each product.

Id	Name	SBO
EC0135	D-Mannose[e]	

Kinetic Law

Derived unit not available

$$v_{238} = \text{FLUX_VALUE} \quad (476)$$

Table 718: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			0.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

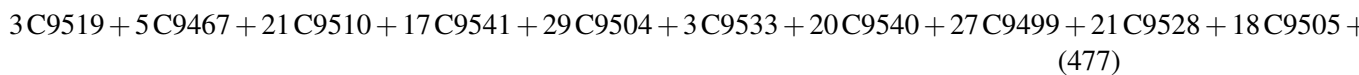
5.239 Reaction IR09909

This is an irreversible reaction of 23 reactants forming 25 products.

Name Protein synthesis (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 719: Properties of each reactant.

Id	Name	SBO
C9519	L-Cysteinyl-tRNA(Cys)	
C9467	L-Methionyl-tRNA(Met)	
C9510	Glycyl-tRNA(Gly)	
C9541	L-Glutaminyl-tRNA(Gln)	
C9504	L-Leucyl-tRNA(Leu)	
C9533	L-Tryptophanyl-tRNA(Trp)	
C9540	L-Asparaginyl-tRNA(Asn)	
C9499	L-Alanyl-tRNA(Ala)	
C9528	L-Seryl-tRNA(Ser)	
C9505	L-Aspartyl-tRNA(Asp)	
C9468	N-Formylmethionyl-tRNA(Met)	
C9508	L-Histidyl-tRNA(His)	
C9512	L-Valyl-tRNA(Val)	
C9503	L-Phenylalanyl-tRNA(Phe)	
C9507	L-Arginyl-tRNA(Arg)	
C0001	H2O	
C9517	L-Threonyl-tRNA(Thr)	
C0002	ATP	
C9506	L-Isoleucyl-tRNA(Ile)	
C9513	L-Prolyl-tRNA(Pro)	
C9516	L-Lysyl-tRNA(Lys)	
C9282	L-Glutamyl-tRNA(Glu)	
C9532	L-Tyrosyl-tRNA(Tyr)	

Products

Table 720: Properties of each product.

Id	Name	SBO
C9539	tRNA(Asn)	
C0047	Formate	
C0008	ADP	
C9526	tRNA(Ser)	

Id	Name	SBO
C9530	tRNA(Tyr)	
C9518	tRNA(Asp)	
C9525	tRNA(His)	
C9536	Protein (Mpn)	
C0065	H+	
C9524	tRNA(Gly)	
C9535	tRNA(Leu)	
C9538	tRNA(Gln)	
C9534	tRNA(Trp)	
C9320	tRNA(Glu)	
C9521	tRNA(Thr)	
C9527	tRNA(Val)	
C0009	Orthophosphate	
C9515	tRNA(Arg)	
C9500	tRNA(Ala)	
C9509	tRNA(Pro)	
C9522	tRNA(Met)	
C9531	tRNA(Ile)	
C9523	tRNA(Cys)	
C9502	tRNA(Phe)	
C9520	tRNA(Lys)	

Kinetic Law

Derived unit not available

$$v_{239} = \text{FLUX_VALUE} \quad (478)$$

Table 721: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

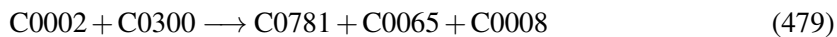
5.240 Reaction IR01174

This is an irreversible reaction of two reactants forming three products.

Name ATP:thiamine phosphotransferase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 722: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0300	Thiamin	

Products

Table 723: Properties of each product.

Id	Name	SBO
C0781	Thiamin monophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{240} = \text{FLUX_VALUE} \quad (480)$$

Table 724: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

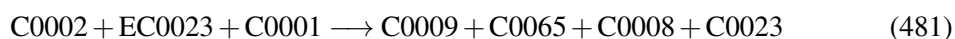
5.241 Reaction IR08747

This is an irreversible reaction of three reactants forming four products.

Name GLUabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 725: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0023	L-Glutamate[e]	
C0001	H2O	

Products

Table 726: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0023	L-Glutamate	

Kinetic Law

Derived unit not available

$$v_{241} = \text{FLUX_VALUE} \quad (482)$$

Table 727: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.242 Reaction IR00688

This is an irreversible reaction of one reactant forming two products.

Name 2-Deoxy-D-ribose-5-phosphate acetaldehyde-lyase

Notes GENE ASSOCIATION: MPN063

Reaction equation



Reactant

Table 728: Properties of each reactant.

Id	Name	SBO
C0504	2-Deoxy-D-ribose 5-phosphate	

Products

Table 729: Properties of each product.

Id	Name	SBO
C0100	D-Glyceraldehyde 3-phosphate	
C0069	Acetaldehyde	

Kinetic Law

Derived unit not available

$$v_{242} = \text{FLUX_VALUE} \quad (484)$$

Table 730: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

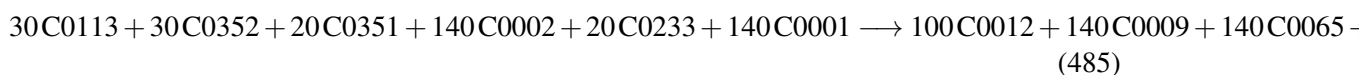
5.243 Reaction IR09861

This is an irreversible reaction of six reactants forming five products.

Name DNA synthesis (Mpn)

Notes GENE ASSOCIATION: MPN034 and MPN378

Reaction equation



Reactants

Table 731: Properties of each reactant.

Id	Name	SBO
C0113	dATP	
C0352	dTTP	
C0351	dCTP	
C0002	ATP	
C0233	dGTP	
C0001	H2O	

Products

Table 732: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0009	Orthophosphate	
C0065	H+	
C9484	DNA (Mpn)	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{243} = \text{FLUX_VALUE} \quad (486)$$

Table 733: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

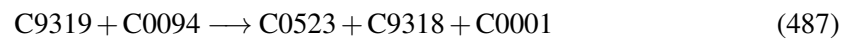
5.244 Reaction IR04343

This is an irreversible reaction of two reactants forming three products.

Name RNDR3

Notes GENE ASSOCIATION: MPN322 and MPN323 and MPN324

Reaction equation



Reactants

Table 734: Properties of each reactant.

Id	Name	SBO
C9319	Reduced thioredoxin	
C0094	CDP	

Products

Table 735: Properties of each product.

Id	Name	SBO
C0523	dCDP	
C9318	Oxidized thioredoxin	
C0001	H2O	

Kinetic Law

Derived unit not available

$$v_{244} = \text{FLUX_VALUE} \quad (488)$$

Table 736: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.245 Reaction IR09859

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Phosphatidylcholine (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 737: Properties of each reactant.

Id	Name	SBO
EC9483_b	Phosphatidylcholine (Mpn)_b[e]	

Product

Table 738: Properties of each product.

Id	Name	SBO
EC9483	Phosphatidylcholine (Mpn)[e]	

Kinetic Law

Derived unit not available

$$v_{245} = \text{FLUX_VALUE} \quad (490)$$

Table 739: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			0.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

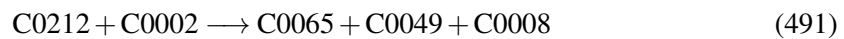
5.246 Reaction IR00293

This is an irreversible reaction of two reactants forming three products.

Name ATP:riboflavin 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN158

Reaction equation



Reactants

Table 740: Properties of each reactant.

Id	Name	SBO
C0212	Riboflavin	
C0002	ATP	

Products

Table 741: Properties of each product.

Id	Name	SBO
C0065	H+	
C0049	FMN	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{246} = \text{FLUX_VALUE} \quad (492)$$

Table 742: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.247 Reaction IR09955

This is an irreversible reaction of 42 reactants forming one product.

Name biomass production (Mpn)

Notes GENE ASSOCIATION:

Reaction equation

167760C0077 + 6837C0126 + 100C0005 + 1981C0175 + 2489C0156 + 148168C9474 + 9318C0041 + 3913C005
(493)

Reactants

Table 743: Properties of each reactant.

Id	Name	SBO
C0077	D-Glucose 6-phosphate	
C0126	L-Proline	
C0005	NADPH	
C0175	Adenosine	
C0156	L-Threonine	
C9474	phosphatidic acid (Mpn)	
C0041	L-Aspartate	
C0050	L-Arginine	
C9536	Protein (Mpn)	
C0082	L-Cysteine	
C0016	Pyridoxal phosphate	
C0055	Thiamin diphosphate	
C0064	L-Phenylalanine	
C0063	L-Tryptophan	
C0009	Orthophosphate	
C0035	L-Alanine	
C2169	5-Formyltetrahydrofolate	

Id	Name	SBO
C0151	L-Valine	
C9484	DNA (Mpn)	
C0052	L-Glutamine	
C0023	L-Glutamate	
C0010	CoA	
C0058	L-Methionine	
C9498	RNA (Mpn)	
C0006	NADP+	
C0129	L-Asparagine	
C9489	glycolipid (Mpn)	
C0017	S-Adenosyl-L-methionine	
C0317	L-Isoleucine	
C0039	L-Lysine	
C0015	FAD	
C0067	L-Tyrosine	
C0117	L-Histidine	
C9262	acyl carrier protein	
C0306	Guanosine	
C0242	Uridine	
C0105	L-Leucine	
C0250	CDPcholine	
C0363	Cytidine	
C0053	L-Serine	
C0177	Thymidine	
C0033	Glycine	

Product

Table 744: Properties of each product.

Id	Name	SBO
C9324	Biomass	

Kinetic Law

Derived unit not available

$$v_{247} = \text{FLUX_VALUE} \quad (494)$$

Table 745: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.248 Reaction IR08779

This is an irreversible reaction of two reactants forming two products.

Name LYST2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 746: Properties of each reactant.

Id	Name	SBO
C0039	L-Lysine	
C0065	H+	

Products

Table 747: Properties of each product.

Id	Name	SBO
EC0039	L-Lysine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{248} = \text{FLUX_VALUE} \quad (496)$$

Table 748: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

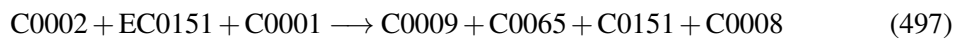
5.249 Reaction IR08770

This is an irreversible reaction of three reactants forming four products.

Name VALabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 749: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0151	L-Valine[e]	
C0001	H2O	

Products

Table 750: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0151	L-Valine	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{249} = \text{FLUX_VALUE} \quad (498)$$

Table 751: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

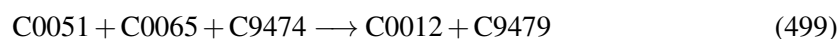
5.250 Reaction IR09826

This is an irreversible reaction of three reactants forming two products.

Name phosphatidate cytidyltransferase

Notes GENE ASSOCIATION: MPN637

Reaction equation



Reactants

Table 752: Properties of each reactant.

Id	Name	SBO
C0051	CTP	
C0065	H+	
C9474	phosphatidic acid (Mpn)	

Products

Table 753: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C9479	CDP-diacylglycerol (Mpn)	

Kinetic Law

Derived unit not available

$$v_{250} = \text{FLUX_VALUE} \quad (500)$$

Table 754: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

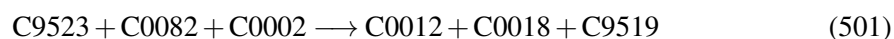
5.251 Reaction IR09903

This is an irreversible reaction of three reactants forming three products.

Name CysteinyI-tRNA synthetase

Notes GENE ASSOCIATION: MPN356

Reaction equation



Reactants

Table 755: Properties of each reactant.

Id	Name	SBO
C9523	tRNA(Cys)	
C0082	L-Cysteine	
C0002	ATP	

Products

Table 756: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9519	L-CysteinyI-tRNA(Cys)	

Kinetic Law

Derived unit not available

$$v_{251} = \text{FLUX_VALUE} \quad (502)$$

Table 757: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.252 Reaction IR08690

This is an irreversible reaction of two reactants forming two products.

Name VALt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 758: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0151	L-Valine	

Products

Table 759: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	
EC0151	L-Valine[e]	

Kinetic Law

Derived unit not available

$$v_{252} = \text{FLUX_VALUE} \quad (504)$$

Table 760: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

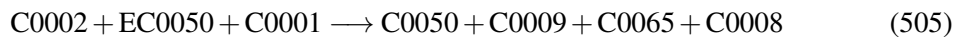
5.253 Reaction IR08753

This is an irreversible reaction of three reactants forming four products.

Name ARGabc

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 761: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0050	L-Arginine[e]	
C0001	H2O	

Products

Table 762: Properties of each product.

Id	Name	SBO
C0050	L-Arginine	
C0009	Orthophosphate	

Id	Name	SBO
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{253} = \text{FLUX_VALUE} \quad (506)$$

Table 763: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.254 Reaction IR00001

This is an irreversible reaction of two reactants forming two products.

Name Pyrophosphate phosphohydrolase

Notes GENE ASSOCIATION: MPN528

Reaction equation



Reactants

Table 764: Properties of each reactant.

Id	Name	SBO
C0012	Pyrophosphate	
C0001	H2O	

Products

Table 765: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	

Kinetic Law**Derived unit** not available

$$v_{254} = \text{FLUX_VALUE} \quad (508)$$

Table 766: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.255 Reaction IR03714

This is an irreversible reaction of two reactants forming three products.

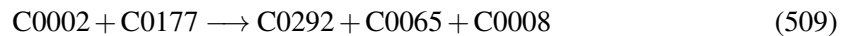
Name ATP:thymidine 5'-phosphotransferase**Notes** GENE ASSOCIATION: MPN044**Reaction equation****Reactants**

Table 767: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0177	Thymidine	

Products

Table 768: Properties of each product.

Id	Name	SBO
C0292	dTMP	
C0065	H+	
C0008	ADP	

Kinetic Law**Derived unit** not available

$$v_{255} = \text{FLUX_VALUE} \quad (510)$$

Table 769: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☑
UPPER_BOUND			999999.0		☑
OBJECTIVE- _COEFFICIENT			0.0		☑

5.256 Reaction IR09904

This is an irreversible reaction of three reactants forming three products.

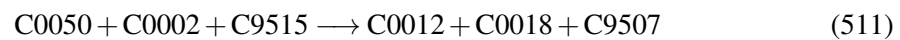
Name Arginyl-tRNA synthetase**Notes** GENE ASSOCIATION: MPN556**Reaction equation****Reactants**

Table 770: Properties of each reactant.

Id	Name	SBO
C0050	L-Arginine	
C0002	ATP	
C9515	tRNA(Arg)	

Products

Table 771: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9507	L-Arginyl-tRNA(Arg)	

Kinetic Law

Derived unit not available

$$v_{256} = \text{FLUX_VALUE} \quad (512)$$

Table 772: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

5.257 Reaction IR08996

This is an irreversible reaction of one reactant forming one product.

Name SINK: Acetate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 773: Properties of each reactant.

Id	Name	SBO
EC0029	Acetate[e]	

Product

Table 774: Properties of each product.

Id	Name	SBO
EC0029_b	Acetate.b[e]	

Kinetic Law

Derived unit not available

$$v_{257} = \text{FLUX_VALUE} \quad (514)$$

Table 775: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.000		☒
UPPER_BOUND			7.444		☒
OBJECTIVE- _COEFFICIENT			0.000		☒

5.258 Reaction IR00909

This is an irreversible reaction of two reactants forming two products.

Name 2'-Deoxycytidine 5'-monophosphate phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 776: Properties of each reactant.

Id	Name	SBO
C0198	dCMP	
C0001	H2O	

Products

Table 777: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0646	Deoxycytidine	

Kinetic Law

Derived unit not available

$$v_{258} = \text{FLUX_VALUE} \quad (516)$$

Table 778: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.259 Reaction IR08828

This is an irreversible reaction of two reactants forming two products.

Name GLCpts

Notes GENE ASSOCIATION: MPN627 and MPN207

Reaction equation



Reactants

Table 779: Properties of each reactant.

Id	Name	SBO
C0059	Phosphoenolpyruvate	
EC0027	D-Glucose[e]	

Products

Table 780: Properties of each product.

Id	Name	SBO
C0077	D-Glucose 6-phosphate	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{259} = \text{FLUX_VALUE} \quad (518)$$

Table 781: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

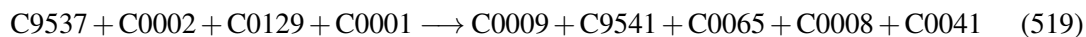
5.260 Reaction IR09912

This is an irreversible reaction of four reactants forming five products.

Name glutaminyl-tRNA synthetase (Asp)

Notes GENE ASSOCIATION: MPN236 and MPN237 and MPN238

Reaction equation



Reactants

Table 782: Properties of each reactant.

Id	Name	SBO
C9537	L-Glutamyl-tRNA(Gln)	
C0002	ATP	
C0129	L-Asparagine	

Id	Name	SBO
C0001	H2O	

Products

Table 783: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C9541	L-Glutaminyl-tRNA(Gln)	
C0065	H+	
C0008	ADP	
C0041	L-Aspartate	

Kinetic Law

Derived unit not available

$$v_{260} = \text{FLUX_VALUE} \quad (520)$$

Table 784: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

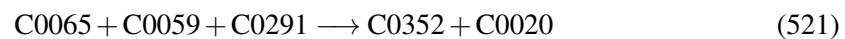
5.261 Reaction IR09793

This is an irreversible reaction of three reactants forming two products.

Name pyruvate kinase 4

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 785: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0059	Phosphoenolpyruvate	
C0291	dTDP	

Products

Table 786: Properties of each product.

Id	Name	SBO
C0352	dTTP	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{261} = \text{FLUX_VALUE} \quad (522)$$

Table 787: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

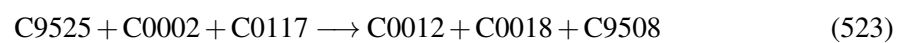
5.262 Reaction IR09892

This is an irreversible reaction of three reactants forming three products.

Name Histidyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN045

Reaction equation



Reactants

Table 788: Properties of each reactant.

Id	Name	SBO
C9525	tRNA(His)	
C0002	ATP	
C0117	L-Histidine	

Products

Table 789: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9508	L-Histidyl-tRNA(His)	

Kinetic Law

Derived unit not available

$$v_{262} = \text{FLUX_VALUE} \quad (524)$$

Table 790: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

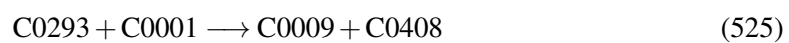
5.263 Reaction IR01159

This is an irreversible reaction of two reactants forming two products.

Name dUMP + H2O $\Rightarrow$ Deoxyuridine + Orthophosphate

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 791: Properties of each reactant.

Id	Name	SBO
C0293	dUMP	
C0001	H2O	

Products

Table 792: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0408	Deoxyuridine	

Kinetic Law

Derived unit not available

$$v_{263} = \text{FLUX_VALUE} \quad (526)$$

Table 793: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

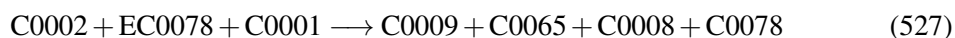
5.264 Reaction IR08849

This is an irreversible reaction of three reactants forming four products.

Name GLYC3Pabc

Notes GENE ASSOCIATION: MPN133 and MPN134 and MPN135 and MPN136

Reaction equation



Reactants

Table 794: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0078	sn-Glycerol 3-phosphate[e]	
C0001	H2O	

Products

Table 795: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0078	sn-Glycerol 3-phosphate	

Kinetic Law

Derived unit not available

$$v_{264} = \text{FLUX_VALUE} \quad (528)$$

Table 796: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.265 Reaction IR00075

This is an irreversible reaction of two reactants forming three products.

Name ATP:NADH 2'-phosphotransferase

Notes GENE ASSOCIATION: MPN267

Reaction equation



Reactants

Table 797: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0004	NADH	

Products

Table 798: Properties of each product.

Id	Name	SBO
C0065	H+	
C0005	NADPH	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{265} = \text{FLUX_VALUE} \quad (530)$$

Table 799: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

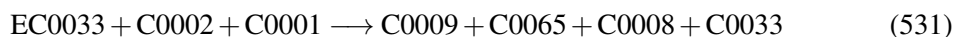
5.266 Reaction IR09960

This is an irreversible reaction of three reactants forming four products.

Name glycine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 800: Properties of each reactant.

Id	Name	SBO
EC0033	Glycine[e]	
C0002	ATP	
C0001	H2O	

Products

Table 801: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0033	Glycine	

Kinetic Law

Derived unit not available

$$v_{266} = \text{FLUX_VALUE} \quad (532)$$

Table 802: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

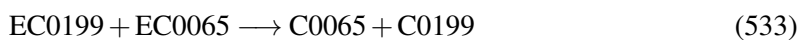
5.267 Reaction IR08645

This is an irreversible reaction of two reactants forming two products.

Name GUA_t2

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 803: Properties of each reactant.

Id	Name	SBO
EC0199	Guanine[e]	
EC0065	H+[e]	

Products

Table 804: Properties of each product.

Id	Name	SBO
C0065	H+	
C0199	Guanine	

Kinetic Law

Derived unit not available

$$v_{267} = \text{FLUX_VALUE} \quad (534)$$

Table 805: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

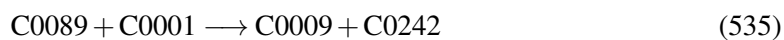
5.268 Reaction IR00614

This is an irreversible reaction of two reactants forming two products.

Name Uridine 5'-monophosphate phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 806: Properties of each reactant.

Id	Name	SBO
C0089	UMP	
C0001	H2O	

Products

Table 807: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0242	Uridine	

Kinetic Law

Derived unit not available

$$v_{268} = \text{FLUX_VALUE} \quad (536)$$

Table 808: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.269 Reaction IR08848

This is an irreversible reaction of two reactants forming two products.

Name L-tyrosine reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 809: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0067	L-Tyrosine	

Products

Table 810: Properties of each product.

Id	Name	SBO
EC0067	L-Tyrosine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{269} = \text{FLUX_VALUE} \quad (538)$$

Table 811: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.270 Reaction IR09864

This is an irreversible reaction of one reactant forming one product.

Name SINK: 5mcDNA (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 812: Properties of each reactant.

Id	Name	SBO
C9485	5mcDNA (Mpn)	

Product

Table 813: Properties of each product.

Id	Name	SBO
C9485_b	5mcDNA (Mpn)_b	

Kinetic Law

Derived unit not available

$$v_{270} = \text{FLUX_VALUE} \quad (540)$$

Table 814: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

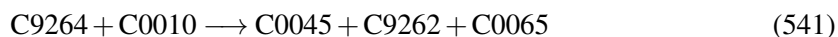
5.271 Reaction IR04278

This is an irreversible reaction of two reactants forming three products.

Name ACPS1

Notes GENE ASSOCIATION: MPN298

Reaction equation



Reactants

Table 815: Properties of each reactant.

Id	Name	SBO
C9264	apoprotein [acyl carrier protein]	
C0010	CoA	

Products

Table 816: Properties of each product.

Id	Name	SBO
C0045	Adenosine 3',5'-bisphosphate	
C9262	acyl carrier protein	
C0065	H+	

Kinetic Law

Derived unit not available

$$v_{271} = \text{FLUX_VALUE} \quad (542)$$

Table 817: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

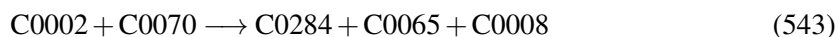
5.272 Reaction IR00450

This is an irreversible reaction of two reactants forming three products.

Name ATP:D-fructose-6-phosphate 1-phosphotransferase

Notes GENE ASSOCIATION: MPN302

Reaction equation



Reactants

Table 818: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0070	D-Fructose 6-phosphate	

Products

Table 819: Properties of each product.

Id	Name	SBO
C0284	D-Fructose 1,6-bisphosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{272} = \text{FLUX_VALUE} \quad (544)$$

Table 820: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓
OBJECTIVE- _COEFFICIENT			0.0		✓

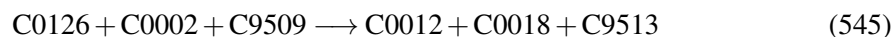
5.273 Reaction IR09896

This is an irreversible reaction of three reactants forming three products.

Name Prolyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN402

Reaction equation



Reactants

Table 821: Properties of each reactant.

Id	Name	SBO
C0126	L-Proline	
C0002	ATP	
C9509	tRNA(Pro)	

Products

Table 822: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9513	L-Prolyl-tRNA(Pro)	

Kinetic Law

Derived unit not available

$$v_{273} = \text{FLUX_VALUE} \quad (546)$$

Table 823: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.274 Reaction IR09923

This is an irreversible reaction of two reactants forming two products.

Name ARGt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 824: Properties of each reactant.

Id	Name	SBO
C0050	L-Arginine	
C0065	H+	

Products

Table 825: Properties of each product.

Id	Name	SBO
EC0050	L-Arginine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{274} = \text{FLUX_VALUE} \quad (548)$$

Table 826: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

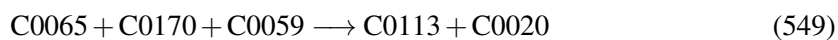
5.275 Reaction IR07969

This is an irreversible reaction of three reactants forming two products.

Name dATP:pyruvate O2-phosphotransferase (R)

Notes GENE ASSOCIATION: MPN303

Reaction equation



Reactants

Table 827: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0170	dADP	
C0059	Phosphoenolpyruvate	

Products

Table 828: Properties of each product.

Id	Name	SBO
C0113	dATP	
C0020	Pyruvate	

Kinetic Law

Derived unit not available

$$v_{275} = \text{FLUX_VALUE} \quad (550)$$

Table 829: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.276 Reaction IR09840

This is an irreversible reaction of one reactant forming one product.

Name SINK: H2O2[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 830: Properties of each reactant.

Id	Name	SBO
EC0025	H2O2[e]	

Product

Table 831: Properties of each product.

Id	Name	SBO
EC0025_b	H2O2_b[e]	

Kinetic Law

Derived unit not available

$$v_{276} = \text{FLUX_VALUE} \quad (552)$$

Table 832: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

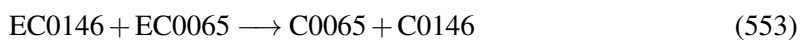
5.277 Reaction IR09855

This is an irreversible reaction of two reactants forming two products.

Name Thymine t2

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 833: Properties of each reactant.

Id	Name	SBO
EC0146	Thymine[e]	
EC0065	H+[e]	

Products

Table 834: Properties of each product.

Id	Name	SBO
C0065	H+	
C0146	Thymine	

Kinetic Law

Derived unit not available

$$v_{277} = \text{FLUX_VALUE} \quad (554)$$

Table 835: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

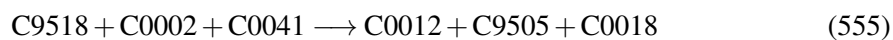
5.278 Reaction IR09894

This is an irreversible reaction of three reactants forming three products.

Name Aspartyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN046

Reaction equation



Reactants

Table 836: Properties of each reactant.

Id	Name	SBO
C9518	tRNA(Asp)	
C0002	ATP	
C0041	L-Aspartate	

Products

Table 837: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C9505	L-Aspartyl-tRNA(Asp)	
C0018	AMP	

Kinetic Law

Derived unit not available

$$v_{278} = \text{FLUX_VALUE} \quad (556)$$

Table 838: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.279 Reaction IR09809

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: dCytidine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 839: Properties of each reactant.

Id	Name	SBO
EC0646_b	Deoxycytidine_b[e]	

Product

Table 840: Properties of each product.

Id	Name	SBO
EC0646	Deoxycytidine[e]	

Kinetic Law

Derived unit not available

$$v_{279} = \text{FLUX_VALUE} \quad (558)$$

Table 841: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.280 Reaction IR08688

This is an irreversible reaction of two reactants forming two products.

Name L-leucine reversible transport via proton symport (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 842: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0105	L-Leucine	

Products

Table 843: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	
EC0105	L-Leucine[e]	

Kinetic Law

Derived unit not available

$$v_{280} = \text{FLUX_VALUE} \quad (560)$$

Table 844: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.281 Reaction IR09505

This is an irreversible reaction of one reactant forming one product.

Name SINK: (S)-Lactate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 845: Properties of each reactant.

Id	Name	SBO
EC0154	(S)-Lactate[e]	

Product

Table 846: Properties of each product.

Id	Name	SBO
EC0154_b	(S)-Lactate_b[e]	

Kinetic Law

Derived unit not available

$$v_{281} = \text{FLUX_VALUE} \quad (562)$$

Table 847: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

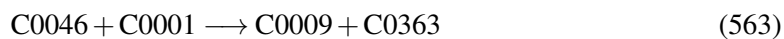
5.282 Reaction IR00267

This is an irreversible reaction of two reactants forming two products.

Name Cytidine-5'-monophosphate phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 848: Properties of each reactant.

Id	Name	SBO
C0046	CMP	
C0001	H2O	

Products

Table 849: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0363	Cytidine	

Kinetic Law

Derived unit not available

$$v_{282} = \text{FLUX_VALUE} \quad (564)$$

Table 850: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

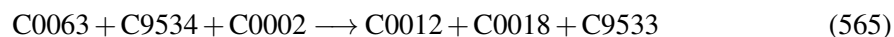
5.283 Reaction IR09900

This is an irreversible reaction of three reactants forming three products.

Name Tryptophanyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN265

Reaction equation



Reactants

Table 851: Properties of each reactant.

Id	Name	SBO
C0063	L-Tryptophan	
C9534	tRNA(Trp)	
C0002	ATP	

Products

Table 852: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0018	AMP	
C9533	L-Tryptophanyl-tRNA(Trp)	

Kinetic Law

Derived unit not available

$$v_{283} = \text{FLUX_VALUE} \quad (566)$$

Table 853: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

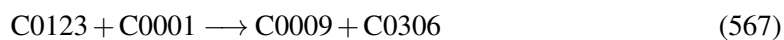
5.284 Reaction IR00777

This is an irreversible reaction of two reactants forming two products.

Name Guanosine 5'-monophosphate phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 854: Properties of each reactant.

Id	Name	SBO
C0123	GMP	
C0001	H2O	

Products

Table 855: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0306	Guanosine	

Kinetic Law

Derived unit not available

$$v_{284} = \text{FLUX_VALUE} \quad (568)$$

Table 856: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

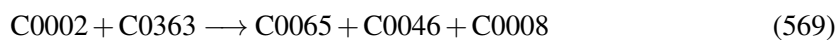
5.285 Reaction IR00269

This is an irreversible reaction of two reactants forming three products.

Name ATP:cytidine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN561

Reaction equation



Reactants

Table 857: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0363	Cytidine	

Products

Table 858: Properties of each product.

Id	Name	SBO
C0065	H+	
C0046	CMP	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{285} = \text{FLUX_VALUE} \quad (570)$$

Table 859: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

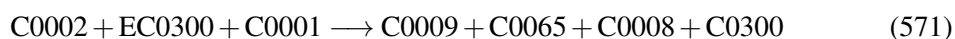
5.286 Reaction IR08836

This is an irreversible reaction of three reactants forming four products.

Name thiamine transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 860: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
EC0300	Thiamin[e]	
C0001	H2O	

Products

Table 861: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	
C0300	Thiamin	

Kinetic Law

Derived unit not available

$$v_{286} = \text{FLUX_VALUE} \quad (572)$$

Table 862: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.287 Reaction IR04231

This is an irreversible reaction of two reactants forming two products.

Name 3-keto-L-gulonate 6-phosphate decarboxylase

Notes GENE ASSOCIATION: MPN493

Reaction equation



Reactants

Table 863: Properties of each reactant.

Id	Name	SBO
C9255	3-keto-L-gulonate-6-phosphate	
C0065	H+	

Products

Table 864: Properties of each product.

Id	Name	SBO
C2070	L-Xylulose 5-phosphate	
C0011	CO ₂	

Kinetic Law

Derived unit not available

$$v_{287} = \text{FLUX_VALUE} \quad (574)$$

Table 865: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
	LOWER_BOUND		0.0		☒
	UPPER_BOUND		999999.0		☒
	OBJECTIVE- _COEFFICIENT		0.0		☒

5.288 Reaction IR09888

This is an irreversible reaction of two reactants forming two products.

Name Acetate export (Mpn)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 866: Properties of each reactant.

Id	Name	SBO
C0029	Acetate	
C0065	H+	

Products

Table 867: Properties of each product.

Id	Name	SBO
EC0065	H+[e]	
EC0029	Acetate[e]	

Kinetic Law

Derived unit not available

$$v_{288} = \text{FLUX_VALUE} \quad (576)$$

Table 868: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.289 Reaction IR09800

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: Nicotinate[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 869: Properties of each reactant.

Id	Name	SBO
EC0210_b	Nicotinate_b[e]	

Product

Table 870: Properties of each product.

Id	Name	SBO
EC0210	Nicotinate[e]	

Kinetic Law

Derived unit not available

$$v_{289} = \text{FLUX_VALUE} \quad (578)$$

Table 871: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

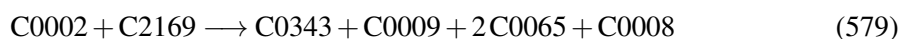
5.290 Reaction IR03832

This is an irreversible reaction of two reactants forming four products.

Name 5-Formyltetrahydrofolate cyclo-ligase (ADP-forming)

Notes GENE ASSOCIATION: MPN348

Reaction equation



Reactants

Table 872: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C2169	5-Formyltetrahydrofolate	

Products

Table 873: Properties of each product.

Id	Name	SBO
C0343	5,10-Methenyltetrahydrofolate	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{290} = \text{FLUX_VALUE} \quad (580)$$

Table 874: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

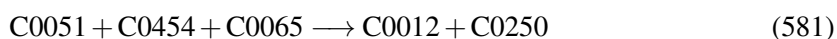
5.291 Reaction IR01064

This is an irreversible reaction of three reactants forming two products.

Name CTP:choline-phosphate cytidyltransferase

Notes GENE ASSOCIATION: MPN336

Reaction equation



Reactants

Table 875: Properties of each reactant.

Id	Name	SBO
C0051	CTP	
C0454	Choline phosphate	
C0065	H+	

Products

Table 876: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C0250	CDPcholine	

Kinetic Law

Derived unit not available

$$v_{291} = \text{FLUX_VALUE} \quad (582)$$

Table 877: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

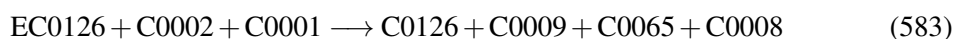
5.292 Reaction IR08823

This is an irreversible reaction of three reactants forming four products.

Name L-proline transport via ABC system

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 878: Properties of each reactant.

Id	Name	SBO
EC0126	L-Proline[e]	
C0002	ATP	
C0001	H2O	

Products

Table 879: Properties of each product.

Id	Name	SBO
C0126	L-Proline	
C0009	Orthophosphate	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{292} = \text{FLUX_VALUE} \quad (584)$$

Table 880: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		✓
UPPER_BOUND			999999.0		✓

Id	Name	SBO	Value	Unit	Constant
OBJECTIVE- _COEFFICIENT			0.0		☒

5.293 Reaction IR09479

This is an irreversible reaction of one reactant forming one product.

Name SINK: L-Ornithine[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 881: Properties of each reactant.

Id	Name	SBO
EC0062	L-Ornithine[e]	

Product

Table 882: Properties of each product.

Id	Name	SBO
EC0062_b	L-Ornithine_b[e]	

Kinetic Law

Derived unit not available

$$v_{293} = \text{FLUX_VALUE} \quad (586)$$

Table 883: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			0.0		☒

Id	Name	SBO	Value	Unit	Constant
OBJECTIVE- _COEFFICIENT			0.0		☒

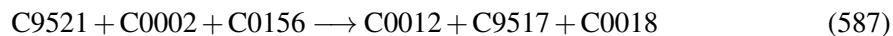
5.294 Reaction IR09901

This is an irreversible reaction of three reactants forming three products.

Name Threonyl-tRNA synthetase

Notes GENE ASSOCIATION: MPN553

Reaction equation



Reactants

Table 884: Properties of each reactant.

Id	Name	SBO
C9521	tRNA(Thr)	
C0002	ATP	
C0156	L-Threonine	

Products

Table 885: Properties of each product.

Id	Name	SBO
C0012	Pyrophosphate	
C9517	L-Threonyl-tRNA(Thr)	
C0018	AMP	

Kinetic Law

Derived unit not available

$$v_{294} = \text{FLUX_VALUE} \quad (588)$$

Table 886: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.295 Reaction IR09811

This is an irreversible reaction of one reactant forming one product.

Name Ribose transporter

Notes GENE ASSOCIATION: MPN258 and MPN259 and MPN260

Reaction equation



Reactant

Table 887: Properties of each reactant.

Id	Name	SBO
EC0103	D-Ribose[e]	

Product

Table 888: Properties of each product.

Id	Name	SBO
C0103	D-Ribose	

Kinetic Law

Derived unit not available

$$v_{295} = \text{FLUX_VALUE} \quad (590)$$

Table 889: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.296 Reaction IR09927

This is an irreversible reaction of two reactants forming two products.

Name MET2t (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 890: Properties of each reactant.

Id	Name	SBO
C0058	L-Methionine	
C0065	H+	

Products

Table 891: Properties of each product.

Id	Name	SBO
EC0058	L-Methionine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{296} = \text{FLUX_VALUE} \quad (592)$$

Table 892: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.297 Reaction IR09817

This is an irreversible reaction of one reactant forming one product.

Name pyridoxal transporter

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 893: Properties of each reactant.

Id	Name	SBO
EC0207	Pyridoxal[e]	

Product

Table 894: Properties of each product.

Id	Name	SBO
C0207	Pyridoxal	

Kinetic Law

Derived unit not available

$$v_{297} = \text{FLUX_VALUE} \quad (594)$$

Table 895: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.298 Reaction IR08842

This is an irreversible reaction of two reactants forming two products.

Name THRt2r (R)

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 896: Properties of each reactant.

Id	Name	SBO
C0065	H+	
C0156	L-Threonine	

Products

Table 897: Properties of each product.

Id	Name	SBO
EC0156	L-Threonine[e]	
EC0065	H+[e]	

Kinetic Law

Derived unit not available

$$v_{298} = \text{FLUX_VALUE} \quad (596)$$

Table 898: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.299 Reaction IR01418

This is an irreversible reaction of two reactants forming two products.

Name Deoxycytidine aminohydrolase

Notes GENE ASSOCIATION: MPN065

Reaction equation



Reactants

Table 899: Properties of each reactant.

Id	Name	SBO
C0646	Deoxycytidine	
C0001	H2O	

Products

Table 900: Properties of each product.

Id	Name	SBO
C0408	Deoxyuridine	
C0013	NH3	

Kinetic Law

Derived unit not available

$$v_{299} = \text{FLUX_VALUE} \quad (598)$$

Table 901: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

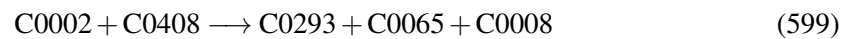
5.300 Reaction IR01156

This is an irreversible reaction of two reactants forming three products.

Name ATP:deoxyuridine 5'-phosphotransferase

Notes GENE ASSOCIATION: MPN044

Reaction equation



Reactants

Table 902: Properties of each reactant.

Id	Name	SBO
C0002	ATP	
C0408	Deoxyuridine	

Products

Table 903: Properties of each product.

Id	Name	SBO
C0293	dUMP	
C0065	H+	
C0008	ADP	

Kinetic Law

Derived unit not available

$$v_{300} = \text{FLUX_VALUE} \quad (600)$$

Table 904: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.301 Reaction IR00094

This is an irreversible reaction of two reactants forming three products.

Name ATP:dephospho-CoA 3'-phosphotransferase

Notes GENE ASSOCIATION: MPN382

Reaction equation



Reactants

Table 905: Properties of each reactant.

Id	Name	SBO
C0647	Dephospho-CoA	
C0002	ATP	

Products

Table 906: Properties of each product.

Id	Name	SBO
C0065	H+	
C0008	ADP	
C0010	CoA	

Kinetic Law

Derived unit not available

$$v_{301} = \text{FLUX_VALUE} \quad (602)$$

Table 907: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.302 Reaction IR09797

This is an irreversible reaction of two reactants forming two products.

Name methionyl-tRNA formyltransferase

Notes GENE ASSOCIATION: MPN543

Reaction equation



Reactants

Table 908: Properties of each reactant.

Id	Name	SBO
C0193	10-Formyltetrahydrofolate	
C9467	L-Methionyl-tRNA(Met)	

Products

Table 909: Properties of each product.

Id	Name	SBO
C9468	N-Formylmethionyl-tRNA(Met)	
C0085	Tetrahydrofolate	

Kinetic Law

Derived unit not available

$$v_{302} = \text{FLUX_VALUE} \quad (604)$$

Table 910: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.303 Reaction IR01486

This is an irreversible reaction of one reactant forming two products.

Name D-Fructose 1-phosphate D-glyceraldehyde-3-phosphate-lyase

Notes GENE ASSOCIATION: MPN025

Reaction equation



Reactant

Table 911: Properties of each reactant.

Id	Name	SBO
C0790	D-Fructose 1-phosphate	

Products

Table 912: Properties of each product.

Id	Name	SBO
C0093	Dihydroxyacetone phosphate	
C0445	D-Glyceraldehyde	

Kinetic Law

Derived unit not available

$$v_{303} = \text{FLUX_VALUE} \quad (606)$$

Table 913: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

5.304 Reaction IR09059

This is an irreversible reaction of one reactant forming one product.

Name SOURCE: D-Fructose[e]

Notes GENE ASSOCIATION:

Reaction equation



Reactant

Table 914: Properties of each reactant.

Id	Name	SBO
EC0080_b	D-Fructose_b[e]	

Product

Table 915: Properties of each product.

Id	Name	SBO
EC0080	D-Fructose[e]	

Kinetic Law

Derived unit not available

$$v_{304} = \text{FLUX_VALUE} \quad (608)$$

Table 916: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			0.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

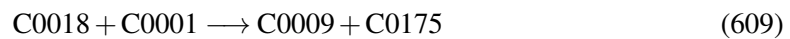
5.305 Reaction IR03505

This is an irreversible reaction of two reactants forming two products.

Name Adenosine 5'-monophosphate phosphohydrolase

Notes GENE ASSOCIATION:

Reaction equation



Reactants

Table 917: Properties of each reactant.

Id	Name	SBO
C0018	AMP	
C0001	H2O	

Products

Table 918: Properties of each product.

Id	Name	SBO
C0009	Orthophosphate	
C0175	Adenosine	

Kinetic Law

Derived unit not available

$$v_{305} = \text{FLUX_VALUE} \quad (610)$$

Table 919: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

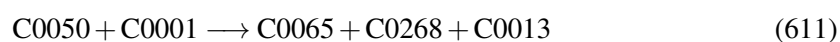
5.306 Reaction IR00296

This is an irreversible reaction of two reactants forming three products.

Name L-Arginine iminohydrolase

Notes GENE ASSOCIATION: (MPN304 and MPN305) or MPN560

Reaction equation



Reactants

Table 920: Properties of each reactant.

Id	Name	SBO
C0050	L-Arginine	
C0001	H2O	

Products

Table 921: Properties of each product.

Id	Name	SBO
C0065	H+	
C0268	L-Citrulline	
C0013	NH3	

Kinetic Law

Derived unit not available

$$v_{306} = \text{FLUX_VALUE} \quad (612)$$

Table 922: Properties of each parameter.

Id	Name	SBO	Value	Unit	Constant
LOWER_BOUND			0.0		☒
UPPER_BOUND			999999.0		☒
OBJECTIVE- _COEFFICIENT			0.0		☒

6 Derived Rate Equations

When interpreted as an ordinary differential equation framework, this model implies the following set of equations for the rates of change of each species.

Identifiers for kinetic laws highlighted in gray cannot be verified to evaluate to units of SBML substance per time. As a result, some SBML interpreters may not be able to verify the consistency of the units on quantities in the model. Please check if

- parameters without an unit definition are involved or
- volume correction is necessary because the `hasOnlySubstanceUnits` flag may be set to `false` and `spacialDimensions` > 0 for certain species.

6.1 Species EC0033

Name Glycine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09076](#), [IR09960](#) and as a product in [IR08707](#)).

$$\frac{d}{dt}\text{EC0033} = v_{158} - v_{51} - v_{266} \quad (613)$$

6.2 Species EC0050_b

Name L-Arginine_b[e]

Charge 1

This species takes part in one reaction (as a product in [RR09050](#)).

$$\frac{d}{dt}\text{EC0050.b} = v_5 \quad (614)$$

6.3 Species [EC0090](#)

Name Uracil[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR08636](#) and as a product in [IR09807](#)).

$$\frac{d}{dt}EC0090 = v_{197} - v_{103} \quad (615)$$

6.4 Species [EC0063](#)

Name L-Tryptophan[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09493](#), [IR09957](#) and as a product in [IR08844](#)).

$$\frac{d}{dt}EC0063 = v_{178} - v_{60} - v_{176} \quad (616)$$

6.5 Species [EC0007](#)

Name Oxygen[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR08670](#) and as a product in [IR09095](#)).

$$\frac{d}{dt}EC0007 = v_{199} - v_{68} \quad (617)$$

6.6 Species [EC0080](#)

Name D-Fructose[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08829](#) and as a product in [IR09059](#)).

$$\frac{d}{dt}EC0080 = v_{304} - v_{163} \quad (618)$$

6.7 Species [EC0646_b](#)

Name Deoxycytidine_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09809](#)).

$$\frac{d}{dt}EC0646_b = -v_{279} \quad (619)$$

6.8 Species EC0210_b

Name Nicotinate_b[e]

Charge -1

This species takes part in one reaction (as a reactant in [IR09800](#)).

$$\frac{d}{dt} \text{EC0210_b} = -v_{289} \quad (620)$$

6.9 Species EC0125

Name Adenine[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09104](#), [RR08870](#)).

$$\frac{d}{dt} \text{EC0125} = -v_{16} - v_{18} \quad (621)$$

6.10 Species EC0098_b

Name Glycerol_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09073](#)).

$$\frac{d}{dt} \text{EC0098_b} = -v_{217} \quad (622)$$

6.11 Species EC0027

Name D-Glucose[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08828](#) and as a product in [IR08970](#)).

$$\frac{d}{dt} \text{EC0027} = v_{181} - v_{259} \quad (623)$$

6.12 Species EC0132_b

Name L-Homocysteine_b[e]

Charge 0

This species takes part in one reaction (as a product in [IR09866](#)).

$$\frac{d}{dt} \text{EC0132_b} = v_{218} \quad (624)$$

6.13 Species [EC0007_b](#)

Name Oxygen_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09095](#)).

$$\frac{d}{dt} \text{EC0007_b} = -v_{199} \quad (625)$$

6.14 Species [EC0052_b](#)

Name L-Glutamine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09084](#)).

$$\frac{d}{dt} \text{EC0052_b} = v_{88} \quad (626)$$

6.15 Species [EC0041_b](#)

Name L-Aspartate_b[e]

Charge -1

This species takes part in one reaction (as a product in [RR09052](#)).

$$\frac{d}{dt} \text{EC0041_b} = v_{30} \quad (627)$$

6.16 Species [EC0151_b](#)

Name L-Valine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09498](#)).

$$\frac{d}{dt} \text{EC0151_b} = v_{87} \quad (628)$$

6.17 Species [EC0057](#)

Name Ascorbate[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09773](#) and as a product in [IR09771](#)).

$$\frac{d}{dt} \text{EC0057} = v_{130} - v_{162} \quad (629)$$

6.18 Species [EC0156_b](#)

Name L-Threonine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09116](#)).

$$\frac{d}{dt}EC0156_b = v_2 \quad (630)$$

6.19 Species [EC0103](#)

Name D-Ribose[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09811](#) and as a product in [IR09109](#)).

$$\frac{d}{dt}EC0103 = v_{149} - v_{295} \quad (631)$$

6.20 Species [EC0151](#)

Name L-Valine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09498](#), [IR08770](#) and as a product in [IR08690](#)).

$$\frac{d}{dt}EC0151 = v_{252} - v_{87} - v_{249} \quad (632)$$

6.21 Species [EC0105_b](#)

Name L-Leucine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09462](#)).

$$\frac{d}{dt}EC0105_b = v_3 \quad (633)$$

6.22 Species EC0035

Name L-Alanine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09082](#), [IR08769](#) and as a product in [IR08635](#)).

$$\frac{d}{dt}EC0035 = v_{192} - v_{46} - v_{226} \quad (634)$$

6.23 Species EC0011

Name CO2[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09054](#), [RR08672](#)).

$$\frac{d}{dt}EC0011 = -v_{47} - v_{59} \quad (635)$$

6.24 Species EC0057_b

Name Ascorbate_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09771](#)).

$$\frac{d}{dt}EC0057_b = -v_{130} \quad (636)$$

6.25 Species EC0207

Name Pyridoxal[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09817](#) and as a product in [IR09801](#)).

$$\frac{d}{dt}EC0207 = v_{118} - v_{297} \quad (637)$$

6.26 Species EC0078

Name sn-Glycerol 3-phosphate[e]

Charge -2

This species takes part in two reactions (as a reactant in [IR08849](#) and as a product in [IR09075](#)).

$$\frac{d}{dt}EC0078 = v_{237} - v_{264} \quad (638)$$

6.27 Species EC0212_b

Name Riboflavin_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09802](#)).

$$\frac{d}{dt} \text{EC0212_b} = -v_{214} \quad (639)$$

6.28 Species EC0610

Name Pantetheine[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09816](#) and as a product in [IR09799](#)).

$$\frac{d}{dt} \text{EC0610} = v_{223} - v_{211} \quad (640)$$

6.29 Species EC0300_b

Name Thiamin_b[e]

Charge 1

This species takes part in one reaction (as a reactant in [IR09803](#)).

$$\frac{d}{dt} \text{EC0300_b} = -v_{156} \quad (641)$$

6.30 Species EC0063_b

Name L-Tryptophan_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09493](#)).

$$\frac{d}{dt} \text{EC0063_b} = v_{60} \quad (642)$$

6.31 Species EC0025_b

Name H2O2_b[e]

Charge 0

This species takes part in one reaction (as a product in [IR09840](#)).

$$\frac{d}{dt} \text{EC0025_b} = v_{276} \quad (643)$$

6.32 Species EC0096_b

Name Choline_b[e]

Charge 1

This species takes part in one reaction (as a product in [RR09424](#)).

$$\frac{d}{dt}EC0096_b = v_{109} \quad (644)$$

6.33 Species EC0146_b

Name Thymine_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09854](#)).

$$\frac{d}{dt}EC0146_b = -v_{232} \quad (645)$$

6.34 Species EC0029

Name Acetate[e]

Charge -1

This species takes part in two reactions (as a reactant in [IR08996](#) and as a product in [IR09888](#)).

$$\frac{d}{dt}EC0029 = v_{288} - v_{257} \quad (646)$$

6.35 Species EC0065

Name H+[e]

Charge 1

This species takes part in 31 reactions (as a reactant in [RR08870](#), [RR08868](#), [RR08593](#), [RR08860](#), [RR08636](#), [IR08807](#), [IR08645](#), [IR09855](#) and as a product in [RR08971](#), [IR08684](#), [IR08692](#), [IR09924](#), [IR08686](#), [IR08822](#), [IR08750](#), [IR08661](#), [IR08707](#), [IR09886](#), [IR08844](#), [IR08635](#), [IR09926](#), [IR09925](#), [IR08835](#), [IR08779](#), [IR08690](#), [IR08848](#), [IR09923](#), [IR08688](#), [IR09888](#), [IR09927](#), [IR08842](#)).

$$\begin{aligned} \frac{d}{dt}EC0065 = & v_{66} + v_{115} + v_{121} + v_{122} + v_{125} + v_{144} + v_{147} + v_{148} + v_{158} + v_{175} + v_{178} \\ & + v_{192} + v_{196} + v_{209} + v_{213} + v_{248} + v_{252} + v_{269} + v_{274} + v_{280} + v_{288} \\ & + v_{296} + v_{298} - v_{18} - v_{28} - 4v_{53} - v_{55} - v_{103} - v_{157} - v_{267} - v_{277} \end{aligned} \quad (647)$$

6.36 Species EC0207_b

Name Pyridoxal_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09801](#)).

$$\frac{d}{dt}EC0207_b = -v_{118} \quad (648)$$

6.37 Species EC0389_b

Name Folic acid_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09798](#)).

$$\frac{d}{dt}EC0389_b = -v_{151} \quad (649)$$

6.38 Species EC0009_b

Name Orthophosphate_b[e]

Charge -2

This species takes part in one reaction (as a product in [RR09100](#)).

$$\frac{d}{dt}EC0009_b = v_{92} \quad (650)$$

6.39 Species EC0065_b

Name H+_b[e]

Charge 1

This species takes part in one reaction (as a reactant in [RR08971](#)).

$$\frac{d}{dt}EC0065_b = -v_{66} \quad (651)$$

6.40 Species EC0050

Name L-Arginine[e]

Charge 1

This species takes part in four reactions (as a reactant in [RR09050](#), [RR08615](#), [IR08753](#) and as a product in [IR09923](#)).

$$\frac{d}{dt}EC0050 = v_{274} - v_5 - v_{31} - v_{253} \quad (652)$$

6.41 Species EC0154_b

Name (S)-Lactate_b[e]

Charge -1

This species takes part in one reaction (as a product in [IR09505](#)).

$$\frac{d}{dt}EC0154_b = v_{281} \quad (653)$$

6.42 Species EC0039

Name L-Lysine[e]

Charge 1

This species takes part in three reactions (as a reactant in [RR09088](#), [IR08755](#) and as a product in [IR08779](#)).

$$\frac{d}{dt}EC0039 = v_{248} - v_{35} - v_{146} \quad (654)$$

6.43 Species EC0210

Name Nicotinate[e]

Charge -1

This species takes part in two reactions (as a reactant in [IR08658](#) and as a product in [IR09800](#)).

$$\frac{d}{dt}EC0210 = v_{289} - v_{216} \quad (655)$$

6.44 Species EC9483_b

Name Phosphatidylcholine (Mpn)_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09859](#)).

$$\frac{d}{dt}EC9483_b = -v_{245} \quad (656)$$

6.45 Species EC0064

Name L-Phenylalanine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09098](#), [IR09959](#) and as a product in [IR08686](#)).

$$\frac{d}{dt}EC0064 = v_{125} - v_{95} - v_{225} \quad (657)$$

6.46 Species EC0199

Name Guanine[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08645](#) and as a product in [IR09079](#)).

$$\frac{d}{dt}EC0199 = v_{145} - v_{267} \quad (658)$$

6.47 Species EC0058

Name L-Methionine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09471](#), [IR08789](#) and as a product in [IR09927](#)).

$$\frac{d}{dt}EC0058 = v_{296} - v_{64} - v_{230} \quad (659)$$

6.48 Species EC0041

Name L-Aspartate[e]

Charge -1

This species takes part in three reactions (as a reactant in [RR09052](#), [IR08746](#) and as a product in [IR09924](#)).

$$\frac{d}{dt}EC0041 = v_{122} - v_{30} - v_{129} \quad (660)$$

6.49 Species [EC0029_b](#)

Name Acetate_b[e]

Charge -1

This species takes part in one reaction (as a product in [IR08996](#)).

$$\frac{d}{dt}EC0029_b = v_{257} \quad (661)$$

6.50 Species [EC0052](#)

Name L-Glutamine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09084](#), [IR08738](#) and as a product in [IR09926](#)).

$$\frac{d}{dt}EC0052 = v_{196} - v_{88} - v_{194} \quad (662)$$

6.51 Species [EC0082_b](#)

Name L-Cysteine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09428](#)).

$$\frac{d}{dt}EC0082_b = v_7 \quad (663)$$

6.52 Species [EC0062](#)

Name L-Ornithine[e]

Charge 1

This species takes part in two reactions (as a reactant in [IR09479](#) and as a product in [RR08615](#)).

$$\frac{d}{dt}EC0062 = v_{31} - v_{293} \quad (664)$$

6.53 Species [EC9471_b](#)

Name Fatty Acid (Mpn)_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [RR09837](#)).

$$\frac{d}{dt}EC9471_b = -v_{83} \quad (665)$$

6.54 Species EC0067

Name L-Tyrosine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09496](#), [IR09958](#) and as a product in [IR08848](#)).

$$\frac{d}{dt}EC0067 = v_{269} - v_{94} - v_{170} \quad (666)$$

6.55 Species EC0309

Name Mannitol[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08826](#) and as a product in [IR09805](#)).

$$\frac{d}{dt}EC0309 = v_{224} - v_{212} \quad (667)$$

6.56 Species EC0389

Name Folic acid[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09815](#) and as a product in [IR09798](#)).

$$\frac{d}{dt}EC0389 = v_{151} - v_{119} \quad (668)$$

6.57 Species EC0129

Name L-Asparagine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09421](#), [IR08659](#) and as a product in [IR08661](#)).

$$\frac{d}{dt}EC0129 = v_{148} - v_{40} - v_{188} \quad (669)$$

6.58 Species EC9483

Name Phosphatidylcholine (Mpn)[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09860](#) and as a product in [IR09859](#)).

$$\frac{d}{dt}EC9483 = v_{245} - v_{168} \quad (670)$$

6.59 Species EC0001_b

Name H2O_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [RR09229](#)).

$$\frac{d}{dt}EC0001_b = -v_{84} \quad (671)$$

6.60 Species EC0001

Name H2O[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR08674](#) and as a product in [RR09229](#)).

$$\frac{d}{dt}EC0001 = v_{84} - v_{85} \quad (672)$$

6.61 Species EC9324

Name Biomass[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09882](#) and as a product in [IR09881](#)).

$$\frac{d}{dt}EC9324 = v_{221} - v_{106} \quad (673)$$

6.62 Species EC9471

Name Fatty Acid (Mpn)[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09830](#) and as a product in [RR09837](#)).

$$\frac{d}{dt}EC9471 = v_{83} - v_{29} \quad (674)$$

6.63 Species EC0035_b

Name L-Alanine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09082](#)).

$$\frac{d}{dt}EC0035_b = v_{46} \quad (675)$$

6.64 Species EC0300

Name Thiamin[e]

Charge 1

This species takes part in two reactions (as a reactant in [IR08836](#) and as a product in [IR09803](#)).

$$\frac{d}{dt}EC0300 = v_{156} - v_{286} \quad (676)$$

6.65 Species EC0082

Name L-Cysteine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09428](#), [IR08663](#) and as a product in [IR09925](#)).

$$\frac{d}{dt}EC0082 = v_{209} - v_7 - v_{113} \quad (677)$$

6.66 Species EC0025

Name H2O2[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09789](#), [IR09840](#)).

$$\frac{d}{dt}EC0025 = -v_{62} - v_{276} \quad (678)$$

6.67 Species EC0156

Name L-Threonine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09116](#), [IR08768](#) and as a product in [IR08842](#)).

$$\frac{d}{dt}EC0156 = v_{298} - v_2 - v_{169} \quad (679)$$

6.68 Species EC0033_b

Name Glycine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09076](#)).

$$\frac{d}{dt}EC0033_b = v_{51} \quad (680)$$

6.69 Species EC0078_b

Name sn-Glycerol 3-phosphate_b[e]

Charge -2

This species takes part in one reaction (as a reactant in [IR09075](#)).

$$\frac{d}{dt}EC0078_b = -v_{237} \quad (681)$$

6.70 Species EC0125_b

Name Adenine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09104](#)).

$$\frac{d}{dt}EC0125_b = v_{16} \quad (682)$$

6.71 Species EC0135

Name D-Mannose[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08786](#) and as a product in [IR09093](#)).

$$\frac{d}{dt}EC0135 = v_{238} - v_{152} \quad (683)$$

6.72 Species EC0309_b

Name Mannitol_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09805](#)).

$$\frac{d}{dt} \text{EC0309_b} = -v_{224} \quad (684)$$

6.73 Species EC0090_b

Name Uracil_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09807](#)).

$$\frac{d}{dt} \text{EC0090_b} = -v_{197} \quad (685)$$

6.74 Species EC0062_b

Name L-Ornithine_b[e]

Charge 1

This species takes part in one reaction (as a product in [IR09479](#)).

$$\frac{d}{dt} \text{EC0062_b} = v_{293} \quad (686)$$

6.75 Species EC0064_b

Name L-Phenylalanine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09098](#)).

$$\frac{d}{dt} \text{EC0064_b} = v_{95} \quad (687)$$

6.76 Species EC0103_b

Name D-Ribose_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09109](#)).

$$\frac{d}{dt} \text{EC0103_b} = -v_{149} \quad (688)$$

6.77 Species EC0610_b

Name Pantetheine_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09799](#)).

$$\frac{d}{dt} \text{EC0610_b} = -v_{223} \quad (689)$$

6.78 Species EC0129_b

Name L-Asparagine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09421](#)).

$$\frac{d}{dt} \text{EC0129_b} = v_{40} \quad (690)$$

6.79 Species EC0646

Name Deoxycytidine[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR08807](#) and as a product in [IR09809](#)).

$$\frac{d}{dt} \text{EC0646} = v_{279} - v_{157} \quad (691)$$

6.80 Species EC0067_b

Name L-Tyrosine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09496](#)).

$$\frac{d}{dt} \text{EC0067_b} = v_{94} \quad (692)$$

6.81 Species EC0363_b

Name Cytidine_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09808](#)).

$$\frac{d}{dt} \text{EC0363_b} = -v_{127} \quad (693)$$

6.82 Species EC0011_b

Name CO2_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09054](#)).

$$\frac{d}{dt}EC0011_b = v_{47} \quad (694)$$

6.83 Species EC0053

Name L-Serine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09484](#), [IR09845](#) and as a product in [IR08835](#)).

$$\frac{d}{dt}EC0053 = v_{213} - v_{80} - v_{128} \quad (695)$$

6.84 Species EC0126_b

Name L-Proline_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09090](#)).

$$\frac{d}{dt}EC0126_b = v_{107} \quad (696)$$

6.85 Species EC0023_b

Name L-Glutamate_b[e]

Charge -1

This species takes part in one reaction (as a product in [RR09070](#)).

$$\frac{d}{dt}EC0023_b = v_{23} \quad (697)$$

6.86 Species EC0363

Name Cytidine[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR08860](#) and as a product in [IR09808](#)).

$$\frac{d}{dt}EC0363 = v_{127} - v_{55} \quad (698)$$

6.87 Species EC0212

Name Riboflavin[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09818](#) and as a product in [IR09802](#)).

$$\frac{d}{dt}EC0212 = v_{214} - v_{207} \quad (699)$$

6.88 Species EC0154

Name (S)-Lactate[e]

Charge -1

This species takes part in two reactions (as a reactant in [IR09505](#) and as a product in [IR09886](#)).

$$\frac{d}{dt}EC0154 = v_{175} - v_{281} \quad (700)$$

6.89 Species EC0027_b

Name D-Glucose_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR08970](#)).

$$\frac{d}{dt}EC0027_b = -v_{181} \quad (701)$$

6.90 Species EC0135_b

Name D-Mannose_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09093](#)).

$$\frac{d}{dt}EC0135_b = -v_{238} \quad (702)$$

6.91 Species EC0023

Name L-Glutamate[e]

Charge -1

This species takes part in three reactions (as a reactant in [RR09070](#), [IR08747](#) and as a product in [IR08750](#)).

$$\frac{d}{dt}EC0023 = v_{147} - v_{23} - v_{241} \quad (703)$$

6.92 Species [EC0013](#)

Name NH3[e]

Charge 0

This species takes part in two reactions (as a reactant in [RR09124](#) and as a product in [RR09841](#)).

$$\frac{d}{dt}\text{EC0013} = v_{111} - v_{97} \quad (704)$$

6.93 Species [EC0096](#)

Name Choline[e]

Charge 1

This species takes part in two reactions (as a reactant in [RR08868](#), [RR09424](#)).

$$\frac{d}{dt}\text{EC0096} = -v_{28} - v_{109} \quad (705)$$

6.94 Species [EC0105](#)

Name L-Leucine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09462](#), [IR08771](#) and as a product in [IR08688](#)).

$$\frac{d}{dt}\text{EC0105} = v_{280} - v_3 - v_{187} \quad (706)$$

6.95 Species [EC0126](#)

Name L-Proline[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09090](#), [IR08823](#) and as a product in [IR08822](#)).

$$\frac{d}{dt}\text{EC0126} = v_{144} - v_{107} - v_{292} \quad (707)$$

6.96 Species [EC0013_b](#)

Name NH3_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09124](#)).

$$\frac{d}{dt}EC0013_b = v_{97} \quad (708)$$

6.97 Species [EC0053_b](#)

Name L-Serine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09484](#)).

$$\frac{d}{dt}EC0053_b = v_{80} \quad (709)$$

6.98 Species [EC0117_b](#)

Name L-Histidine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09086](#)).

$$\frac{d}{dt}EC0117_b = v_{86} \quad (710)$$

6.99 Species [EC0146](#)

Name Thymine[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09855](#) and as a product in [IR09854](#)).

$$\frac{d}{dt}EC0146 = v_{232} - v_{277} \quad (711)$$

6.100 Species [EC0009](#)

Name Orthophosphate[e]

Charge -2

This species takes part in three reactions (as a reactant in [RR09100](#), [IR08824](#) and as a product in [IR09590](#)).

$$\frac{d}{dt}EC0009 = v_{141} - v_{92} - v_{190} \quad (712)$$

6.101 Species EC0199_b

Name Guanine_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09079](#)).

$$\frac{d}{dt} \text{EC0199_b} = -v_{145} \quad (713)$$

6.102 Species EC0039_b

Name L-Lysine_b[e]

Charge 1

This species takes part in one reaction (as a product in [RR09088](#)).

$$\frac{d}{dt} \text{EC0039_b} = v_{35} \quad (714)$$

6.103 Species EC0058_b

Name L-Methionine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09471](#)).

$$\frac{d}{dt} \text{EC0058_b} = v_{64} \quad (715)$$

6.104 Species EC9324_b

Name Biomass_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09882](#)).

$$\frac{d}{dt} \text{EC9324_b} = v_{106} \quad (716)$$

6.105 Species EC0080_b

Name D-Fructose_b[e]

Charge 0

This species takes part in one reaction (as a reactant in [IR09059](#)).

$$\frac{d}{dt} \text{EC0080_b} = -v_{304} \quad (717)$$

6.106 Species [EC0098](#)

Name Glycerol[e]

Charge 0

This species takes part in two reactions (as a product in [RR08739](#), [IR09073](#)).

$$\frac{d}{dt}\text{EC0098} = v_{24} + v_{217} \quad (718)$$

6.107 Species [EC0317](#)

Name L-Isoleucine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09458](#), [IR08767](#) and as a product in [IR08692](#)).

$$\frac{d}{dt}\text{EC0317} = v_{121} - v_{48} - v_{229} \quad (719)$$

6.108 Species [EC0132](#)

Name L-Homocysteine[e]

Charge 0

This species takes part in two reactions (as a reactant in [IR09866](#) and as a product in [IR09865](#)).

$$\frac{d}{dt}\text{EC0132} = v_{180} - v_{218} \quad (720)$$

6.109 Species [EC0117](#)

Name L-Histidine[e]

Charge 0

This species takes part in three reactions (as a reactant in [RR09086](#), [IR08754](#) and as a product in [IR08684](#)).

$$\frac{d}{dt}\text{EC0117} = v_{115} - v_{86} - v_{160} \quad (721)$$

6.110 Species EC0317_b

Name L-Isoleucine_b[e]

Charge 0

This species takes part in one reaction (as a product in [RR09458](#)).

$$\frac{d}{dt}EC0317_b = v_{48} \quad (722)$$

6.111 Species C0090

Name Uracil

Charge 0

This species takes part in four reactions (as a product in [RR01416](#), [RR01052](#), [RR00617](#), [RR08636](#)).

$$\frac{d}{dt}C0090 = v_{79} + v_{81} + v_{90} + v_{103} \quad (723)$$

6.112 Species C0445

Name D-Glyceraldehyde

Charge 0

This species takes part in two reactions (as a product in [RR00667](#), [IR01486](#)).

$$\frac{d}{dt}C0445 = v_{19} + v_{303} \quad (724)$$

6.113 Species C0796

Name L-Ribulose 5-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR01850](#), [RR03195](#)).

$$\frac{d}{dt}C0796 = -v_4 - v_{63} \quad (725)$$

6.114 Species C9537

Name L-Glutamyl-tRNA(Gln)

Charge 0

This species takes part in three reactions (as a reactant in [IR09913](#), [IR09912](#) and as a product in [IR09911](#)).

$$\frac{d}{dt}C9537 = v_{228} - v_{182} - v_{260} \quad (726)$$

6.115 Species C9510

Name Glycyl-tRNA(Gly)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09899](#)).

$$\frac{d}{dt}C9510 = v_{172} - v_{124} - 21 v_{239} \quad (727)$$

6.116 Species C0018

Name AMP

Charge -2

This species takes part in 28 reactions (as a reactant in [RR03512](#), [RR08879](#), [IR03505](#) and as a product in [IR09820](#), [IR09910](#), [IR09898](#), [IR09906](#), [IR03510](#), [IR09895](#), [IR09905](#), [IR09899](#), [IR09902](#), [IR09891](#), [IR09893](#), [IR04284](#), [IR00674](#), [IR03511](#), [IR09908](#), [IR09911](#), [IR09885](#), [IR09907](#), [IR09903](#), [IR09904](#), [IR09892](#), [IR09896](#), [IR09894](#), [IR09900](#), [IR09901](#)).

$$\begin{aligned} \frac{d}{dt}C0018 = & v_{112} + v_{123} + v_{126} + v_{150} + v_{153} + v_{155} + v_{161} + v_{172} + v_{173} \\ & + v_{177} + v_{183} + v_{184} + v_{186} + v_{202} + v_{203} + v_{228} + 29 v_{234} + v_{235} \\ & + v_{251} + v_{256} + v_{262} + v_{273} + v_{278} + v_{283} + v_{294} - v_8 - v_{67} - v_{305} \end{aligned} \quad (728)$$

6.117 Species C0125

Name Adenine

Charge 0

This species takes part in four reactions (as a product in [RR03512](#), [RR03709](#), [RR01475](#), [RR08870](#)).

$$\frac{d}{dt}C0125 = v_8 + v_{12} + v_{13} + v_{18} \quad (729)$$

6.118 Species C0016

Name Pyridoxal phosphate

Charge -2

This species takes part in two reactions (as a reactant in [IR09955](#) and as a product in [IR03497](#)).

$$\frac{d}{dt}C0016 = v_{138} - 100 v_{247} \quad (730)$$

6.119 Species C2070

Name L-Xylulose 5-phosphate

Charge -2

This species takes part in two reactions (as a product in [RR01850](#), [IR04231](#)).

$$\frac{d}{dt}C2070 = v_4 + v_{287} \quad (731)$$

6.120 Species C0055

Name Thiamin diphosphate

Charge -2

This species takes part in two reactions (as a reactant in [IR09955](#) and as a product in [IR00341](#)).

$$\frac{d}{dt}C0055 = v_{231} - 100 v_{247} \quad (732)$$

6.121 Species C0190

Name D-Xylulose 5-phosphate

Charge -2

This species takes part in four reactions (as a product in [RR03769](#), [RR00689](#), [RR03195](#), [RR03691](#)).

$$\frac{d}{dt}C0190 = v_{43} + v_{45} + v_{63} + v_{105} \quad (733)$$

6.122 Species C9531

Name tRNA(Ile)

Charge 0

This species takes part in three reactions (as a reactant in [IR09905](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9531 = 8 v_{124} + 22 v_{239} - v_{161} \quad (734)$$

6.123 Species C0233

Name dGTP

Charge -4

This species takes part in three reactions (as a reactant in [RR09796](#), [IR09861](#) and as a product in [IR05416](#)).

$$\frac{d}{dt}C0233 = v_{204} - v_{20} - 20 v_{243} \quad (735)$$

6.124 Species C0454

Name Choline phosphate

Charge -1

This species takes part in two reactions (as a reactant in [IR01064](#) and as a product in [IR00656](#)).

$$\frac{d}{dt}C0454 = v_{171} - v_{291} \quad (736)$$

6.125 Species C0822

Name Pantetheine 4'-phosphate

Charge -2

This species takes part in three reactions (as a reactant in [IR03894](#) and as a product in [IR09847](#), [IR01710](#)).

$$\frac{d}{dt}C0822 = v_{166} + v_{201} - v_{136} \quad (737)$$

6.126 Species C0020

Name Pyruvate

Charge -1

This species takes part in 15 reactions (as a reactant in [IR00932](#) and as a product in [RR00404](#), [IR07938](#), [IR07898](#), [IR04690](#), [IR08786](#), [IR09773](#), [IR08829](#), [IR09792](#), [IR05416](#), [IR08826](#), [IR04740](#), [IR08828](#), [IR09793](#), [IR07969](#)).

$$\begin{aligned} \frac{d}{dt}C0020 = & v_{82} + v_{120} + v_{131} + v_{137} + v_{152} + v_{162} + v_{163} + v_{200} \\ & + v_{204} + v_{212} + v_{220} + v_{259} + v_{261} + v_{275} - v_{142} \end{aligned} \quad (738)$$

6.127 Species C0163

Name 3-Phospho-D-glycerate

Charge -3

This species takes part in five reactions (as a reactant in [RR09796](#), [RR09795](#), [RR03675](#), [RR03680](#) and as a product in [RR03681](#)).

$$\frac{d}{dt}C0163 = v_{44} - v_{20} - v_{57} - v_{71} - v_{78} \quad (739)$$

6.128 Species C9524

Name tRNA(Gly)

Charge 0

This species takes part in three reactions (as a reactant in [IR09899](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9524 = v_{124} + 21 v_{239} - v_{172} \quad (740)$$

6.129 Species C0089

Name UMP

Charge -2

This species takes part in five reactions (as a reactant in [RR00617](#), [RR00111](#), [IR00614](#) and as a product in [IR00615](#), [IR09885](#)).

$$\frac{d}{dt}C0089 = v_{233} + 25 v_{234} - v_{90} - v_{100} - v_{268} \quad (741)$$

6.130 Species C0151

Name L-Valine

Charge 0

This species takes part in five reactions (as a reactant in [IR09902](#), [IR09955](#), [IR08690](#) and as a product in [IR09954](#), [IR08770](#)).

$$\frac{d}{dt}C0151 = 25 v_{143} + v_{249} - v_{173} - 2793 v_{247} - v_{252} \quad (742)$$

6.131 Species C0175

Name Adenosine

Charge 0

This species takes part in four reactions (as a reactant in [RR03709](#), [IR09955](#) and as a product in [IR03514](#), [IR03505](#)).

$$\frac{d}{dt}C0175 = v_{114} + v_{305} - v_{12} - 1981 v_{247} \quad (743)$$

6.132 Species C0031

Name GDP

Charge -3

This species takes part in four reactions (as a reactant in [IR07938](#), [IR04341](#) and as a product in [RR03680](#), [RR00171](#)).

$$\frac{d}{dt}C0031 = v_{78} + v_{110} - v_{120} - v_{210} \quad (744)$$

6.133 Species C9319

Name Reduced thioredoxin

Charge 0

This species takes part in five reactions (as a reactant in [IR04342](#), [IR04340](#), [IR04341](#), [IR04343](#) and as a product in [RR04369](#)).

$$\frac{d}{dt}C9319 = v_6 - v_{135} - v_{140} - v_{210} - v_{244} \quad (745)$$

6.134 Species C0504

Name 2-Deoxy-D-ribose 5-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [IR00688](#) and as a product in [RR01592](#)).

$$\frac{d}{dt}C0504 = v_{17} - v_{242} \quad (746)$$

6.135 Species C0343

Name 5,10-Methenyltetrahydrofolate

Charge -1

This species takes part in four reactions (as a reactant in [RR00902](#), [IR08208](#) and as a product in [RR00771](#), [IR03832](#)).

$$\frac{d}{dt}C0343 = v_{56} + v_{290} - v_{33} - v_{198} \quad (747)$$

6.136 Species C0035

Name L-Alanine

Charge 0

This species takes part in five reactions (as a reactant in [IR09891](#), [IR08635](#), [IR09955](#) and as a product in [IR09954](#), [IR08769](#)).

$$\frac{d}{dt}C0035 = 27 v_{143} + v_{226} - v_{177} - v_{192} - 9824 v_{247} \quad (748)$$

6.137 Species C0011

Name CO2

Charge 0

This species takes part in five reactions (as a reactant in [RR00106](#) and as a product in [RR00275](#), [RR08672](#), [IR00932](#), [IR04231](#)).

$$\frac{d}{dt}C0011 = v_{38} + v_{59} + v_{142} + v_{287} - v_{36} \quad (749)$$

6.138 Species C0078

Name sn-Glycerol 3-phosphate

Charge -2

This species takes part in six reactions (as a reactant in [RR00521](#), [IR09827](#), [IR09821](#) and as a product in [RR00662](#), [IR00522](#), [IR08849](#)).

$$\frac{d}{dt}C0078 = v_{70} + v_{116} + v_{264} - v_9 - v_{189} - v_{195} \quad (750)$$

6.139 Species C0093

Name Dihydroxyacetone phosphate

Charge -2

This species takes part in six reactions (as a reactant in [IR00648](#) and as a product in [RR00521](#), [RR00690](#), [RR00652](#), [IR00649](#), [IR01486](#)).

$$\frac{d}{dt}C0093 = v_9 + v_{21} + v_{104} + v_{159} + v_{303} - v_{165} \quad (751)$$

6.140 Species C9528

Name L-Seryl-tRNA(Ser)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09907](#)).

$$\frac{d}{dt}C9528 = v_{235} - 5 v_{124} - 21 v_{239} \quad (752)$$

6.141 Species C0292

Name dTMP

Charge -2

This species takes part in four reactions (as a reactant in [RR01151](#), [IR03716](#) and as a product in [RR01158](#), [IR03714](#)).

$$\frac{d}{dt}C0292 = v_{50} + v_{255} - v_{49} - v_{236} \quad (753)$$

6.142 Species C9540

Name L-Asparaginyl-tRNA(Asn)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09910](#)).

$$\frac{d}{dt}C9540 = v_{123} - 3 v_{124} - 20 v_{239} \quad (754)$$

6.143 Species C9262

Name acyl carrier protein

Charge -4

This species takes part in six reactions (as a reactant in [IR09820](#), [IR09847](#), [IR09955](#) and as a product in [IR09822](#), [IR09821](#), [IR04278](#)).

$$\frac{d}{dt}C9262 = v_{117} + v_{195} + v_{271} - v_{112} - v_{166} - 3 v_{247} \quad (755)$$

6.144 Species C0029

Name Acetate

Charge -1

This species takes part in two reactions (as a reactant in [RR00157](#), [IR09888](#)).

$$\frac{d}{dt}C0029 = -v_{39} - v_{288} \quad (756)$$

6.145 Species C0325

Name Dihydrofolate

Charge -1

This species takes part in three reactions (as a reactant in [RR01236](#) and as a product in [RR01158](#), [RR00590](#)).

$$\frac{d}{dt}C0325 = v_{50} + v_{72} - v_{54} \quad (757)$$

6.146 Species C0065

Name H+

Charge 1

This species takes part in 133 reactions (as a reactant in [RR04369](#), [RR03709](#), [RR01184](#), [RR03873](#), [RR01052](#), [RR09889](#), [RR00146](#), [IR08684](#), [IR07938](#), [IR08692](#), [IR09924](#), [IR08686](#), [IR07898](#), [IR03894](#), [IR04690](#), [IR00932](#), [IR08822](#), [IR08750](#), [IR08661](#), [IR00114](#), [IR08707](#), [IR09773](#), [IR09886](#), [IR08844](#), [IR08635](#), [IR09926](#), [IR08208](#), [IR09792](#), [IR05416](#), [IR09925](#), [IR08835](#), [IR04740](#), [IR08779](#), [IR09826](#), [IR08690](#), [IR09793](#), [IR08848](#), [IR09923](#), [IR07969](#), [IR08688](#), [IR04231](#), [IR09888](#), [IR01064](#), [IR09927](#), [IR08842](#) and as a product in [RR00685](#), [RR09769](#), [RR08870](#), [RR00667](#), [RR03544](#), [RR01146](#), [RR03808](#), [RR08868](#), [RR00452](#), [RR00106](#), [RR08593](#), [RR08860](#), [RR00771](#), [RR00682](#), [RR03649](#), [RR00662](#), [RR00590](#), [RR00404](#), [RR00952](#), [RR09342](#),

RR08636, IR09820, IR08663, IR00522, IR09934, IR09845, IR08746, IR09956, IR08910, IR03497, IR09829, IR09954, IR08755, IR08807, IR00649, IR08754, IR09847, IR09884, IR09860, IR08768, IR09958, IR00656, IR09871, IR09957, IR00911, IR09913, IR08984, IR00674, IR08771, IR08659, IR09827, IR08824, IR09961, IR08738, IR01710, IR00074, IR09881, IR00676, IR09959, IR08769, IR08767, IR08789, IR00615, IR09885, IR09909, IR01174, IR08747, IR09861, IR00293, IR08770, IR08753, IR00001, IR03714, IR09912, IR08849, IR00075, IR09960, IR08645, IR04278, IR00450, IR09855, IR00269, IR08836, IR03832, IR08823, IR01156, IR00094, IR00296).

$$\begin{aligned} \frac{d}{dt}C0065 = & v_1 + 3 v_{15} + v_{18} + v_{19} + v_{22} + v_{26} + v_{27} + v_{28} + v_{32} + v_{36} + 3 v_{53} + v_{55} \\ & + v_{56} + 2 v_{58} + v_{61} + v_{70} + v_{72} + v_{82} + v_{93} + v_{99} + v_{103} + v_{112} + v_{113} \\ & + v_{116} + 282 v_{124} + v_{128} + v_{129} + v_{133} + v_{134} + v_{138} + v_{139} + 688 v_{143} \\ & + v_{146} + v_{157} + v_{159} + v_{160} + v_{166} + 40 v_{167} + v_{168} + v_{169} + v_{170} + v_{171} \\ & + 6 v_{174} + v_{176} + v_{179} + v_{182} + v_{185} + v_{186} + v_{187} + v_{188} + v_{189} + v_{190} \\ & + 7 v_{193} + v_{194} + v_{201} + v_{219} + 25 v_{221} + v_{222} + v_{225} + v_{226} + v_{229} + v_{230} \\ & + v_{233} + 100 v_{234} + 826 v_{239} + v_{240} + v_{241} + 140 v_{243} + v_{246} + v_{249} + v_{253} \\ & + v_{254} + v_{255} + v_{260} + v_{264} + v_{265} + v_{266} + v_{267} + v_{271} + v_{272} + v_{277} \\ & + v_{285} + v_{286} + 2 v_{290} + v_{292} + v_{300} + v_{301} + v_{306} - v_6 - 2 v_{12} - 2 v_{37} - v_{42} \\ & - 2 v_{81} - 2 v_{89} - v_{98} - v_{115} - v_{120} - v_{121} - v_{122} - v_{125} - v_{131} - v_{136} - v_{137} \\ & - v_{142} - v_{144} - v_{147} - v_{148} - v_{154} - v_{158} - 2 v_{162} - v_{175} - v_{178} - v_{192} \\ & - v_{196} - v_{198} - v_{200} - v_{204} - v_{209} - v_{213} - v_{220} - v_{248} - v_{250} - v_{252} \\ & - v_{261} - v_{269} - v_{274} - v_{275} - v_{280} - v_{287} - v_{288} - v_{291} - v_{296} - v_{298} \end{aligned} \quad (758)$$

6.147 Species C0291

Name dTDP

Charge -3

This species takes part in two reactions (as a reactant in [IR09793](#) and as a product in [RR01151](#)).

$$\frac{d}{dt}C0291 = v_{49} - v_{261} \quad (759)$$

6.148 Species C0199

Name Guanine

Charge 0

This species takes part in four reactions (as a product in [RR00779](#), [RR03810](#), [RR01184](#), [IR08645](#)).

$$\frac{d}{dt}C0199 = v_{25} + v_{34} + v_{37} + v_{267} \quad (760)$$

6.149 Species C0058

Name L-Methionine

Charge 0

This species takes part in six reactions (as a reactant in [IR09893](#), [IR07876](#), [IR09955](#), [IR09927](#) and as a product in [IR09954](#), [IR08789](#)).

$$\frac{d}{dt}C0058 = 6 v_{143} + v_{230} - v_{183} - v_{215} - 1370 v_{247} - v_{296} \quad (761)$$

6.150 Species C0015

Name FAD

Charge -2

This species takes part in two reactions (as a reactant in [IR09955](#) and as a product in [IR00114](#)).

$$\frac{d}{dt}C0015 = v_{154} - 100 v_{247} \quad (762)$$

6.151 Species C0060

Name UTP

Charge -4

This species takes part in three reactions (as a reactant in [RR00146](#), [IR09884](#) and as a product in [IR04740](#)).

$$\frac{d}{dt}C0060 = v_{220} - v_{98} - 25 v_{167} \quad (763)$$

6.152 Species C9282

Name L-Glutamyl-tRNA(Glu)

Charge 0

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR04284](#)).

$$\frac{d}{dt}C9282 = v_{184} - 11 v_{124} - 22 v_{239} \quad (764)$$

6.153 Species C0101

Name 5-Phospho-alpha-D-ribose 1-diphosphate

Charge -5

This species takes part in five reactions (as a product in [RR03512](#), [RR00779](#), [RR00617](#), [RR00952](#), [IR00674](#)).

$$\frac{d}{dt}C0101 = v_8 + v_{25} + v_{90} + v_{93} + v_{186} \quad (765)$$

6.154 Species C0284

Name D-Fructose 1,6-bisphosphate

Charge -4

This species takes part in three reactions (as a reactant in [RR00690](#) and as a product in [IR08910](#), [IR00450](#)).

$$\frac{d}{dt}C0284 = v_{134} + v_{272} - v_{21} \quad (766)$$

6.155 Species C0062

Name L-Ornithine

Charge 1

This species takes part in two reactions (as a reactant in [RR08615](#), [RR03649](#)).

$$\frac{d}{dt}C0062 = -v_{31} - v_{61} \quad (767)$$

6.156 Species C0006_b

Name NADP+_b

Charge -3

This species takes part in one reaction (as a product in [IR09271](#)).

$$\frac{d}{dt}C0006_b = v_{205} \quad (768)$$

6.157 Species C0067

Name L-Tyrosine

Charge 0

This species takes part in five reactions (as a reactant in [IR09895](#), [IR09955](#), [IR08848](#) and as a product in [IR09954](#), [IR09958](#)).

$$\frac{d}{dt}C0067 = 10 v_{143} + v_{170} - v_{155} - 1366 v_{247} - v_{269} \quad (769)$$

6.158 Species C0165

Name D-Ribulose 5-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR03691](#) and as a product in [RR00681](#)).

$$\frac{d}{dt}C0165 = v_{101} - v_{105} \quad (770)$$

6.159 Species C9482

Name Cardiolipin (Mpn)

Charge -2

This species takes part in one reaction (as a product in [IR09829](#)).

$$\frac{d}{dt}C9482 = v_{139} \quad (771)$$

6.160 Species C0289

Name dGDP

Charge -3

This species takes part in four reactions (as a reactant in [IR05416](#) and as a product in [RR01147](#), [RR09796](#), [IR04341](#)).

$$\frac{d}{dt}C0289 = v_{10} + v_{20} + v_{210} - v_{204} \quad (772)$$

6.161 Species C9535

Name tRNA(Leu)

Charge 0

This species takes part in three reactions (as a reactant in [IR09898](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9535 = 12 v_{124} + 29 v_{239} - v_{126} \quad (773)$$

6.162 Species C9515

Name tRNA(Arg)

Charge 0

This species takes part in three reactions (as a reactant in [IR09904](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9515 = 2 v_{124} + 13 v_{239} - v_{256} \quad (774)$$

6.163 Species C0198

Name dCMP

Charge -2

This species takes part in three reactions (as a reactant in [RR00910](#), [IR00909](#) and as a product in [IR00911](#)).

$$\frac{d}{dt}C0198 = v_{179} - v_{52} - v_{258} \quad (775)$$

6.164 Species C9473

Name 1-acyl-Glycerol 3-phosphate (Mpn)

Charge -2

This species takes part in two reactions (as a reactant in [IR09822](#) and as a product in [IR09821](#)).

$$\frac{d}{dt}C9473 = v_{195} - v_{117} \quad (776)$$

6.165 Species C9485_b

Name 5mcDNA (Mpn)_b

Charge -99

This species takes part in one reaction (as a product in [IR09864](#)).

$$\frac{d}{dt}C9485_b = v_{270} \quad (777)$$

6.166 Species C0389

Name Folic acid

Charge 0

This species takes part in two reactions (as a product in [RR01236](#), [IR09815](#)).

$$\frac{d}{dt}C0389 = v_{54} + v_{119} \quad (778)$$

6.167 Species C0129

Name L-Asparagine

Charge 0

This species takes part in six reactions (as a reactant in [IR09910](#), [IR08661](#), [IR09955](#), [IR09912](#) and as a product in [IR09954](#), [IR08659](#)).

$$\frac{d}{dt}C0129 = 20 v_{143} + v_{188} - v_{123} - v_{148} - 54 v_{247} - v_{260} \quad (779)$$

6.168 Species C0017

Name S-Adenosyl-L-methionine

Charge 1

This species takes part in three reactions (as a reactant in [IR09863](#), [IR09955](#) and as a product in [IR07876](#)).

$$\frac{d}{dt}C0017 = v_{215} - v_{164} - 100 v_{247} \quad (780)$$

6.169 Species C0002

Name ATP

Charge -4

This species takes part in 99 reactions (as a reactant in [RR01147](#), [RR01146](#), [RR03808](#), [RR00106](#), [RR00157](#), [RR03873](#), [RR01151](#), [RR00910](#), [RR01155](#), [RR08879](#), [RR03675](#), [RR00268](#), [RR03698](#), [RR00111](#), [RR00171](#), [IR09820](#), [IR08663](#), [IR00522](#), [IR09910](#), [IR09934](#), [IR09898](#), [IR09845](#), [IR08746](#), [IR09956](#), [IR08910](#), [IR03894](#), [IR03497](#), [IR09954](#), [IR08755](#), [IR09906](#), [IR00114](#), [IR09895](#), [IR00649](#), [IR08754](#), [IR09905](#), [IR09884](#), [IR09860](#), [IR08768](#), [IR09958](#), [IR00656](#), [IR09899](#), [IR09902](#), [IR09957](#), [IR09891](#), [IR00911](#), [IR09913](#), [IR09893](#), [IR04284](#), [IR08984](#), [IR00674](#), [IR08771](#), [IR08659](#), [IR08824](#), [IR00596](#), [IR09961](#), [IR08738](#), [IR01710](#), [IR03511](#), [IR09908](#), [IR07876](#), [IR00074](#), [IR09881](#), [IR00676](#), [IR09959](#), [IR08769](#), [IR09911](#), [IR08767](#), [IR08789](#), [IR00341](#), [IR00615](#), [IR09907](#), [IR09909](#), [IR01174](#), [IR08747](#), [IR09861](#), [IR00293](#), [IR08770](#), [IR09903](#), [IR08753](#), [IR03714](#), [IR09904](#), [IR09912](#), [IR09892](#), [IR08849](#), [IR00075](#), [IR09960](#), [IR00450](#), [IR09896](#), [IR09894](#), [IR09900](#), [IR00269](#), [IR08836](#), [IR03832](#), [IR08823](#), [IR09901](#), [IR01156](#), [IR00094](#) and as a product in [RR08593](#), [IR07898](#)).

$$\begin{aligned} \frac{d}{dt}C0002 = & v_{53} + v_{131} - v_{10} - v_{26} - v_{27} - v_{36} - v_{39} - v_{42} - v_{49} - v_{52} - v_{65} \\ & - v_{67} - v_{71} - v_{74} - v_{75} - v_{100} - v_{110} - v_{112} - v_{113} - v_{116} - v_{123} \\ & - 199 v_{124} - v_{126} - v_{128} - v_{129} - v_{133} - v_{134} - v_{136} - v_{138} - 688 v_{143} \\ & - v_{146} - v_{150} - v_{154} - v_{155} - v_{159} - v_{160} - v_{161} - 69 v_{167} - v_{168} - v_{169} \\ & - v_{170} - v_{171} - v_{172} - v_{173} - v_{176} - v_{177} - v_{179} - v_{182} - v_{183} - v_{184} \\ & - v_{185} - v_{186} - v_{187} - v_{188} - v_{190} - v_{191} - 7 v_{193} - v_{194} - v_{201} - v_{202} \\ & - v_{203} - v_{215} - v_{219} - 25 v_{221} - v_{222} - v_{225} - v_{226} - v_{228} - v_{229} - v_{230} \\ & - v_{231} - v_{233} - v_{235} - 826 v_{239} - v_{240} - v_{241} - 140 v_{243} - v_{246} - v_{249} \\ & - v_{251} - v_{253} - v_{255} - v_{256} - v_{260} - v_{262} - v_{264} - v_{265} - v_{266} - v_{272} \\ & - v_{273} - v_{278} - v_{283} - v_{285} - v_{286} - v_{290} - v_{292} - v_{294} - v_{300} - v_{301} \end{aligned} \quad (781)$$

6.170 Species C9481

Name Phosphatidyl Glycerol (Mpn)

Charge -1

This species takes part in two reactions (as a reactant in [IR09829](#) and as a product in [IR09828](#)).

$$\frac{d}{dt}C9481 = v_{227} - v_{139} \quad (782)$$

6.171 Species C9520

Name tRNA(Lys)

Charge 0

This species takes part in three reactions (as a reactant in [IR09906](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9520 = 11 v_{124} + 32 v_{239} - v_{150} \quad (783)$$

6.172 Species C9532

Name L-Tyrosyl-tRNA(Tyr)

Charge 1

This species takes part in two reactions (as a reactant in [IR09909](#) and as a product in [IR09895](#)).

$$\frac{d}{dt}C9532 = v_{155} - 10 v_{239} \quad (784)$$

6.173 Species C0859

Name Nicotinate D-ribonucleotide

Charge -2

This species takes part in two reactions (as a reactant in [RR03873](#), [RR00952](#)).

$$\frac{d}{dt}C0859 = -v_{42} - v_{93} \quad (785)$$

6.174 Species C9471

Name Fatty Acid (Mpn)

Charge 0

This species takes part in three reactions (as a reactant in [IR09820](#) and as a product in [RR09830](#), [RR09857](#)).

$$\frac{d}{dt}C9471 = v_{29} + 2 v_{41} - v_{112} \quad (786)$$

6.175 Species C0082

Name L-Cysteine

Charge 0

This species takes part in five reactions (as a reactant in [IR09925](#), [IR09955](#), [IR09903](#) and as a product in [IR08663](#), [IR09954](#)).

$$\frac{d}{dt}C0082 = v_{113} + 3 v_{143} - v_{209} - 67 v_{247} - v_{251} \quad (787)$$

6.176 Species C0228

Name D-Erythrose 4-phosphate

Charge -2

This species takes part in two reactions (as a product in [RR01018](#), [RR00689](#)).

$$\frac{d}{dt}C0228 = v_{14} + v_{45} \quad (788)$$

6.177 Species C0268

Name L-Citrulline

Charge 0

This species takes part in two reactions (as a product in [RR03649](#), [IR00296](#)).

$$\frac{d}{dt}C0268 = v_{61} + v_{306} \quad (789)$$

6.178 Species C0014

Name UDP

Charge -3

This species takes part in four reactions (as a reactant in [IR04342](#), [IR04740](#) and as a product in [RR00111](#), [IR09871](#)).

$$\frac{d}{dt}C0014 = v_{100} + 6 v_{174} - v_{135} - v_{220} \quad (790)$$

6.179 Species C9538

Name tRNA(Gln)

Charge 0

This species takes part in three reactions (as a reactant in [IR09911](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9538 = 4 v_{124} + 17 v_{239} - v_{228} \quad (791)$$

6.180 Species C0069

Name Acetaldehyde

Charge 0

This species takes part in two reactions (as a reactant in [RR03544](#) and as a product in [IR00688](#)).

$$\frac{d}{dt}C0069 = v_{242} - v_{22} \quad (792)$$

6.181 Species C9502

Name tRNA(Phe)

Charge 0

This species takes part in three reactions (as a reactant in [IR09908](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9502 = 5 v_{124} + 15 v_{239} - v_{203} \quad (793)$$

6.182 Species C0049

Name FMN

Charge -2

This species takes part in two reactions (as a reactant in [IR00114](#) and as a product in [IR00293](#)).

$$\frac{d}{dt}C0049 = v_{246} - v_{154} \quad (794)$$

6.183 Species C9484

Name DNA (Mpn)

Charge -100

This species takes part in three reactions (as a reactant in [IR09863](#), [IR09955](#) and as a product in [IR09861](#)).

$$\frac{d}{dt}C9484 = v_{243} - v_{164} - 1681 v_{247} \quad (795)$$

6.184 Species C9475

Name Diacylglycerol (Mpn)

Charge 0

This species takes part in two reactions (as a reactant in [IR09871](#) and as a product in [IR09824](#)).

$$\frac{d}{dt}C9475 = v_{208} - v_{174} \quad (796)$$

6.185 Species C0646

Name Deoxycytidine

Charge 0

This species takes part in four reactions (as a reactant in [IR00911](#), [IR01418](#) and as a product in [IR08807](#), [IR00909](#)).

$$\frac{d}{dt}C0646 = v_{157} + v_{258} - v_{179} - v_{299} \quad (797)$$

6.186 Species C9512

Name L-Valyl-tRNA(Val)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09902](#)).

$$\frac{d}{dt}C9512 = v_{173} - 5 v_{124} - 25 v_{239} \quad (798)$$

6.187 Species C0446

Name Dihydrolipoamide

Charge 0

This species takes part in two reactions (as a reactant in [RR09342](#), [RR01487](#)).

$$\frac{d}{dt}C0446 = -v_{99} - v_{108} \quad (799)$$

6.188 Species C9498

Name RNA (Mpn)

Charge -100

This species takes part in three reactions (as a reactant in [IR09885](#), [IR09955](#) and as a product in [IR09884](#)).

$$\frac{d}{dt}C9498 = v_{167} - v_{234} - 2015 v_{247} \quad (800)$$

6.189 Species C9489

Name glycolipid (Mpn)

Charge 0

This species takes part in two reactions (as a reactant in [IR09955](#) and as a product in [IR09871](#)).

$$\frac{d}{dt}C9489 = v_{174} - 63702 v_{247} \quad (801)$$

6.190 Species C9506

Name L-Isoleucyl-tRNA(Ile)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09905](#)).

$$\frac{d}{dt}C9506 = v_{161} - 8 v_{124} - 22 v_{239} \quad (802)$$

6.191 Species C0212

Name Riboflavin

Charge 0

This species takes part in two reactions (as a reactant in [IR00293](#) and as a product in [IR09818](#)).

$$\frac{d}{dt}C0212 = v_{207} - v_{246} \quad (803)$$

6.192 Species C0154

Name (S)-Lactate

Charge -1

This species takes part in two reactions (as a reactant in [RR00404](#), [IR09886](#)).

$$\frac{d}{dt}C0154 = -v_{82} - v_{175} \quad (804)$$

6.193 Species C9499

Name L-Alanyl-tRNA(Ala)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09891](#)).

$$\frac{d}{dt}C9499 = v_{177} - 4 v_{124} - 27 v_{239} \quad (805)$$

6.194 Species C0352

Name dTTP

Charge -4

This species takes part in two reactions (as a reactant in [IR09861](#) and as a product in [IR09793](#)).

$$\frac{d}{dt}C0352 = v_{261} - 30 v_{243} \quad (806)$$

6.195 Species C9480

Name Phosphatidyl Glycerol-3-Phosphate (Mpn)

Charge -3

This species takes part in two reactions (as a reactant in [IR09828](#) and as a product in [IR09827](#)).

$$\frac{d}{dt}C9480 = v_{189} - v_{227} \quad (807)$$

6.196 Species C9527

Name tRNA(Val)

Charge 0

This species takes part in three reactions (as a reactant in [IR09902](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9527 = 5 v_{124} + 25 v_{239} - v_{173} \quad (808)$$

6.197 Species C9517

Name L-Threonyl-tRNA(Thr)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09901](#)).

$$\frac{d}{dt}C9517 = v_{294} - v_{124} - 21 v_{239} \quad (809)$$

6.198 Species C0023

Name L-Glutamate

Charge -1

This species takes part in seven reactions (as a reactant in [IR08750](#), [IR04284](#), [IR09911](#), [IR09955](#) and as a product in [IR09954](#), [IR09913](#), [IR08747](#)).

$$\frac{d}{dt}C0023 = 22 v_{143} + v_{182} + v_{241} - v_{147} - v_{184} - v_{228} - 18651 v_{247} \quad (810)$$

6.199 Species C0096

Name Choline

Charge 1

This species takes part in three reactions (as a reactant in [IR00656](#) and as a product in [RR08868](#), [RR00662](#)).

$$\frac{d}{dt}C0096 = v_{28} + v_{70} - v_{171} \quad (811)$$

6.200 Species C0126

Name L-Proline

Charge 0

This species takes part in five reactions (as a reactant in [IR08822](#), [IR09955](#), [IR09896](#) and as a product in [IR09954](#), [IR08823](#)).

$$\frac{d}{dt}C0126 = 14 v_{143} + v_{292} - v_{144} - 6837 v_{247} - v_{273} \quad (812)$$

6.201 Species C9508

Name L-Histidyl-tRNA(His)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09892](#)).

$$\frac{d}{dt}C9508 = v_{262} - v_{124} - 6 v_{239} \quad (813)$$

6.202 Species C0047

Name Formate

Charge -1

This species takes part in three reactions (as a reactant in [RR00275](#), [IR00596](#) and as a product in [IR09909](#)).

$$\frac{d}{dt}C0047 = v_{239} - v_{38} - v_{191} \quad (814)$$

6.203 Species C9519

Name L-Cysteinyl-tRNA(Cys)

Charge 1

This species takes part in two reactions (as a reactant in [IR09909](#) and as a product in [IR09903](#)).

$$\frac{d}{dt}C9519 = v_{251} - 3 v_{239} \quad (815)$$

6.204 Species C0008

Name ADP

Charge -3

This species takes part in 73 reactions (as a reactant in [RR08593](#), [IR07898](#), [IR04340](#) and as a product in [RR01147](#), [RR01146](#), [RR03808](#), [RR00106](#), [RR00157](#), [RR01151](#), [RR00910](#), [RR01155](#), [RR08879](#), [RR03675](#), [RR00268](#), [RR03698](#), [RR00111](#), [RR00171](#), [IR08663](#), [IR00522](#), [IR09934](#), [IR09845](#), [IR08746](#), [IR09956](#), [IR08910](#), [IR03497](#), [IR09954](#), [IR08755](#), [IR00649](#), [IR08754](#), [IR09884](#), [IR09860](#), [IR08768](#), [IR09958](#), [IR00656](#), [IR09957](#), [IR00911](#), [IR09913](#), [IR08984](#), [IR08771](#), [IR08659](#), [IR08824](#), [IR00596](#), [IR09961](#), [IR08738](#), [IR01710](#), [IR00074](#), [IR09881](#), [IR00676](#), [IR09959](#), [IR08769](#), [IR08767](#), [IR08789](#), [IR00341](#), [IR00615](#), [IR09909](#), [IR01174](#), [IR08747](#), [IR09861](#), [IR00293](#), [IR08770](#), [IR08753](#), [IR03714](#), [IR09912](#), [IR08849](#), [IR00075](#), [IR09960](#), [IR00450](#), [IR00269](#), [IR08836](#), [IR03832](#), [IR08823](#), [IR01156](#), [IR00094](#)).

$$\begin{aligned} \frac{d}{dt}C0008 = & v_{10} + v_{26} + v_{27} + v_{36} + v_{39} + v_{49} + v_{52} + v_{65} + 2 v_{67} + v_{71} + v_{74} + v_{75} \\ & + v_{100} + v_{110} + v_{113} + v_{116} + 199 v_{124} + v_{128} + v_{129} + v_{133} + v_{134} + v_{138} \\ & + 688 v_{143} + v_{146} + v_{159} + v_{160} + 40 v_{167} + v_{168} + v_{169} + v_{170} + v_{171} + v_{176} \\ & + v_{179} + v_{182} + v_{185} + v_{187} + v_{188} + v_{190} + v_{191} + 7 v_{193} + v_{194} + v_{201} \\ & + v_{219} + 25 v_{221} + v_{222} + v_{225} + v_{226} + v_{229} + v_{230} + v_{231} + v_{233} + 826 v_{239} \\ & + v_{240} + v_{241} + 140 v_{243} + v_{246} + v_{249} + v_{253} + v_{255} + v_{260} + v_{264} + v_{265} \\ & + v_{266} + v_{272} + v_{285} + v_{286} + v_{290} + v_{292} + v_{300} + v_{301} - v_{53} - v_{131} - v_{140} \end{aligned} \quad (816)$$

6.205 Species C0098

Name Glycerol

Charge 0

This species takes part in three reactions (as a reactant in [RR00667](#), [RR08739](#), [IR00522](#)).

$$\frac{d}{dt}C0098 = -v_{19} - v_{24} - v_{116} \quad (817)$$

6.206 Species C9255

Name 3-keto-L-gulonate-6-phosphate

Charge -3

This species takes part in two reactions (as a reactant in [IR04231](#) and as a product in [RR09769](#)).

$$\frac{d}{dt}C9255 = v_{15} - v_{287} \quad (818)$$

6.207 Species C9539

Name tRNA(Asn)

Charge 0

This species takes part in three reactions (as a reactant in [IR09910](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9539 = 3 v_{124} + 20 v_{239} - v_{123} \quad (819)$$

6.208 Species C0043

Name UDP-D-galactose

Charge -2

This species takes part in two reactions (as a reactant in [IR09871](#) and as a product in [RR00147](#)).

$$\frac{d}{dt}C0043 = v_{69} - 3 v_{174} \quad (820)$$

6.209 Species C0117

Name L-Histidine

Charge 0

This species takes part in five reactions (as a reactant in [IR08684](#), [IR09955](#), [IR09892](#) and as a product in [IR09954](#), [IR08754](#)).

$$\frac{d}{dt}C0117 = 6 v_{143} + v_{160} - v_{115} - 2422 v_{247} - v_{262} \quad (821)$$

6.210 Species C9504

Name L-Leucyl-tRNA(Leu)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09898](#)).

$$\frac{d}{dt}C9504 = v_{126} - 12 v_{124} - 29 v_{239} \quad (822)$$

6.211 Species C0485

Name D-Mannitol 1-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR00452](#) and as a product in [IR08826](#)).

$$\frac{d}{dt}C0485 = v_{212} - v_{32} \quad (823)$$

6.212 Species C0195

Name 1,3-Bisphospho-D-glycerate

Charge -4

This species takes part in five reactions (as a product in [RR00685](#), [RR09796](#), [RR09795](#), [RR03675](#), [RR03680](#)).

$$\frac{d}{dt}C0195 = v_1 + v_{20} + v_{57} + v_{71} + v_{78} \quad (824)$$

6.213 Species C0005

Name NADPH

Charge -4

This species takes part in four reactions (as a reactant in [RR04369](#), [IR09955](#) and as a product in [RR00771](#), [IR00075](#)).

$$\frac{d}{dt}C0005 = v_{56} + v_{265} - v_6 - 100 v_{247} \quad (825)$$

6.214 Species C9534

Name tRNA(Trp)

Charge 0

This species takes part in two reactions (as a reactant in [IR09900](#) and as a product in [IR09909](#)).

$$\frac{d}{dt}C9534 = 3 v_{239} - v_{283} \quad (826)$$

6.215 Species C0170

Name dADP

Charge -3

This species takes part in four reactions (as a reactant in [IR07969](#) and as a product in [RR09795](#), [RR03698](#), [IR04340](#)).

$$\frac{d}{dt}C0170 = v_{57} + v_{75} + v_{140} - v_{275} \quad (827)$$

6.216 Species C0033

Name Glycine

Charge 0

This species takes part in six reactions (as a reactant in [RR00598](#), [IR08707](#), [IR09899](#), [IR09955](#) and as a product in [IR09954](#), [IR09960](#)).

$$\frac{d}{dt}C0033 = 21 v_{143} + v_{266} - v_{11} - v_{158} - v_{172} - 9220 v_{247} \quad (828)$$

6.217 Species C0063

Name L-Tryptophan

Charge 0

This species takes part in five reactions (as a reactant in [IR08844](#), [IR09955](#), [IR09900](#) and as a product in [IR09954](#), [IR09957](#)).

$$\frac{d}{dt}C0063 = 3 v_{143} + v_{176} - v_{178} - 1864 v_{247} - v_{283} \quad (829)$$

6.218 Species C9462

Name L-ascorbate 6 phosphate

Charge 0

This species takes part in two reactions (as a reactant in [RR09769](#) and as a product in [IR09773](#)).

$$\frac{d}{dt}C9462 = v_{162} - v_{15} \quad (830)$$

6.219 Species C9522

Name tRNA(Met)

Charge 0

This species takes part in three reactions (as a reactant in [IR09893](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9522 = 4 v_{124} + 6 v_{239} - v_{183} \quad (831)$$

6.220 Species C0007

Name Oxygen

Charge 0

This species takes part in three reactions (as a reactant in [RR00521](#), [RR09889](#) and as a product in [RR08670](#)).

$$\frac{d}{dt}C0007 = v_{68} - v_9 - v_{89} \quad (832)$$

6.221 Species C0523

Name dCDP

Charge -3

This species takes part in three reactions (as a reactant in [IR09792](#) and as a product in [RR00910](#), [IR04343](#)).

$$\frac{d}{dt}C0523 = v_{52} + v_{244} - v_{200} \quad (833)$$

6.222 Species C9468

Name N-Formylmethionyl-tRNA(Met)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09797](#)).

$$\frac{d}{dt}C9468 = v_{302} - v_{124} - v_{239} \quad (834)$$

6.223 Species C0026

Name UDPglucose

Charge -2

This species takes part in three reactions (as a reactant in [RR00147](#), [IR09871](#) and as a product in [RR00146](#)).

$$\frac{d}{dt}C0026 = v_{98} - v_{69} - 3 v_{174} \quad (835)$$

6.224 Species C0227

Name D-Mannose 6-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR00465](#) and as a product in [IR08786](#)).

$$\frac{d}{dt}C0227 = v_{152} - v_{73} \quad (836)$$

6.225 Species C9557

Name DnaK-folded protein (Mpn)

Charge 5

This species takes part in one reaction (as a product in [IR09956](#)).

$$\frac{d}{dt}C9557 = v_{133} \quad (837)$$

6.226 Species C0193

Name 10-Formyltetrahydrofolate

Charge -1

This species takes part in three reactions (as a reactant in [IR09797](#) and as a product in [RR00902](#), [IR00596](#)).

$$\frac{d}{dt}C0193 = v_{33} + v_{191} - v_{302} \quad (838)$$

6.227 Species C9500

Name tRNA(Ala)

Charge 0

This species takes part in three reactions (as a reactant in [IR09891](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9500 = 4 v_{124} + 27 v_{239} - v_{177} \quad (839)$$

6.228 Species C0288

Name dAMP

Charge -2

This species takes part in two reactions (as a reactant in [RR03698](#) and as a product in [RR01146](#)).

$$\frac{d}{dt}C0288 = v_{26} - v_{75} \quad (840)$$

6.229 Species C0099

Name D-Ribose 5-phosphate

Charge -2

This species takes part in five reactions (as a reactant in [RR00681](#), [IR00674](#) and as a product in [RR03769](#), [RR00682](#), [IR00676](#)).

$$\frac{d}{dt}C0099 = v_{43} + v_{58} + v_{222} - v_{101} - v_{186} \quad (841)$$

6.230 Species C0433

Name Deoxyadenosine

Charge 0

This species takes part in two reactions (as a reactant in [RR01475](#), [RR01146](#)).

$$\frac{d}{dt}C0433 = -v_{13} - v_{26} \quad (842)$$

6.231 Species C0351

Name dCTP

Charge -4

This species takes part in two reactions (as a reactant in [IR09861](#) and as a product in [IR09792](#)).

$$\frac{d}{dt}C0351 = v_{200} - 20 v_{243} \quad (843)$$

6.232 Species C0647

Name Dephospho-CoA

Charge -2

This species takes part in two reactions (as a reactant in [IR00094](#) and as a product in [IR03894](#)).

$$\frac{d}{dt}C0647 = v_{136} - v_{301} \quad (844)$$

6.233 Species C0790

Name D-Fructose 1-phosphate

Charge -2

This species takes part in three reactions (as a reactant in [IR08910](#), [IR01486](#) and as a product in [IR08829](#)).

$$\frac{d}{dt}C0790 = v_{163} - v_{134} - v_{303} \quad (845)$$

6.234 Species C9264

Name apoprotein [acyl carrier protein]

Charge -3

This species takes part in three reactions (as a reactant in [IR04278](#) and as a product in [IR09934](#), [IR09847](#)).

$$\frac{d}{dt}C9264 = v_{124} + v_{166} - v_{271} \quad (846)$$

6.235 Species C0230

Name Sedoheptulose 7-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR01018](#), [RR03769](#)).

$$\frac{d}{dt}C0230 = -v_{14} - v_{43} \quad (847)$$

6.236 Species C2169

Name 5-Formyltetrahydrofolate

Charge 0

This species takes part in three reactions (as a reactant in [IR09955](#), [IR03832](#) and as a product in [IR08208](#)).

$$\frac{d}{dt}C2169 = v_{198} - 100 v_{247} - v_{290} \quad (848)$$

6.237 Species C0103

Name D-Ribose

Charge 0

This species takes part in two reactions (as a reactant in [IR00676](#) and as a product in [IR09811](#)).

$$\frac{d}{dt}C0103 = v_{295} - v_{222} \quad (849)$$

6.238 Species C0477

Name 2-Phospho-D-glycerate

Charge -3

This species takes part in two reactions (as a reactant in [RR03681](#), [RR00362](#)).

$$\frac{d}{dt}C0477 = -v_{44} - v_{102} \quad (850)$$

6.239 Species C0010

Name CoA

Charge -4

This species takes part in six reactions (as a reactant in [RR03544](#), [IR09955](#), [IR04278](#) and as a product in [RR03546](#), [RR01487](#), [IR00094](#)).

$$\frac{d}{dt}C0010 = v_{96} + v_{108} + v_{301} - v_{22} - 100 v_{247} - v_{271} \quad (851)$$

6.240 Species C0113

Name dATP

Charge -4

This species takes part in three reactions (as a reactant in [RR09795](#), [IR09861](#) and as a product in [IR07969](#)).

$$\frac{d}{dt}C0113 = v_{275} - v_{57} - 30 v_{243} \quad (852)$$

6.241 Species C0152

Name Glycerone

Charge 0

This species takes part in two reactions (as a reactant in [IR00649](#) and as a product in [IR00648](#)).

$$\frac{d}{dt}C0152 = v_{165} - v_{159} \quad (853)$$

6.242 Species C9530

Name tRNA(Tyr)

Charge 0

This species takes part in two reactions (as a reactant in [IR09895](#) and as a product in [IR09909](#)).

$$\frac{d}{dt}C9530 = 10 v_{239} - v_{155} \quad (854)$$

6.243 Species C0142

Name Carbamoyl phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR03649](#) and as a product in [RR00106](#)).

$$\frac{d}{dt}C0142 = v_{36} - v_{61} \quad (855)$$

6.244 Species C0207

Name Pyridoxal

Charge 0

This species takes part in two reactions (as a reactant in [IR03497](#) and as a product in [IR09817](#)).

$$\frac{d}{dt}C0207 = v_{297} - v_{138} \quad (856)$$

6.245 Species C9525

Name tRNA(His)

Charge 0

This species takes part in three reactions (as a reactant in [IR09892](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9525 = v_{124} + 6 v_{239} - v_{262} \quad (857)$$

6.246 Species C0087

Name D-Glucose 1-phosphate

Charge -2

This species takes part in two reactions (as a reactant in [RR08939](#), [RR00146](#)).

$$\frac{d}{dt}C0087 = -v_{76} - v_{98} \quad (858)$$

6.247 Species C0610

Name Pantetheine

Charge 0

This species takes part in two reactions (as a reactant in [IR01710](#) and as a product in [IR09816](#)).

$$\frac{d}{dt}C0610 = v_{211} - v_{201} \quad (859)$$

6.248 Species C0077

Name D-Glucose 6-phosphate

Charge -2

This species takes part in four reactions (as a reactant in [RR00464](#), [IR09955](#) and as a product in [RR08939](#), [IR08828](#)).

$$\frac{d}{dt}C0077 = v_{76} + v_{259} - v_{91} - 167760 v_{247} \quad (860)$$

6.249 Species C0290

Name dGMP

Charge -2

This species takes part in two reactions (as a reactant in [RR01147](#) and as a product in [RR03808](#)).

$$\frac{d}{dt}C0290 = v_{27} - v_{10} \quad (861)$$

6.250 Species C0123

Name GMP

Charge -2

This species takes part in four reactions (as a reactant in [RR00779](#), [RR00171](#), [IR00777](#) and as a product in [IR09885](#)).

$$\frac{d}{dt}C0123 = 28 v_{234} - v_{25} - v_{110} - v_{284} \quad (862)$$

6.251 Species C0050

Name L-Arginine

Charge 1

This species takes part in seven reactions (as a reactant in [IR09955](#), [IR09904](#), [IR09923](#), [IR00296](#) and as a product in [RR08615](#), [IR09954](#), [IR08753](#)).

$$\frac{d}{dt}C0050 = v_{31} + 13 v_{143} + v_{253} - 3913 v_{247} - v_{256} - v_{274} - v_{306} \quad (863)$$

6.252 Species C9558

Name GroEL-folded protein (Mpn)

Charge 5

This species takes part in one reaction (as a product in [IR09961](#)).

$$\frac{d}{dt}C9558 = v_{193} \quad (864)$$

6.253 Species C0039

Name L-Lysine

Charge 1

This species takes part in five reactions (as a reactant in [IR09906](#), [IR09955](#), [IR08779](#) and as a product in [IR09954](#), [IR08755](#)).

$$\frac{d}{dt}C0039 = 32 v_{143} + v_{146} - v_{150} - 1741 v_{247} - v_{248} \quad (865)$$

6.254 Species C0210

Name Nicotinate

Charge -1

This species takes part in two reactions (as a product in [RR00952](#), [IR08658](#)).

$$\frac{d}{dt}C0210 = v_{93} + v_{216} \quad (866)$$

6.255 Species C0064

Name L-Phenylalanine

Charge 0

This species takes part in five reactions (as a reactant in [IR08686](#), [IR09908](#), [IR09955](#) and as a product in [IR09954](#), [IR09959](#)).

$$\frac{d}{dt}C0064 = 15 v_{143} + v_{225} - v_{125} - v_{203} - 5122 v_{247} \quad (867)$$

6.256 Species C0012

Name Pyrophosphate

Charge -3

This species takes part in 36 reactions (as a reactant in [RR03512](#), [RR00779](#), [RR00617](#), [RR00952](#), [IR00001](#) and as a product in [RR03873](#), [RR00146](#), [IR09820](#), [IR09910](#), [IR09898](#), [IR03894](#), [IR09906](#), [IR00114](#), [IR09895](#), [IR09905](#), [IR09884](#), [IR09899](#), [IR09902](#), [IR09891](#), [IR09893](#), [IR04284](#), [IR03511](#), [IR09908](#), [IR07876](#), [IR09911](#), [IR09907](#), [IR09861](#), [IR09826](#), [IR09903](#), [IR09904](#), [IR09892](#), [IR09896](#), [IR09894](#), [IR09900](#), [IR01064](#), [IR09901](#)).

$$\begin{aligned} \frac{d}{dt}C0012 = & v_{42} + v_{98} + v_{112} + v_{123} + v_{126} + v_{136} + v_{150} + v_{154} + v_{155} \\ & + v_{161} + 100 v_{167} + v_{172} + v_{173} + v_{177} + v_{183} + v_{184} + v_{202} + v_{203} \\ & + v_{215} + v_{228} + v_{235} + 100 v_{243} + v_{250} + v_{251} + v_{256} + v_{262} + v_{273} \\ & + v_{278} + v_{283} + v_{291} + v_{294} - v_8 - v_{25} - v_{90} - v_{93} - v_{254} \end{aligned} \quad (868)$$

6.257 Species C0041

Name L-Aspartate

Charge -1

This species takes part in six reactions (as a reactant in [IR09924](#), [IR09955](#), [IR09894](#) and as a product in [IR08746](#), [IR09954](#), [IR09912](#)).

$$\frac{d}{dt}C0041 = v_{129} + 18 v_{143} + v_{260} - v_{122} - 9318 v_{247} - v_{278} \quad (869)$$

6.258 Species C0052

Name L-Glutamine

Charge 0

This species takes part in five reactions (as a reactant in [IR09913](#), [IR09926](#), [IR09955](#) and as a product in [IR09954](#), [IR08738](#)).

$$\frac{d}{dt}C0052 = 17 v_{143} + v_{194} - v_{182} - v_{196} - 294 v_{247} \quad (870)$$

6.259 Species C9485

Name 5mcDNA (Mpn)

Charge -99

This species takes part in two reactions (as a reactant in [IR09864](#) and as a product in [IR09863](#)).

$$\frac{d}{dt}C9485 = v_{164} - v_{270} \quad (871)$$

6.260 Species C9474

Name phosphatidic acid (Mpn)

Charge -2

This species takes part in four reactions (as a reactant in [IR09824](#), [IR09955](#), [IR09826](#) and as a product in [IR09822](#)).

$$\frac{d}{dt}C9474 = v_{117} - v_{208} - 148168 v_{247} - v_{250} \quad (872)$$

6.261 Species C0045

Name Adenosine 3',5'-bisphosphate

Charge -4

This species takes part in three reactions (as a reactant in [IR09198](#), [IR03510](#) and as a product in [IR04278](#)).

$$\frac{d}{dt}C0045 = v_{271} - v_{132} - v_{153} \quad (873)$$

6.262 Species C0019

Name S-Adenosyl-L-homocysteine

Charge 0

This species takes part in two reactions (as a reactant in [IR03514](#) and as a product in [IR09863](#)).

$$\frac{d}{dt}C0019 = v_{164} - v_{114} \quad (874)$$

6.263 Species C9521

Name tRNA(Thr)

Charge 0

This species takes part in three reactions (as a reactant in [IR09901](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9521 = v_{124} + 21 v_{239} - v_{294} \quad (875)$$

6.264 Species C0293

Name dUMP

Charge -2

This species takes part in four reactions (as a reactant in [RR01158](#), [RR01155](#), [IR01159](#) and as a product in [IR01156](#)).

$$\frac{d}{dt}C0293 = v_{300} - v_{50} - v_{65} - v_{263} \quad (876)$$

6.265 Species C0306

Name Guanosine

Charge 0

This species takes part in three reactions (as a reactant in [RR01184](#), [IR09955](#) and as a product in [IR00777](#)).

$$\frac{d}{dt}C0306 = v_{284} - v_{37} - 1657 v_{247} \quad (877)$$

6.266 Species C0503

Name 2-Deoxy-D-ribose 1-phosphate

Charge -2

This species takes part in five reactions (as a reactant in [RR01592](#) and as a product in [RR01475](#), [RR03810](#), [RR03717](#), [RR01416](#)).

$$\frac{d}{dt}C0503 = v_{13} + v_{34} + v_{77} + v_{79} - v_{17} \quad (878)$$

6.267 Species C9320

Name tRNA(Glu)

Charge 0

This species takes part in three reactions (as a reactant in [IR04284](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9320 = 11 v_{124} + 22 v_{239} - v_{184} \quad (879)$$

6.268 Species C0070

Name D-Fructose 6-phosphate

Charge -2

This species takes part in six reactions (as a reactant in [RR00689](#), [IR00450](#) and as a product in [RR01018](#), [RR00452](#), [RR00465](#), [RR00464](#)).

$$\frac{d}{dt}C0070 = v_{14} + v_{32} + v_{73} + v_{91} - v_{45} - v_{272} \quad (880)$$

6.269 Species C0470

Name D-Ribose 1-phosphate

Charge 0

This species takes part in four reactions (as a reactant in [RR00682](#) and as a product in [RR03709](#), [RR01184](#), [RR01052](#)).

$$\frac{d}{dt}C0470 = v_{12} + v_{37} + v_{81} - v_{58} \quad (881)$$

6.270 Species C0188

Name Acetyl phosphate

Charge -2

This species takes part in two reactions (as a product in [RR00157](#), [RR03546](#)).

$$\frac{d}{dt}C0188 = v_{39} + v_{96} \quad (882)$$

6.271 Species C9541

Name L-Glutaminyl-tRNA(Gln)

Charge 1

This species takes part in four reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09913](#), [IR09912](#)).

$$\frac{d}{dt}C9541 = v_{182} + v_{260} - 4 v_{124} - 17 v_{239} \quad (883)$$

6.272 Species C9483

Name Phosphatidylcholine (Mpn)

Charge 0

This species takes part in two reactions (as a reactant in [RR09857](#) and as a product in [IR09860](#)).

$$\frac{d}{dt}C9483 = v_{168} - v_{41} \quad (884)$$

6.273 Species C9324

Name Biomass

Charge 0

This species takes part in two reactions (as a reactant in [IR09881](#) and as a product in [IR09955](#)).

$$\frac{d}{dt}C9324 = 1000000 v_{247} - v_{221} \quad (885)$$

6.274 Species C0001

Name H2O

Charge 0

This species takes part in 68 reactions (as a reactant in [RR00598](#), [RR09769](#), [RR00902](#), [RR09857](#), [RR00662](#), [IR08663](#), [IR03514](#), [IR09934](#), [IR09845](#), [IR08746](#), [IR09956](#), [IR09954](#), [IR08755](#), [IR03510](#), [IR08754](#), [IR00648](#), [IR09847](#), [IR09884](#), [IR09860](#), [IR08768](#), [IR09958](#), [IR09957](#), [IR09913](#), [IR08984](#), [IR08771](#), [IR08659](#), [IR08824](#), [IR09961](#), [IR08738](#), [IR08208](#), [IR01054](#), [IR09824](#), [IR07876](#), [IR09881](#), [IR09959](#), [IR08769](#), [IR09828](#), [IR08767](#), [IR08789](#), [IR09885](#), [IR03716](#), [IR09909](#), [IR08747](#), [IR09861](#), [IR08770](#), [IR08753](#), [IR00001](#), [IR00909](#), [IR09912](#), [IR01159](#), [IR08849](#), [IR09960](#), [IR00614](#), [IR00267](#), [IR00777](#), [IR08836](#), [IR08823](#), [IR01418](#), [IR03505](#), [IR00296](#) and as a product in [RR08593](#), [RR08674](#), [RR09889](#), [RR00362](#), [IR04342](#), [IR04340](#), [IR04341](#), [IR04343](#)).

$$\begin{aligned} \frac{d}{dt}C0001 = & v_{53} + v_{85} + 2 v_{89} + v_{102} + v_{135} + v_{140} + v_{210} + v_{244} - v_{11} - v_{15} - v_{33} \\ & - 2 v_{41} - v_{70} - v_{113} - v_{114} - 199 v_{124} - v_{128} - v_{129} - v_{133} - 1032 v_{143} \\ & - v_{146} - v_{153} - v_{160} - v_{165} - v_{166} - 40 v_{167} - v_{168} - v_{169} - v_{170} - v_{176} \\ & - v_{182} - v_{185} - v_{187} - v_{188} - v_{190} - 7 v_{193} - v_{194} - v_{198} - v_{206} - v_{208} \\ & - v_{215} - 25 v_{221} - v_{225} - v_{226} - v_{227} - v_{229} - v_{230} - 100 v_{234} - v_{236} \\ & - 482 v_{239} - v_{241} - 140 v_{243} - v_{249} - v_{253} - v_{254} - v_{258} - v_{260} - v_{263} \\ & - v_{264} - v_{266} - v_{268} - v_{282} - v_{284} - v_{286} - v_{292} - v_{299} - v_{305} - v_{306} \end{aligned} \quad (886)$$

6.275 Species C9518

Name tRNA(Asp)

Charge 0

This species takes part in three reactions (as a reactant in [IR09894](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9518 = 6 v_{124} + 18 v_{239} - v_{278} \quad (887)$$

6.276 Species C0100

Name D-Glyceraldehyde 3-phosphate

Charge -2

This species takes part in seven reactions (as a reactant in [RR00685](#), [RR01018](#), [RR03769](#), [RR00689](#), [RR00652](#) and as a product in [RR00690](#), [IR00688](#)).

$$\frac{d}{dt}C0100 = v_{21} + v_{242} - v_1 - v_{14} - v_{43} - v_{45} - v_{104} \quad (888)$$

6.277 Species C0300

Name Thiamin

Charge 1

This species takes part in two reactions (as a reactant in [IR01174](#) and as a product in [IR08836](#)).

$$\frac{d}{dt}C0300 = v_{286} - v_{240} \quad (889)$$

6.278 Species C0824

Name S-Acetyldihydrolipoamide

Charge 0

This species takes part in two reactions (as a product in [RR01487](#), [IR00932](#)).

$$\frac{d}{dt}C0824 = v_{108} + v_{142} \quad (890)$$

6.279 Species C0025

Name H2O2

Charge 0

This species takes part in two reactions (as a product in [RR00521](#), [RR09789](#)).

$$\frac{d}{dt}C0025 = v_9 + v_{62} \quad (891)$$

6.280 Species C9472

Name ACP-R (Mpn)

Charge -4

This species takes part in three reactions (as a reactant in [IR09822](#), [IR09821](#) and as a product in [IR09820](#)).

$$\frac{d}{dt}C9472 = v_{112} - v_{117} - v_{195} \quad (892)$$

6.281 Species C0156

Name L-Threonine

Charge 0

This species takes part in five reactions (as a reactant in [IR09955](#), [IR09901](#), [IR08842](#) and as a product in [IR09954](#), [IR08768](#)).

$$\frac{d}{dt}C0156 = 21 v_{143} + v_{169} - 2489 v_{247} - v_{294} - v_{298} \quad (893)$$

6.282 Species C0250

Name CDPcholine

Charge -1

This species takes part in two reactions (as a reactant in [IR09955](#) and as a product in [IR01064](#)).

$$\frac{d}{dt}C0250 = v_{291} - 100 v_{247} \quad (894)$$

6.283 Species C0059

Name Phosphoenolpyruvate

Charge -3

This species takes part in 14 reactions (as a reactant in [IR07938](#), [IR07898](#), [IR04690](#), [IR08786](#), [IR09773](#), [IR08829](#), [IR09792](#), [IR05416](#), [IR08826](#), [IR04740](#), [IR08828](#), [IR09793](#), [IR07969](#) and as a product in [RR00362](#)).

$$\begin{aligned} \frac{d}{dt}C0059 = & v_{102} - v_{120} - v_{131} - v_{137} - v_{152} - v_{162} - v_{163} \\ & - v_{200} - v_{204} - v_{212} - v_{220} - v_{259} - v_{261} - v_{275} \end{aligned} \quad (895)$$

6.284 Species C0501

Name sn-Glycero-3-phosphocholine

Charge 0

This species takes part in two reactions (as a reactant in [RR00662](#) and as a product in [RR09857](#)).

$$\frac{d}{dt}C0501 = v_{41} - v_{70} \quad (896)$$

6.285 Species C0960

Name dUDP

Charge -3

This species takes part in two reactions (as a product in [RR01155](#), [IR04342](#)).

$$\frac{d}{dt}C0960 = v_{65} + v_{135} \quad (897)$$

6.286 Species C0177

Name Thymidine

Charge 0

This species takes part in four reactions (as a reactant in [RR03717](#), [IR09955](#), [IR03714](#) and as a product in [IR03716](#)).

$$\frac{d}{dt}C0177 = v_{236} - v_{77} - 1242 v_{247} - v_{255} \quad (898)$$

6.287 Species C0271

Name Deoxyguanosine

Charge 0

This species takes part in two reactions (as a reactant in [RR03808](#), [RR03810](#)).

$$\frac{d}{dt}C0271 = -v_{27} - v_{34} \quad (899)$$

6.288 Species C9509

Name tRNA(Pro)

Charge 0

This species takes part in three reactions (as a reactant in [IR09896](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9509 = v_{124} + 14 v_{239} - v_{273} \quad (900)$$

6.289 Species C0053

Name L-Serine

Charge 0

This species takes part in six reactions (as a reactant in [IR08835](#), [IR09907](#), [IR09955](#) and as a product in [RR00598](#), [IR09845](#), [IR09954](#)).

$$\frac{d}{dt}C0053 = v_{11} + v_{128} + 21 v_{143} - v_{213} - v_{235} - 3202 v_{247} \quad (901)$$

6.290 Species C9505

Name L-Aspartyl-tRNA(Asp)

Charge 0

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09894](#)).

$$\frac{d}{dt}C9505 = v_{278} - 6 v_{124} - 18 v_{239} \quad (902)$$

6.291 Species C9318

Name Oxidized thioredoxin

Charge 0

This species takes part in five reactions (as a reactant in [RR04369](#) and as a product in [IR04342](#), [IR04340](#), [IR04341](#), [IR04343](#)).

$$\frac{d}{dt}C9318 = v_{135} + v_{140} + v_{210} + v_{244} - v_6 \quad (903)$$

6.292 Species C9507

Name L-Arginyl-tRNA(Arg)

Charge 2

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09904](#)).

$$\frac{d}{dt}C9507 = v_{256} - 2 v_{124} - 13 v_{239} \quad (904)$$

6.293 Species C0122

Name 5,10-Methylenetetrahydrofolate

Charge -1

This species takes part in three reactions (as a reactant in [RR00598](#), [RR01158](#), [RR00771](#)).

$$\frac{d}{dt}C0122 = -v_{11} - v_{50} - v_{56} \quad (905)$$

6.294 Species C9523

Name tRNA(Cys)

Charge 0

This species takes part in two reactions (as a reactant in [IR09903](#) and as a product in [IR09909](#)).

$$\frac{d}{dt}C9523 = 3 v_{239} - v_{251} \quad (906)$$

6.295 Species C9516

Name L-Lysyl-tRNA(Lys)

Charge 2

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09906](#)).

$$\frac{d}{dt}C9516 = v_{150} - 11 v_{124} - 32 v_{239} \quad (907)$$

6.296 Species C0205

Name Lipoamide

Charge 0

This species takes part in two reactions (as a reactant in [IR00932](#) and as a product in [RR09342](#)).

$$\frac{d}{dt}C0205 = v_{99} - v_{142} \quad (908)$$

6.297 Species C0022

Name Acetyl-CoA

Charge -4

This species takes part in three reactions (as a reactant in [RR03546](#), [RR01487](#) and as a product in [RR03544](#)).

$$\frac{d}{dt}C0022 = v_{22} - v_{96} - v_{108} \quad (909)$$

6.298 Species C0045_b

Name Adenosine 3',5'-bisphosphate_b

Charge -4

This species takes part in one reaction (as a product in [IR09198](#)).

$$\frac{d}{dt}C0045_b = v_{132} \quad (910)$$

6.299 Species C0363

Name Cytidine

Charge 0

This species takes part in five reactions (as a reactant in [IR01054](#), [IR09955](#), [IR00269](#) and as a product in [RR08860](#), [IR00267](#)).

$$\frac{d}{dt}C0363 = v_{55} + v_{282} - v_{206} - 503 v_{247} - v_{285} \quad (911)$$

6.300 Species C9513

Name L-Prolyl-tRNA(Pro)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09896](#)).

$$\frac{d}{dt}C9513 = v_{273} - v_{124} - 14 v_{239} \quad (912)$$

6.301 Species C0046

Name CMP

Charge -2

This species takes part in six reactions (as a reactant in [RR00268](#), [IR00267](#) and as a product in [IR09829](#), [IR09827](#), [IR09885](#), [IR00269](#)).

$$\frac{d}{dt}C0046 = v_{139} + v_{189} + 18 v_{234} + v_{285} - v_{74} - v_{282} \quad (913)$$

6.302 Species C0013

Name NH3

Charge 0

This species takes part in six reactions (as a reactant in [RR00106](#), [RR09841](#), [IR03511](#) and as a product in [IR01054](#), [IR01418](#), [IR00296](#)).

$$\frac{d}{dt}C0013 = v_{206} + v_{299} + v_{306} - v_{36} - v_{111} - v_{202} \quad (914)$$

6.303 Species C9479

Name CDP-diacylglycerol (Mpn)

Charge -2

This species takes part in three reactions (as a reactant in [IR09829](#), [IR09827](#) and as a product in [IR09826](#)).

$$\frac{d}{dt}C9479 = v_{250} - v_{139} - v_{189} \quad (915)$$

6.304 Species C0105

Name L-Leucine

Charge 0

This species takes part in five reactions (as a reactant in [IR09898](#), [IR09955](#), [IR08688](#) and as a product in [IR09954](#), [IR08771](#)).

$$\frac{d}{dt}C0105 = 29 v_{143} + v_{187} - v_{126} - 21313 v_{247} - v_{280} \quad (916)$$

6.305 Species C9467

Name L-Methionyl-tRNA(Met)

Charge 1

This species takes part in four reactions (as a reactant in [IR09934](#), [IR09909](#), [IR09797](#) and as a product in [IR09893](#)).

$$\frac{d}{dt}C9467 = v_{183} - 3 v_{124} - 5 v_{239} - v_{302} \quad (917)$$

6.306 Species C0006

Name NADP+

Charge -3

This species takes part in five reactions (as a reactant in [RR00771](#), [IR09271](#), [IR09955](#) and as a product in [RR04369](#), [IR00074](#)).

$$\frac{d}{dt}C0006 = v_6 + v_{219} - v_{56} - v_{205} - 100 v_{247} \quad (918)$$

6.307 Species C0085

Name Tetrahydrofolate

Charge -1

This species takes part in four reactions (as a reactant in [RR00590](#), [IR00596](#) and as a product in [RR00598](#), [IR09797](#)).

$$\frac{d}{dt}C0085 = v_{11} + v_{302} - v_{72} - v_{191} \quad (919)$$

6.308 Species C9526

Name tRNA(Ser)

Charge 0

This species takes part in three reactions (as a reactant in [IR09907](#) and as a product in [IR09934](#), [IR09909](#)).

$$\frac{d}{dt}C9526 = 5 v_{124} + 21 v_{239} - v_{235} \quad (920)$$

6.309 Species C0630

Name Deamino-NAD+

Charge -2

This species takes part in two reactions (as a reactant in [IR03511](#) and as a product in [RR03873](#)).

$$\frac{d}{dt}C0630 = v_{42} - v_{202} \quad (921)$$

6.310 Species C0003

Name NAD+

Charge -1

This species takes part in twelve reactions (as a reactant in [RR00685](#), [RR00667](#), [RR03544](#), [RR00452](#), [RR00275](#), [RR01236](#), [RR00590](#), [RR00404](#), [RR09342](#), [IR00074](#) and as a product in [RR09889](#), [IR03511](#)).

$$\frac{d}{dt}C0003 = 2 v_{89} + v_{202} - v_1 - v_{19} - v_{22} - v_{32} - v_{38} - v_{54} - v_{72} - v_{82} - v_{99} - v_{219} \quad (922)$$

6.311 Species C0094

Name CDP

Charge -3

This species takes part in three reactions (as a reactant in [IR04690](#), [IR04343](#) and as a product in [RR00268](#)).

$$\frac{d}{dt}C0094 = v_{74} - v_{137} - v_{244} \quad (923)$$

6.312 Species C0146

Name Thymine

Charge 0

This species takes part in two reactions (as a product in [RR03717](#), [IR09855](#)).

$$\frac{d}{dt}C0146 = v_{77} + v_{277} \quad (924)$$

6.313 Species C0051

Name CTP

Charge -4

This species takes part in four reactions (as a reactant in [IR09884](#), [IR09826](#), [IR01064](#) and as a product in [IR04690](#)).

$$\frac{d}{dt}C0051 = v_{137} - 18 v_{167} - v_{250} - v_{291} \quad (925)$$

6.314 Species C0009

Name Orthophosphate

Charge -2

This species takes part in 63 reactions (as a reactant in [RR00685](#), [RR03709](#), [RR01475](#), [RR03810](#), [RR01184](#), [RR08593](#), [RR03717](#), [RR01416](#), [RR01052](#), [RR03546](#), [IR09590](#), [IR09955](#) and as a product in [RR03649](#), [IR08663](#), [IR09934](#), [IR09845](#), [IR08746](#), [IR09956](#), [IR09954](#), [IR08755](#), [IR03510](#), [IR08754](#), [IR00648](#), [IR09884](#), [IR09860](#), [IR08768](#), [IR09958](#), [IR09957](#), [IR09913](#), [IR08984](#), [IR08771](#), [IR08659](#), [IR08824](#), [IR00596](#), [IR09961](#), [IR08738](#), [IR09824](#), [IR07876](#), [IR09881](#), [IR09959](#), [IR08769](#), [IR09828](#), [IR08767](#), [IR08789](#), [IR03716](#), [IR09909](#), [IR08747](#), [IR09861](#), [IR08770](#), [IR08753](#), [IR00001](#), [IR00909](#), [IR09912](#), [IR01159](#), [IR08849](#), [IR09960](#), [IR00614](#), [IR00267](#), [IR00777](#), [IR08836](#), [IR03832](#), [IR08823](#), [IR03505](#)).

$$\begin{aligned} \frac{d}{dt}C0009 = & v_{61} + v_{113} + 199 v_{124} + v_{128} + v_{129} + v_{133} + 688 v_{143} + v_{146} + v_{153} \\ & + v_{160} + v_{165} + 40 v_{167} + v_{168} + v_{169} + v_{170} + v_{176} + v_{182} + v_{185} \\ & + v_{187} + v_{188} + 2 v_{190} + v_{191} + 7 v_{193} + v_{194} + v_{208} + v_{215} + 25 v_{221} \\ & + v_{225} + v_{226} + v_{227} + v_{229} + v_{230} + v_{236} + 826 v_{239} + v_{241} + 140 v_{243} \\ & + v_{249} + v_{253} + 2 v_{254} + v_{258} + v_{260} + v_{263} + v_{264} + v_{266} + v_{268} \\ & + v_{282} + v_{284} + v_{286} + v_{290} + v_{292} + v_{305} - v_1 - v_{12} - v_{13} \\ & - v_{34} - v_{37} - v_{53} - v_{77} - v_{79} - v_{81} - v_{96} - v_{141} - 41537 v_{247} \end{aligned} \quad (926)$$

6.315 Species C0408

Name Deoxyuridine

Charge 0

This species takes part in four reactions (as a reactant in [RR01416](#), [IR01156](#) and as a product in [IR01159](#), [IR01418](#)).

$$\frac{d}{dt}C0408 = v_{263} + v_{299} - v_{79} - v_{300} \quad (927)$$

6.316 Species C0038

Name GTP

Charge -4

This species takes part in three reactions (as a reactant in [RR03680](#), [IR09884](#) and as a product in [IR07938](#)).

$$\frac{d}{dt}C0038 = v_{120} - v_{78} - 28 v_{167} \quad (928)$$

6.317 Species C0004

Name NADH

Charge -2

This species takes part in eleven reactions (as a reactant in [RR09889](#), [IR00075](#) and as a product in [RR00685](#), [RR00667](#), [RR03544](#), [RR00452](#), [RR00275](#), [RR01236](#), [RR00590](#), [RR00404](#), [RR09342](#)).

$$\frac{d}{dt}C0004 = v_1 + v_{19} + v_{22} + v_{32} + v_{38} + v_{54} + v_{72} + v_{82} + v_{99} - 2 v_{89} - v_{265} \quad (929)$$

6.318 Species C0317

Name L-Isoleucine

Charge 0

This species takes part in five reactions (as a reactant in [IR08692](#), [IR09905](#), [IR09955](#) and as a product in [IR09954](#), [IR08767](#)).

$$\frac{d}{dt}C0317 = 22 v_{143} + v_{229} - v_{121} - v_{161} - 1858 v_{247} \quad (930)$$

6.319 Species C9533

Name L-Tryptophanyl-tRNA(Trp)

Charge 1

This species takes part in two reactions (as a reactant in [IR09909](#) and as a product in [IR09900](#)).

$$\frac{d}{dt}C9533 = v_{283} - 3 v_{239} \quad (931)$$

6.320 Species C9503

Name L-Phenylalanyl-tRNA(Phe)

Charge 1

This species takes part in three reactions (as a reactant in [IR09934](#), [IR09909](#) and as a product in [IR09908](#)).

$$\frac{d}{dt}C9503 = v_{203} - 5 v_{124} - 15 v_{239} \quad (932)$$

6.321 Species C0132

Name L-Homocysteine

Charge 0

This species takes part in two reactions (as a reactant in [IR09865](#) and as a product in [IR03514](#)).

$$\frac{d}{dt}C0132 = v_{114} - v_{180} \quad (933)$$

6.322 Species C9536

Name Protein (Mpn)

Charge 5

This species takes part in five reactions (as a reactant in [IR09956](#), [IR09954](#), [IR09961](#), [IR09955](#) and as a product in [IR09909](#)).

$$\frac{d}{dt}C9536 = v_{239} - v_{133} - v_{143} - v_{193} - 16049 v_{247} \quad (934)$$

6.323 Species C0781

Name Thiamin monophosphate

Charge -1

This species takes part in two reactions (as a reactant in [IR00341](#) and as a product in [IR01174](#)).

$$\frac{d}{dt}C0781 = v_{240} - v_{231} \quad (935)$$

6.324 Species C0242

Name Uridine

Charge 0

This species takes part in five reactions (as a reactant in [RR01052](#), [IR00615](#), [IR09955](#) and as a product in [IR01054](#), [IR00614](#)).

$$\frac{d}{dt}C0242 = v_{206} + v_{268} - v_{81} - v_{233} - 2541 v_{247} \quad (936)$$

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